

Chapter 9

Carbene and Carbyne Complexes

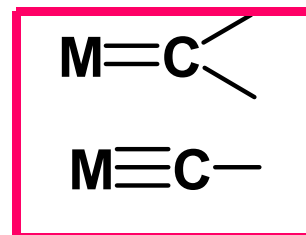
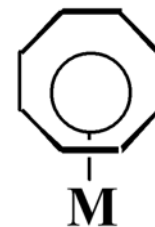
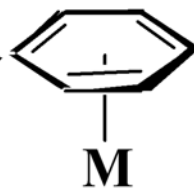
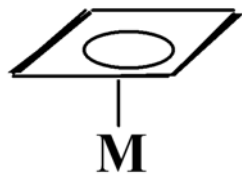
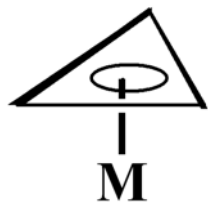
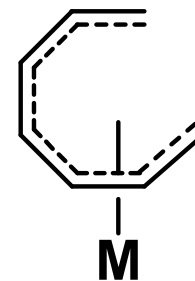
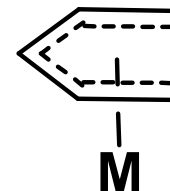
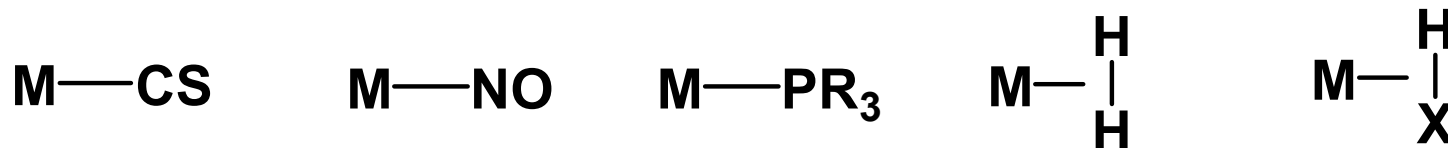
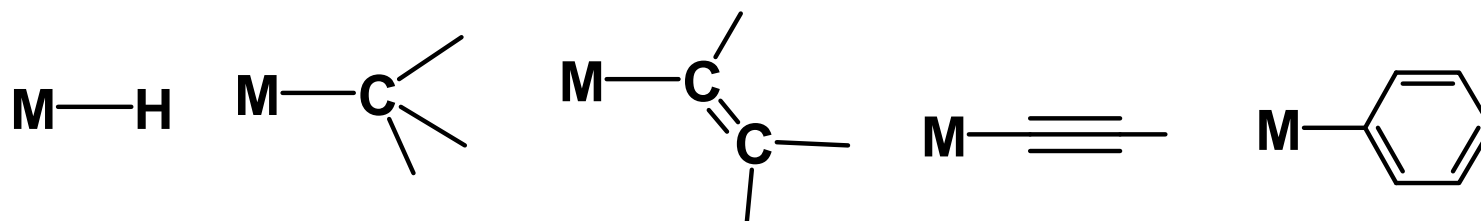
Outline

- A. Introduction to M-C multiple bonds**
- B. Common synthetic routes to carbene complexes**
- C. Reactivity of carbene complexes**
- D. Common synthetic routes to carbyne complexes**
- E. Reactivity of carbyne complexes**

References and suggested readings:

1. *The Organometallic Chemistry of the Transition Metals*, Robert H. Crabtree, 6th Edition, **2014**, Chapter 11.
2. *Organotransition Metal Chemistry*, Akio Yamamoto, **1986**, 124-128.
3. Dotz, K. H.; Fisher, H.; Hofmann, P.; Kreissl, F. R. Schubert, U.; Weiss, K. *Transition Metal Carbene Complexes*, Verlag: Weinheim, **1983**.
4. Kim, H.P.; Angelici, R. J. *Adv. Organomet. Chem.* **1987**. 27, 51.

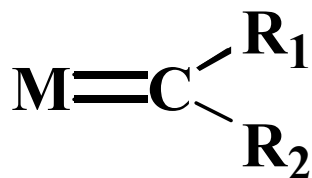
Common organometallic ligands



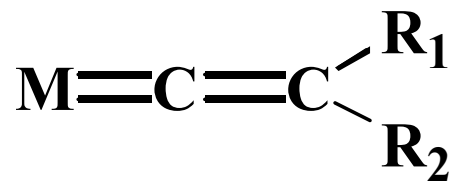
A. General introduction to metal-carbon multiple bonds

There are many organometallic compounds containing metal carbon multiple bonds. e.g.

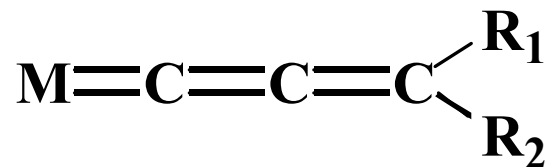
Carbene (or alkylidene) complexes



Vinylidene (or alkenylidene) complexes



Allenylidene complexes

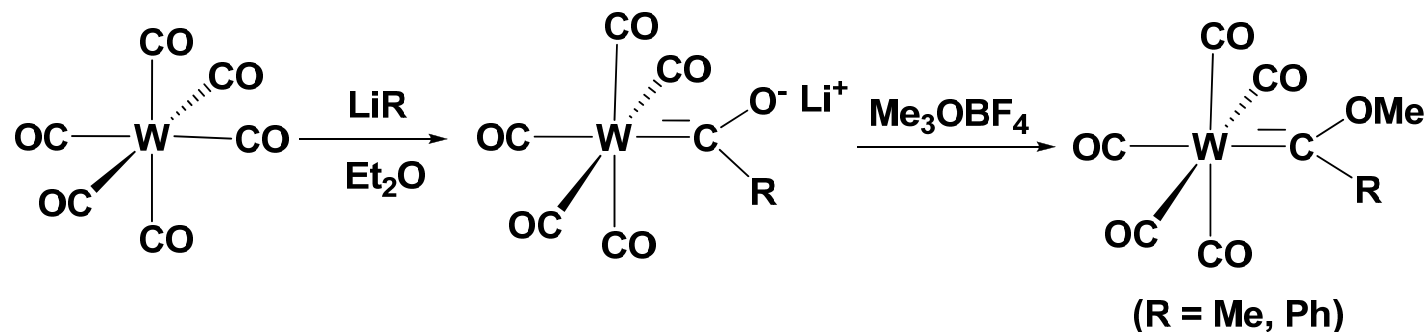


Carbyne (or alkylidyne) complexes



First Metal Carbene Complex (*E. O. Fischer, 1964*)

Addition of an RLi to M-CO



A new metal-carbon bond

(Fischer, E. O. and Maasböl, A. *Angew. Chem. Int. Ed.*, **1964**, 3, 580.)



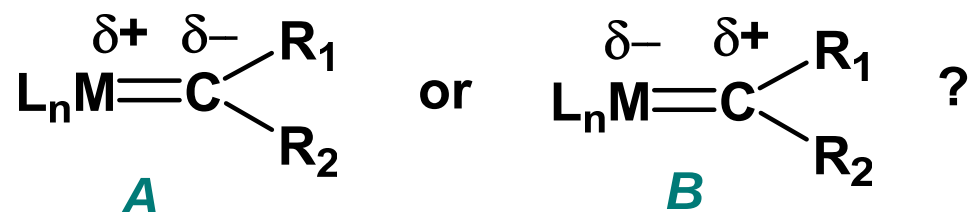
Ernst Otto Fischer

The Nobel Prize in Chemistry 1973 was awarded jointly to *Ernst Otto Fischer* and *Geoffrey Wilkinson* "for their pioneering work, performed independently, on the chemistry of the organometallic, so called *sandwich compounds*".



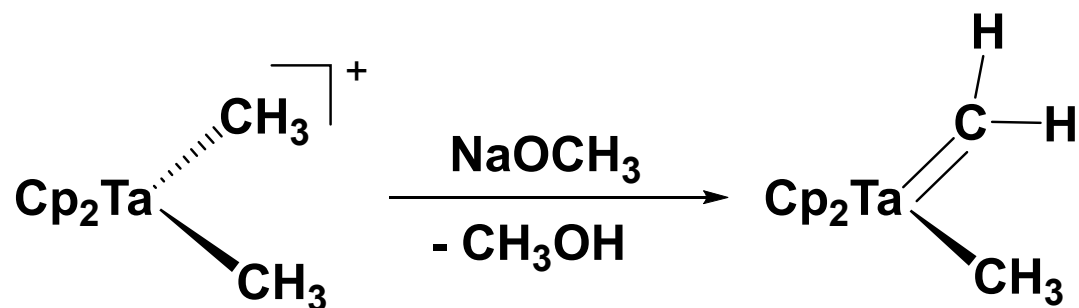
Geoffrey Wilkinson

Question 1.



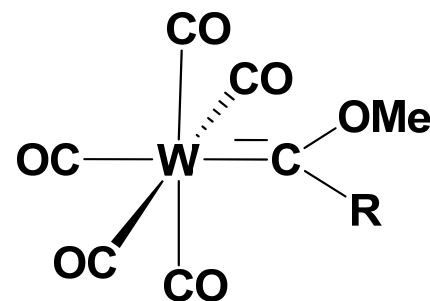
Answer: *Both are possible, depending on metals and ligands.*

A is Schrock carbenes and B is Fischer carbenes.



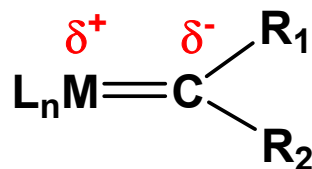
(Schrock, R. R., et al. *J. Am. Chem. Soc.*, **1975**, 97, 6577.)

Schrock carbene



Fischer carbene

Schrock carbenes

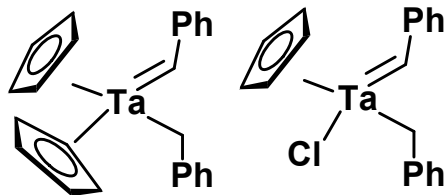


non- π -donor substituent $\rightarrow$

$L_nM=C$

alkyl, π donor or non- π -acceptor ligands $\rightarrow$

early TM in high O.S.



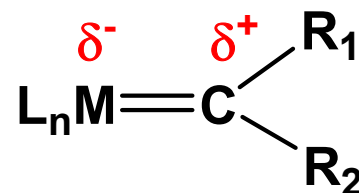
Central M in higher oxidation state;
Carry C or H substituents at carbene Carbon;
Normally with non π -acceptor ligands.

The carbene carbon will be attacked by Nu^- or E^+ ?

E^+

If no heteroatom are bound directly to the carbene carbon, the term “alkylidene complex” is also used.

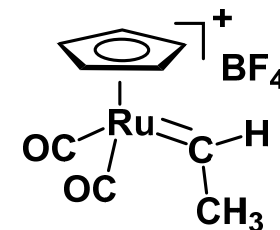
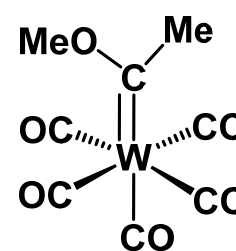
Fisher carbenes



$L_nM=C$

π acceptor e.g. CO $\rightarrow$

late TM low O.S.



Central M in low oxidation state;
Bear heteroatoms at carbene carbon;
Normally with π -acceptor ligands.

The carbene carbon will be attacked by Nu^- or E^+ ?

Nu^-

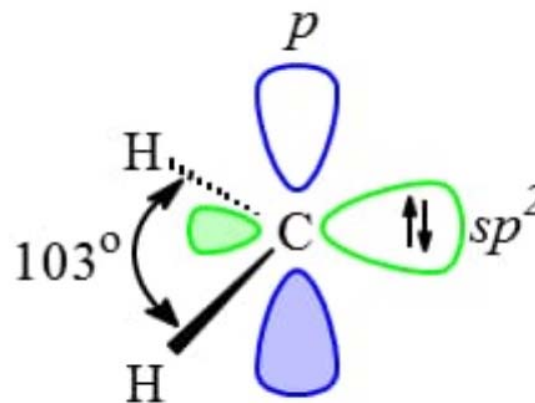
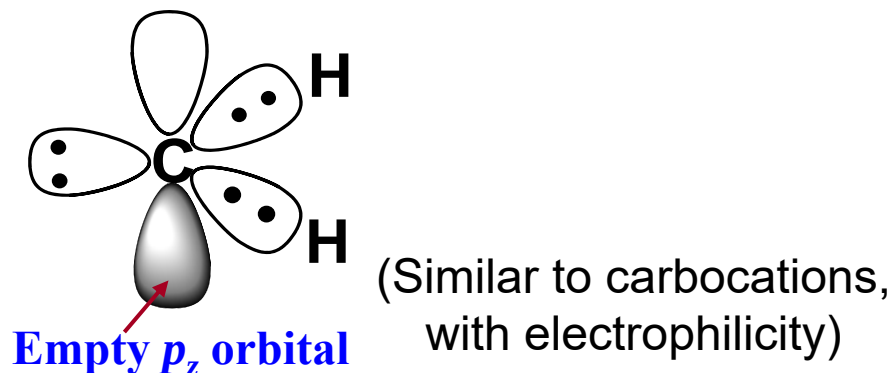
The classification is justified by their different reactivities!

Fischer- and Schrock-Type Carbene Complexes

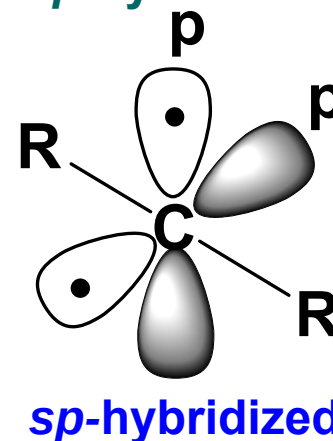
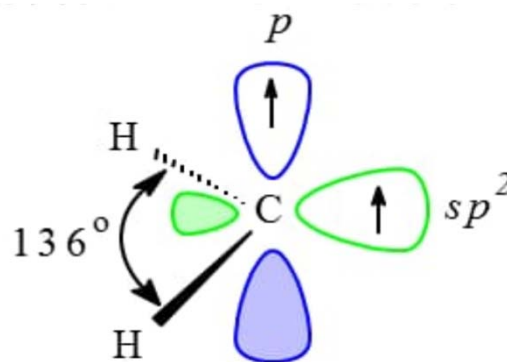
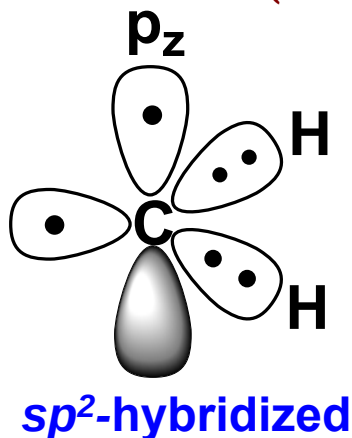
Characteristic	Fischer-Type	Schrock-Type (Alkylidenes)
Typical metal [ox. state]	Middle to late transition metal [Fe(0), Mo(0), Cr(0)]	Early transition metal [Ti(IV), Ta(V)]
Substituents attached to C _{carbene}	At least one electronegative heteroatom, e.g., O or N	H or alkyl
Typical ligands also attached to metal	Good π acceptor, e.g., CO	Good σ or π donor, e.g., Cp, Cl, alkyl
Electron count	18 e ⁻	10-18 e ⁻
Typical chemical behavior	Nucleophile attacks at C _{carbene}	Electrophile attacks at C _{carbene}
Ligand type	Normally with π -acceptor ligands	Normally with non π -acceptor ligands

Bonding in free carbenes: CR_2

Singlet Carbene (sp^2 hybridized)



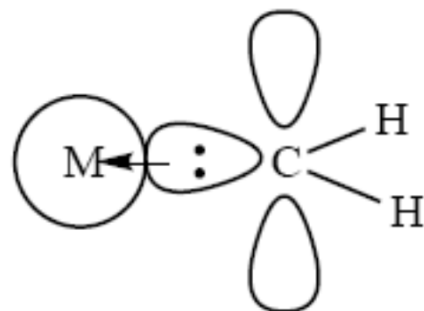
Triplet Carbene (*Diradical, Ground State*) (can be sp^2 or sp hybridized)



Most carbenes are in non linear triplet ground states, except for dihalogenated carbenes and carbenes attached to nitrogen, oxygen, or sulfur atoms, which may have a singlet state as their main form of ground state due to the contribution of electrons from the lone pair of these donating substituents to the empty π -orbitals of the singlet carbenes, thereby increasing their stability and lowering the energy of singlet states compared to triplet states.

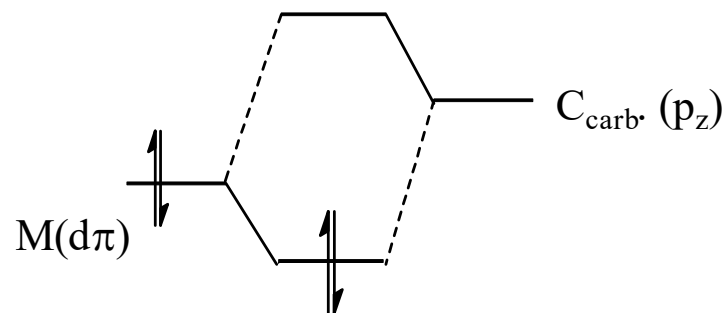
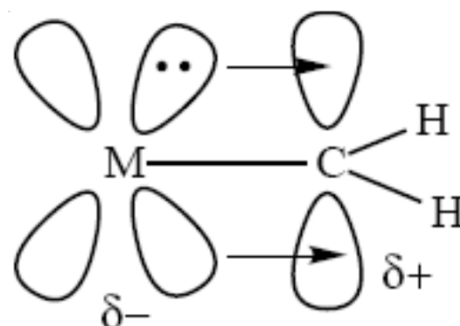
Bonding in metal carbene complexes

$M \leftarrow L$ σ -Bonding

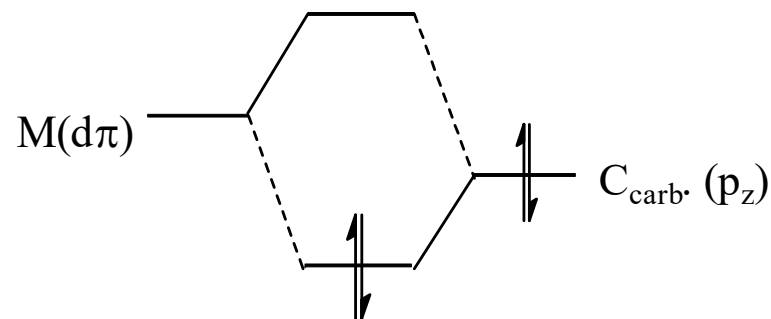
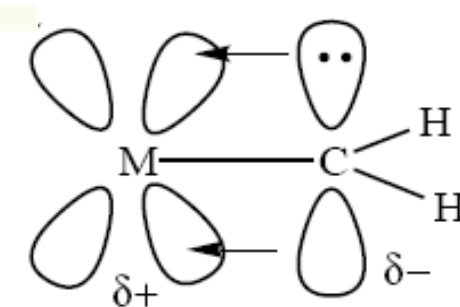


π -Bonding: *The relative energies of the $M(d\pi)$ and The $C_{carb.}(p_z)$ orbitals control the electrophilic or nucleophilic character of the carbene.*

(a) $M(d\pi)$ levels are lower in energy, a **Fischer carbene** will result.

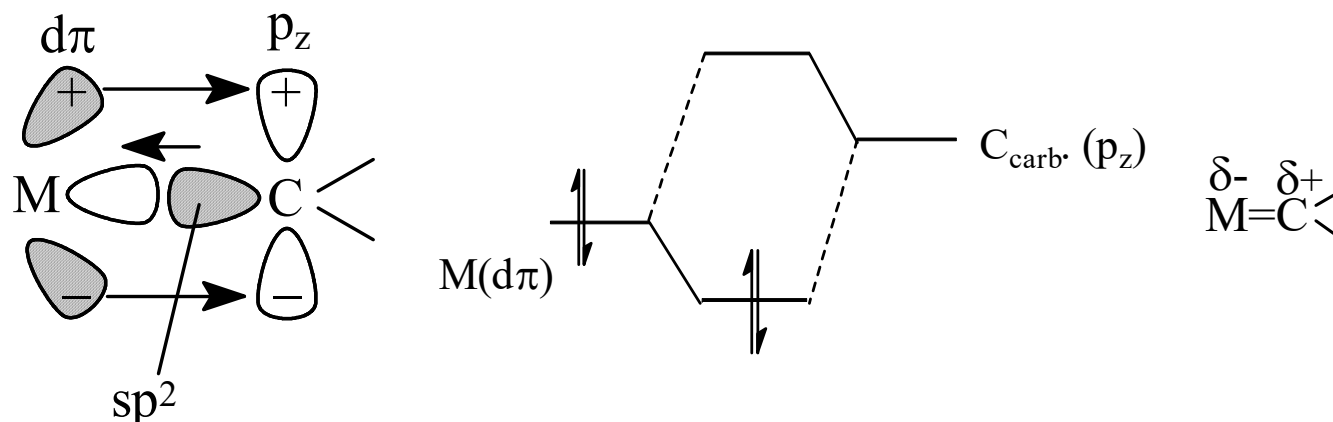


(b) $M(d\pi)$ levels are higher in energy, a **Schrock carbene** will result.



Overall bonding in metal carbene complexes

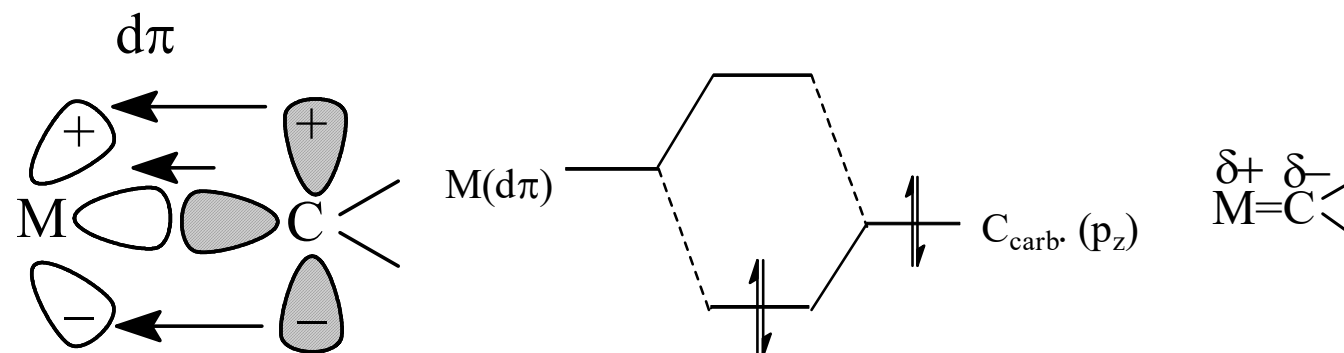
(a) Fischer Carbene



Late Metal: more electronegative, lower $M(d\pi)$, π -acid ligand will stabilize $M(d\pi)$, π -electrons remain on $M(d\pi)$, low oxidation State, limited $M(d\pi)$ - C_{carb} backdonation.

Fischer carbene ligand is considered as a CO-like neutral 2e ligand (L-type), but is a stronger σ -donor and a weaker π -acceptor than CO.

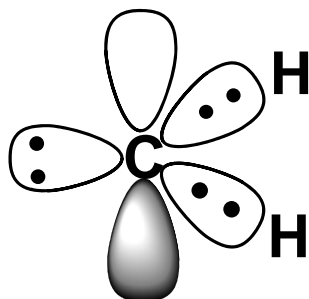
(b) Schrock Carbene



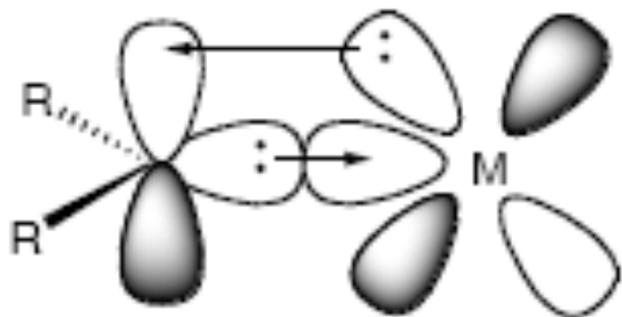
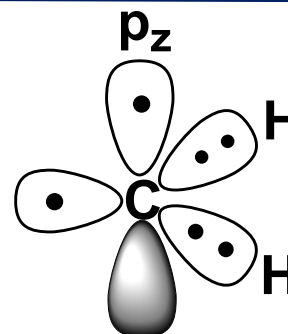
Early Metal: higher $M(d\pi)$, π -electrons move from $M(d\pi)$ to C_{carb} to give a CR_2^{2-} ligand, M is oxidized, high oxidation State. Schrock carbene ligand is considered as a 4e ligand X_2 -type ligand, and is a σ -donor and a π -donor ligand.

Bonding in metal carbenes

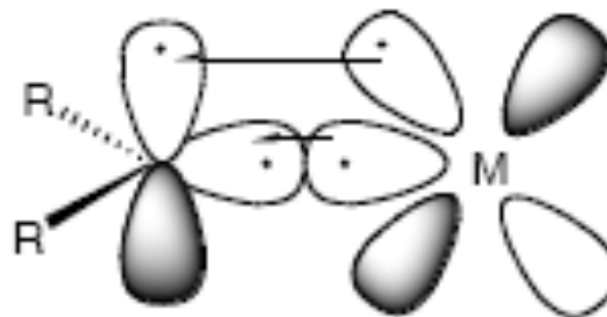
Singlet Carbene
(sp^2 hybridized)



Triplet Carbene
(sp^2 hybridized)



Electrophilic (Fischer) carbene



Nucleophilic (Schrock) carbene

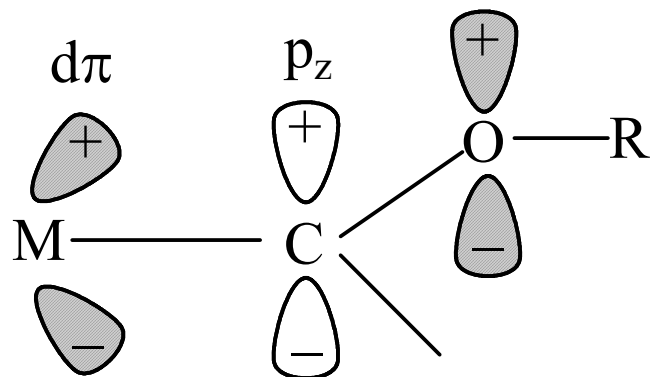


Fischer carbenes can be considered as neutral 2 e ligands (L-type) derived from singlet carbenes.

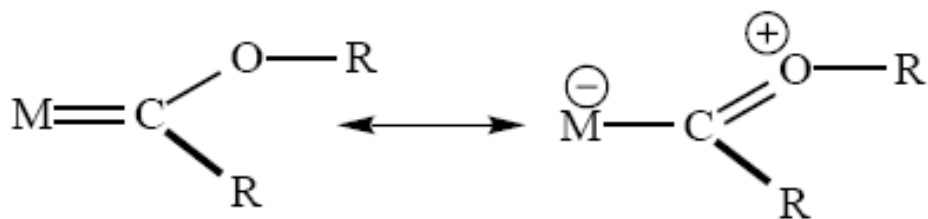
Schrock carbenes are formally considered as X_2 -type ligands derived from triplet carbenes.

How electron-deficient carbene (Fischer Carbene) is stabilized?

- By two lone pairs

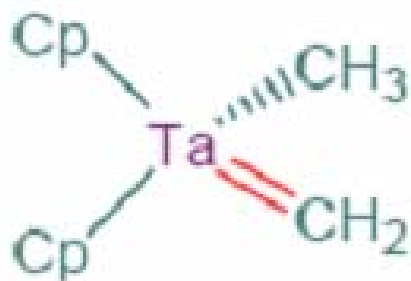


- Also resonance stabilisation



Oxidation states of carbene complexes

What are the oxidation states of M in the following complexes?

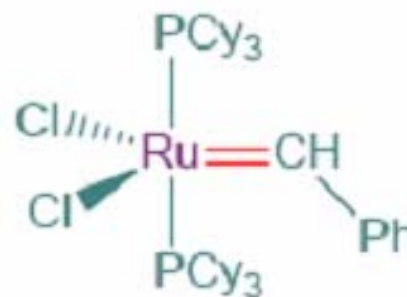


$R_2C:\rightarrow M$

Ta(III)

$R_2C=M$

Ta(V)



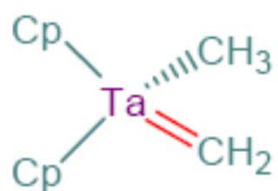
Ru(II)

Ru(IV)

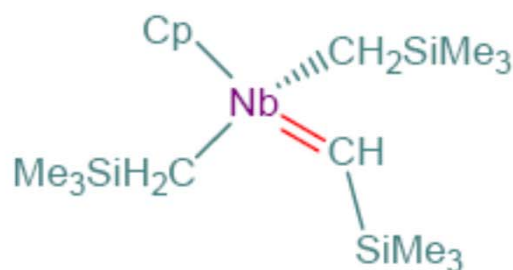
Can be ambiguous !

Oxidation states of carbene complexes

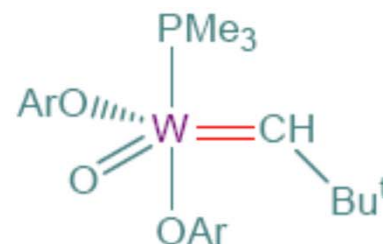
Note that the metal-carbon bond can be shown as double ($M=C$), single ($M-C$, or MC), or dative ($M\leftarrow C$) bond, depending on personal preferences.



18 e⁻, Ta (V)

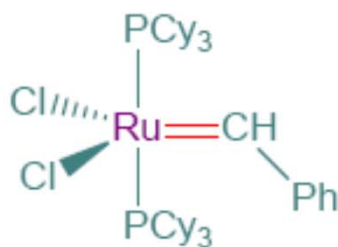


14 e⁻, Nb (V)

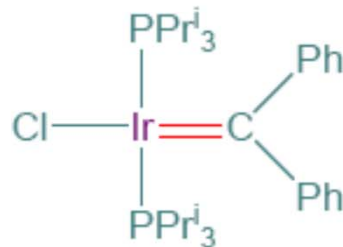


14 e⁻, W (VI)

With the less electronegative early to middle transition metals (groups 4 – 7) Schrock carbenes are usually treated as 2⁻ ligands.



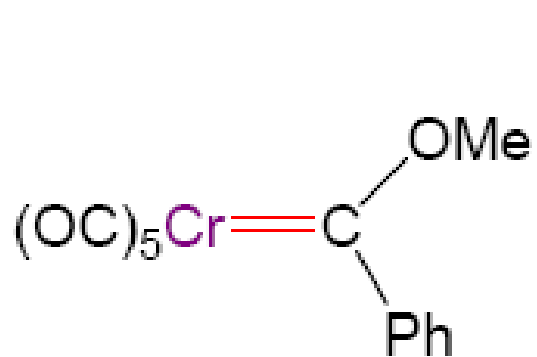
16 e⁻, Ru (II)



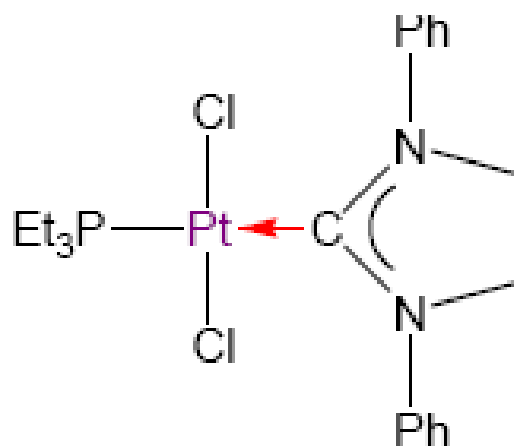
16 e⁻, Ir (I)

With the late transition metals, carbenes are best viewed as *neutral* 2e⁻ σ-donors ligands!

$C(OR)R$ or $C(NR_2)_2$ carbenes are viewed as *neutral* $2e^-$ σ -donors ligands

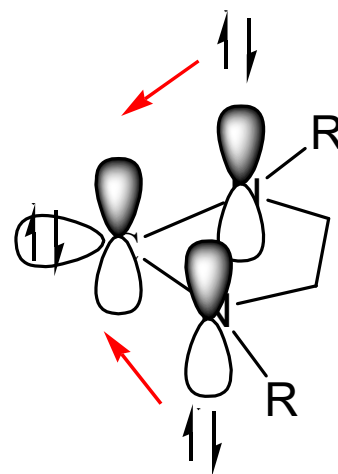
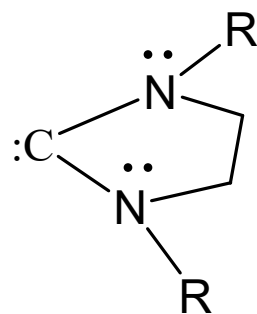
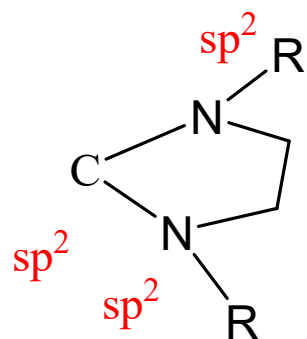


18 e^- , Cr(0)



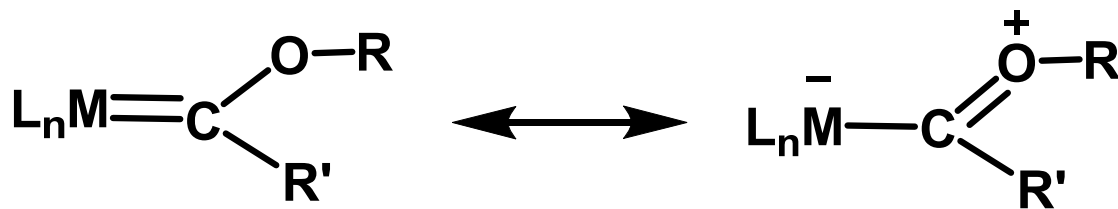
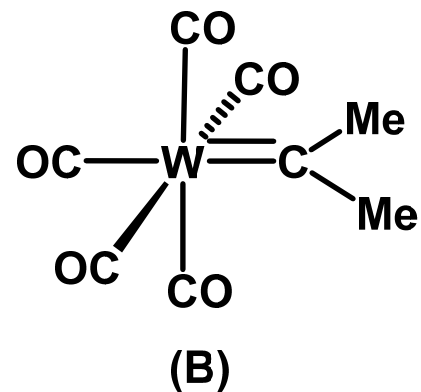
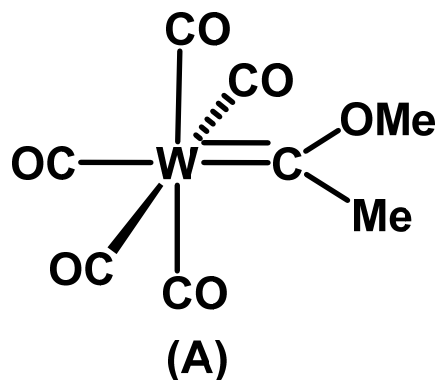
16 e^- , Pt (II)

N-heterocyclic carbenes are very good σ -donors, like trialkylphosphines, PR_3 . There is practically no M-C π -bonding in such complexes.



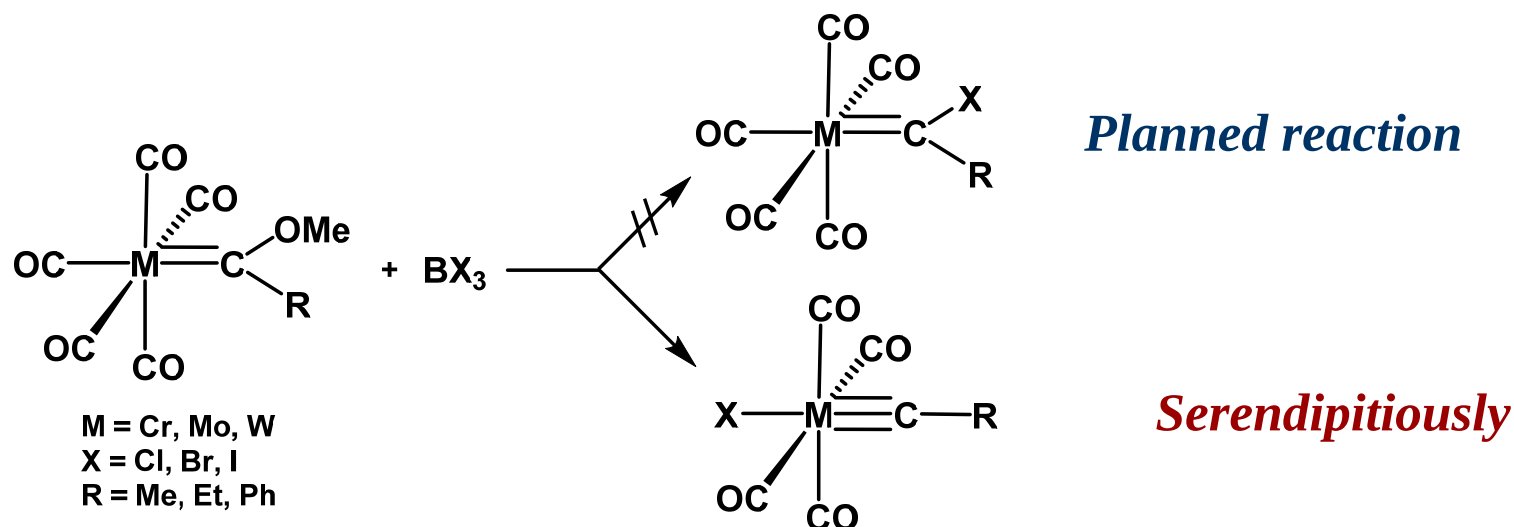
Practice Problem

Which one is thermodynamically more stable?



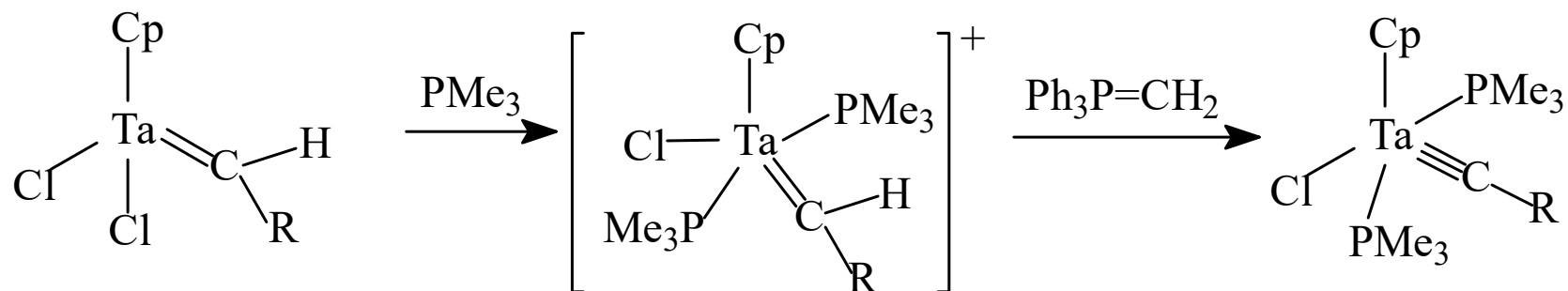
(A). Due to the resonance effect, heteroatom can stabilize the carbene form.

First Metal Carbyne Complex (*E. O. Fischer, 1973*)



(Fischer, E. O. and Maasböl, A. *Angew. Chem. Int. Ed.*, **1973**, 12, 564.)

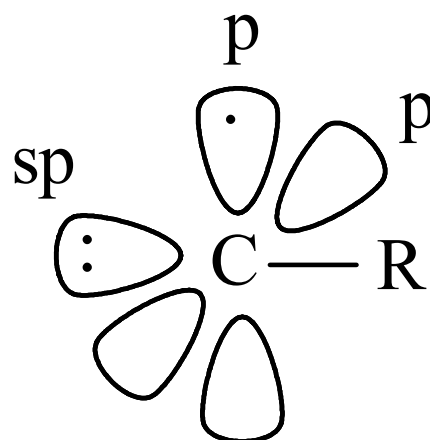
Schrock Carbyne Complex



(Schrock, R. R., *et al. J. Am. Chem. Soc.*, **1975**, 97, 6577.)

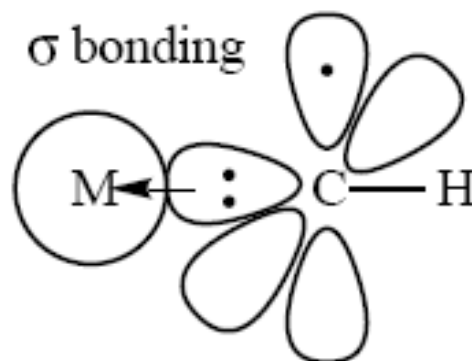
Bonding in metal carbyne complexes:

Carbyne Ligand:



(doublet carbyne)

$M \leftarrow L$ σ -Bonding

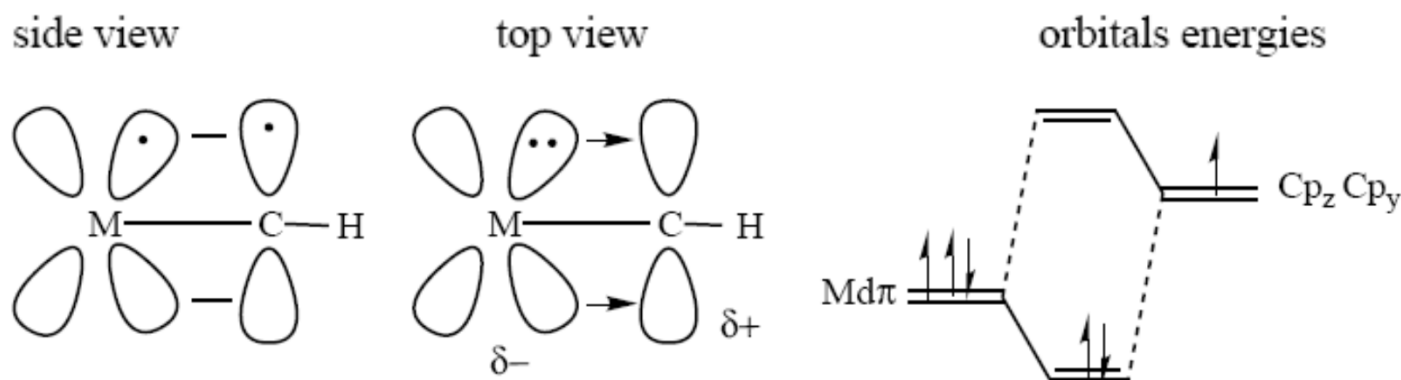


π -Bonding: *The relative energies of the $M(d\pi)$ and the $C(p_y$ and $p_z)$ orbitals control the Fisher- or Schrock-type character of the carbyne.*

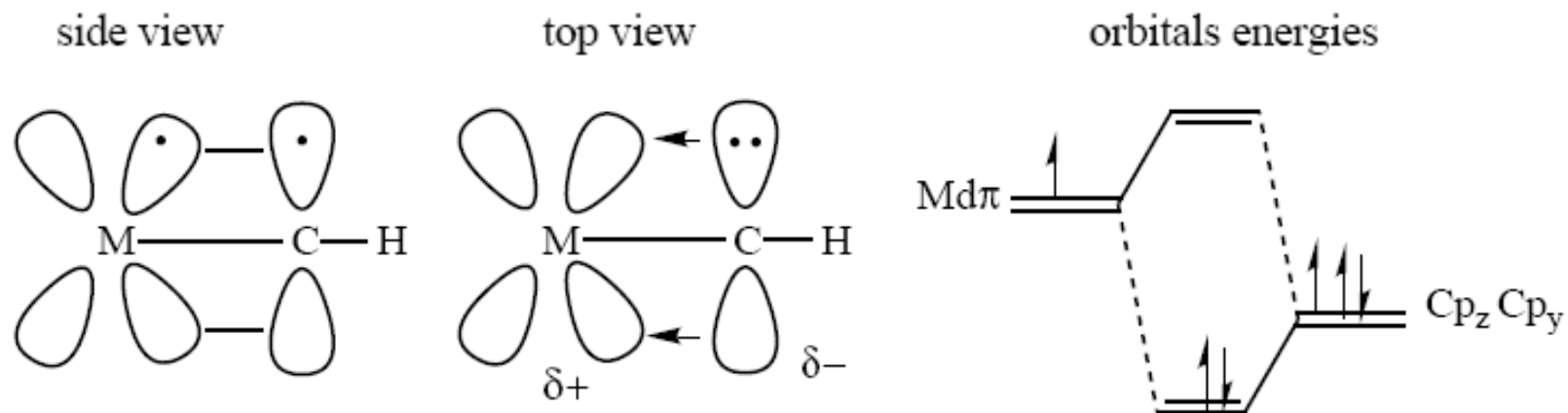
Bonding in metal carbyne complexes:

π -Bonding: *The relative energies of the $M(d\pi)$ and the $C(p_y$ and $p_z)$ orbitals control the Fisher- or Schrock-type character of the carbyne.*

(a) If the $M(d\pi)$ levels are lower in energy, a **Fischer carbyne** will result (CR^- , 4e).



(b) If the $M(d\pi)$ levels are higher in energy, a **Schrock carbyne** will result (CR^{3-} , 6e).

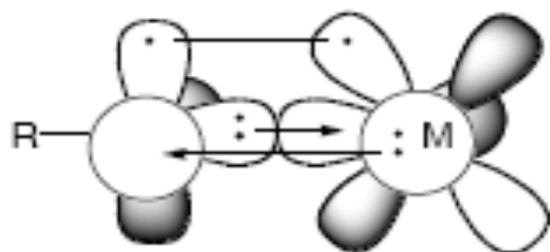




Answer: The carbyne carbon is usually negatively charged.

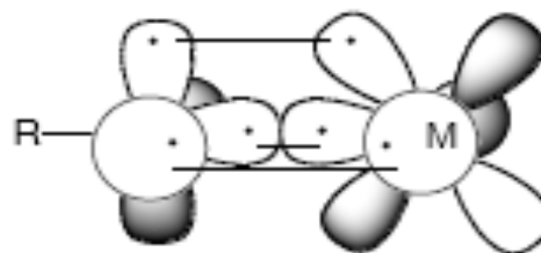
The distinction between **Fischer** and **Schrock carbynes** is not nearly as distinct as in the case of carbenes.

Bonding in carbynes:



Electrophilic (Fischer) carbyne

(Derived from a **doublet carbyne**)



Nucleophilic (Schrock) carbyne

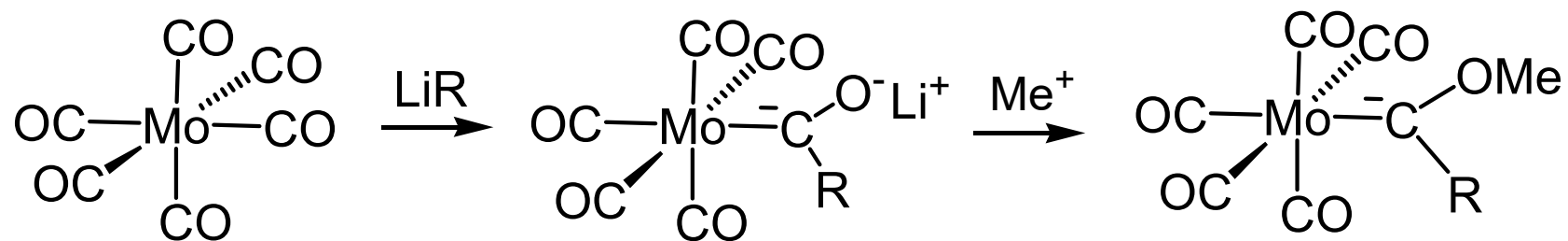
(Derived from a **quartet carbyne**)

Like carbenes, electrophilic (Fischer) and nucleophilic (Schrock) extremes can be envisioned for the bonding model.

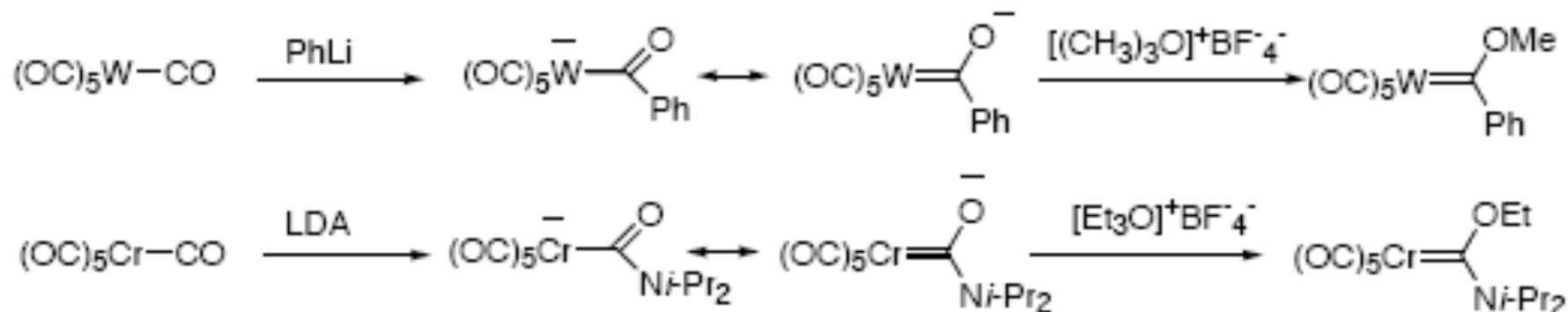
The electrophilic carbyne complex is derived from a **doublet carbyne**, while the nucleophilic complex is derived from a **quartet carbyne**.

B. Common synthetic routes to carbene complexes

(1). From metal carbonyls, e.g.

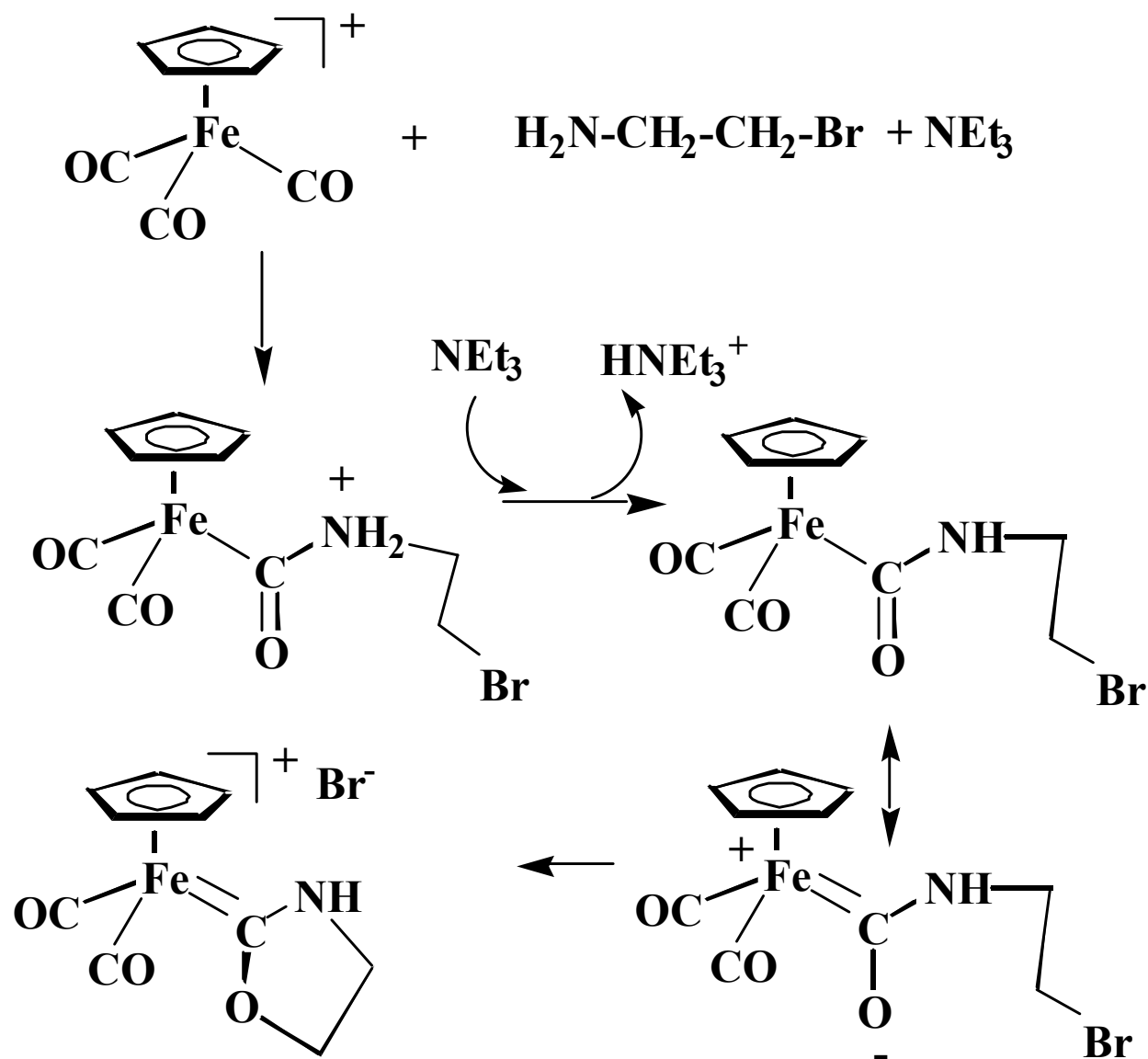


Me^+ : e.g. MeI , Me_3OBF_4

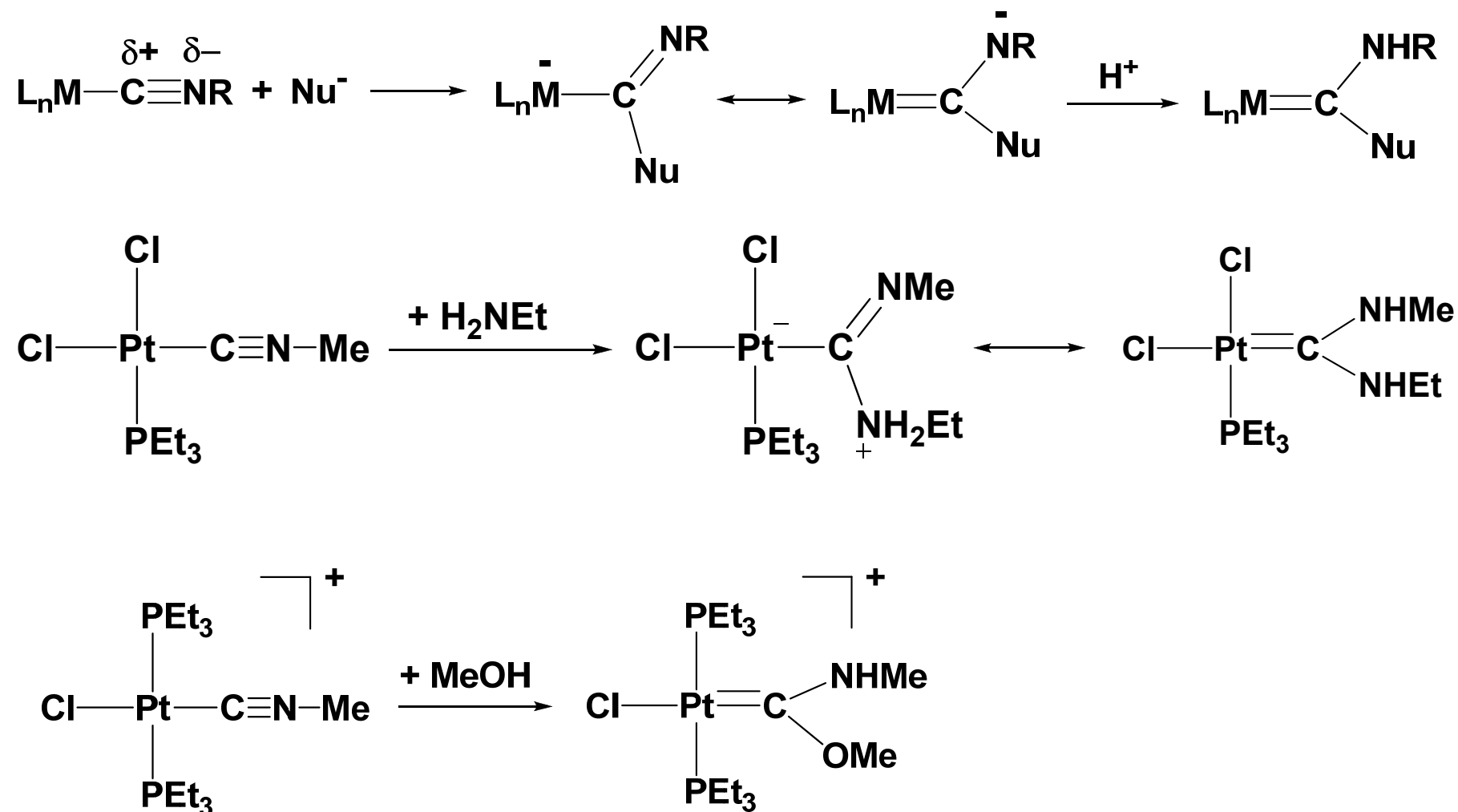


LDA = $\text{LiN}(i\text{-Pr})_2$

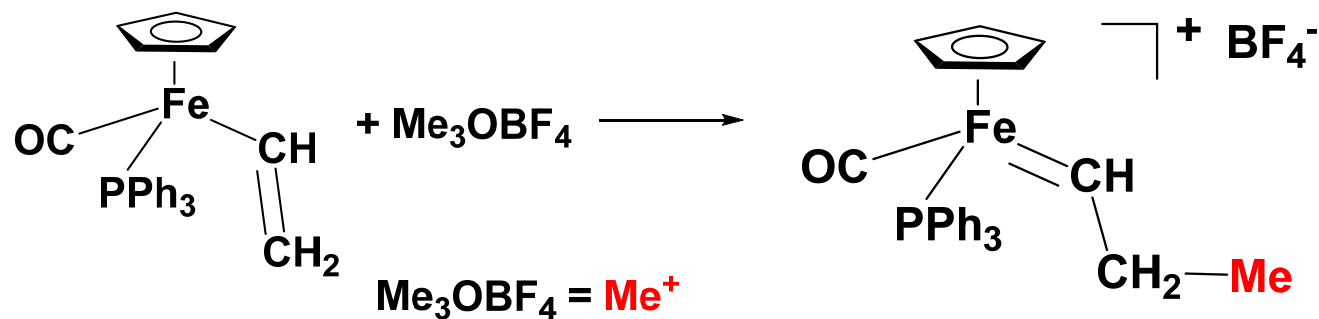
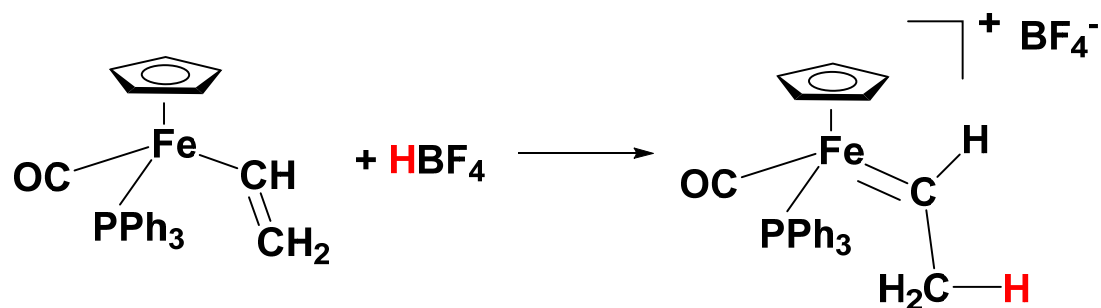
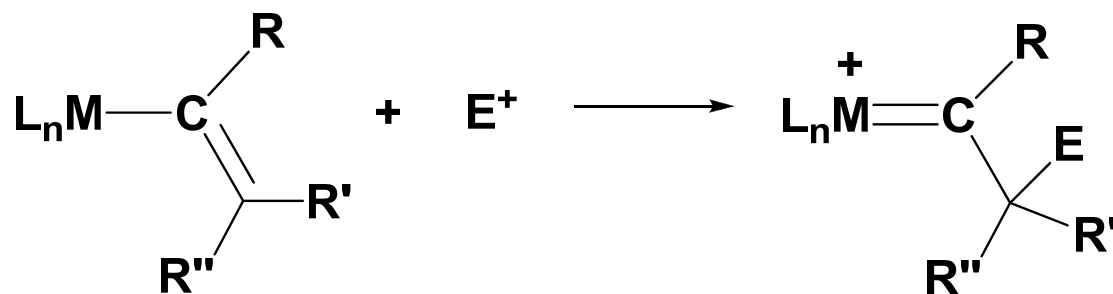
In some cases, a weak nucleophile can also attack CO, especially those in cationic complexes, e.g.



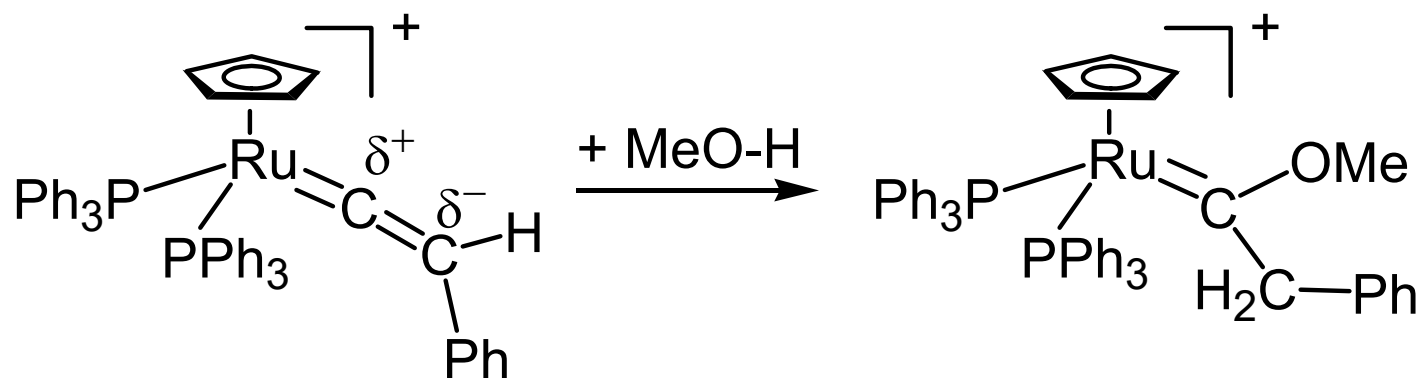
(2) From isonitrile complexes.



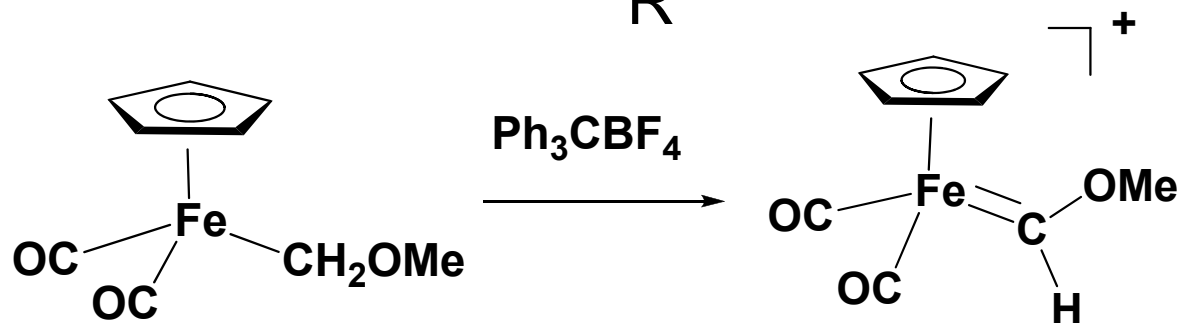
(3) From $L_nM-CR=CR'R''$ (Electrophilic attack of M-Vinyl Complex)

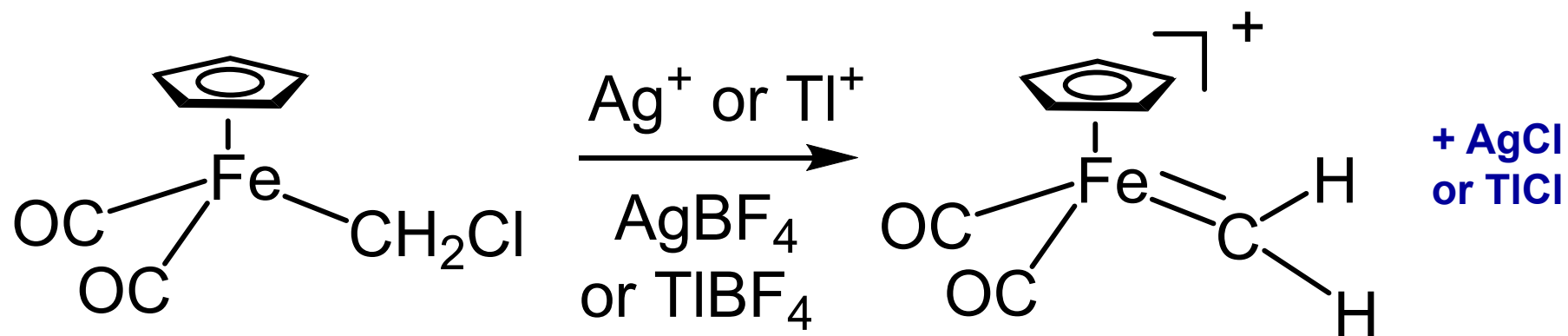
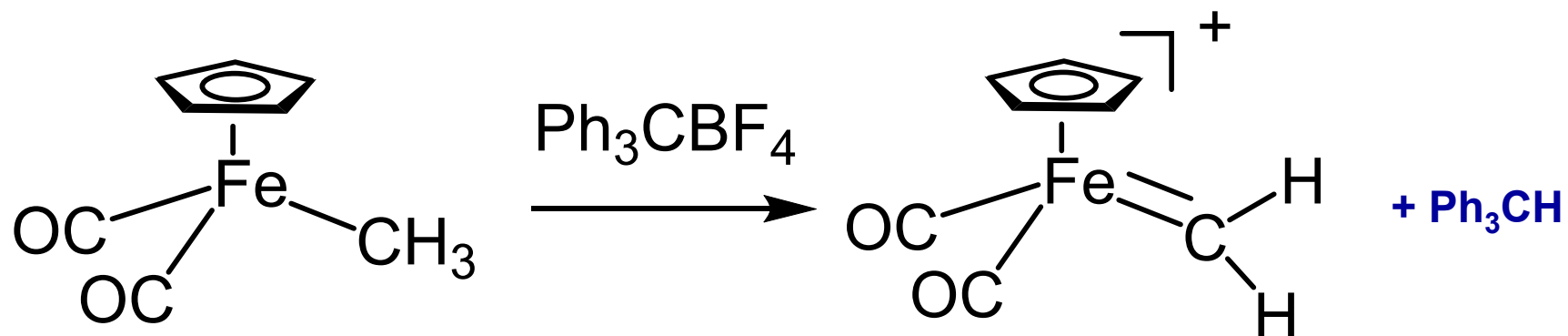


(4) From $L_nM=C=CRR'$ (Nucleophilic addition of M-Vinylidene Complex)

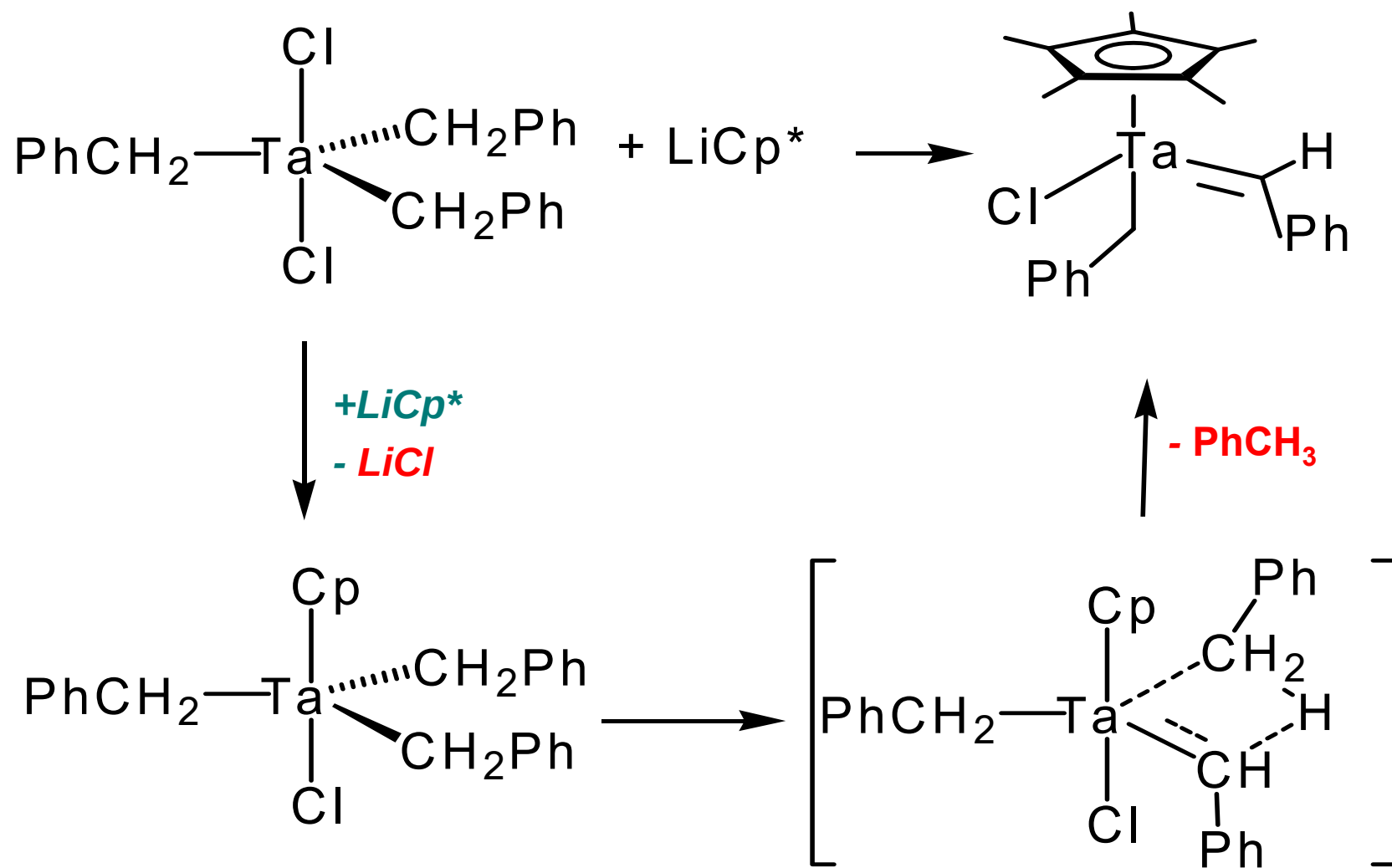


(5) From $LM-CXR_2$ (Electrophilic abstraction of M-alkyl)

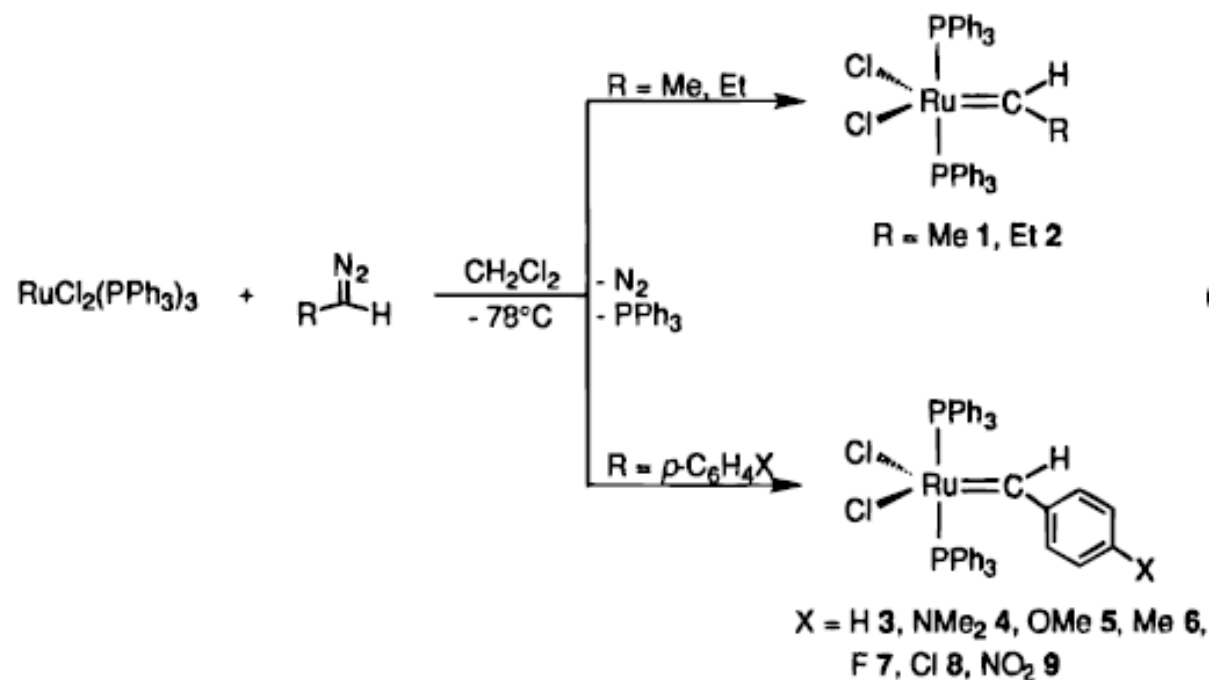
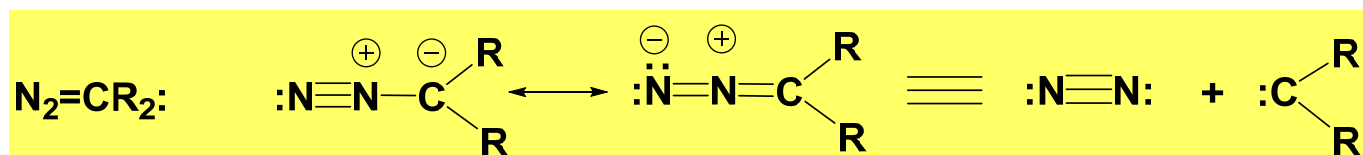
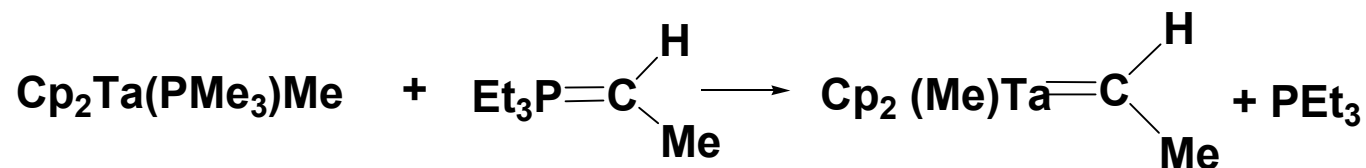
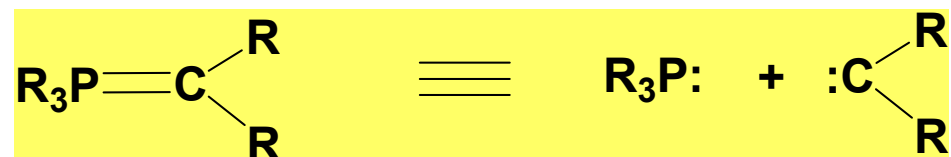




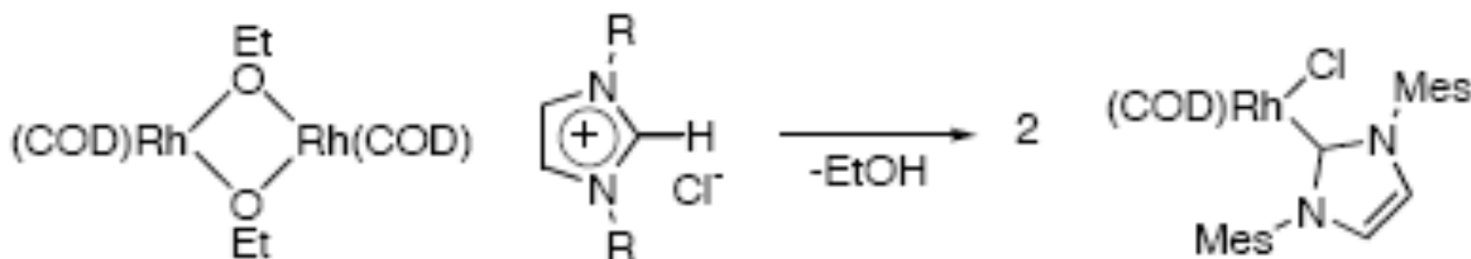
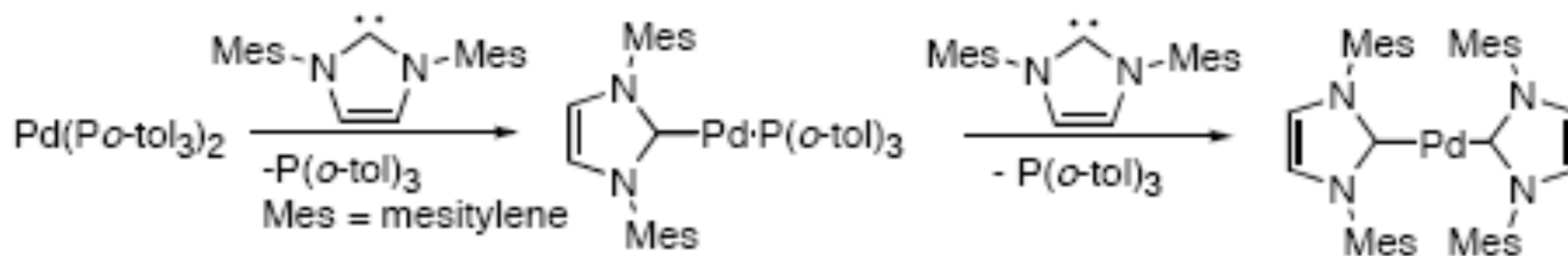
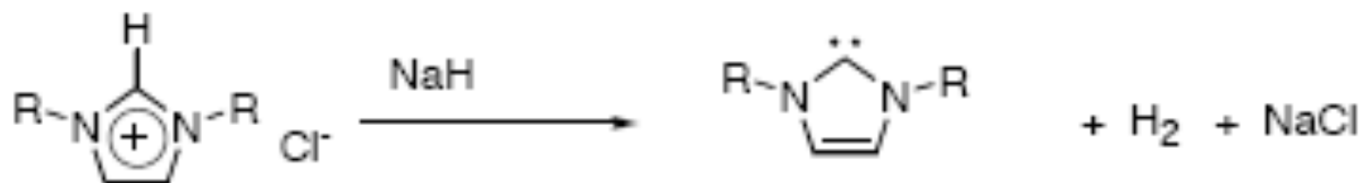
(6) By α -elimination.



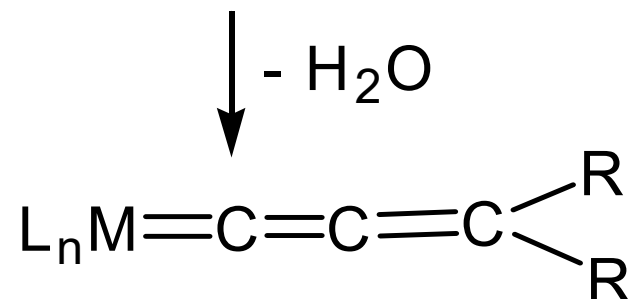
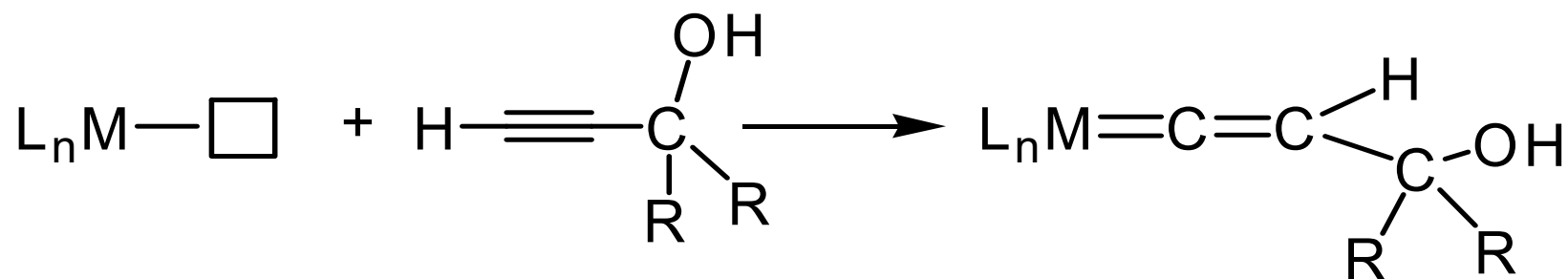
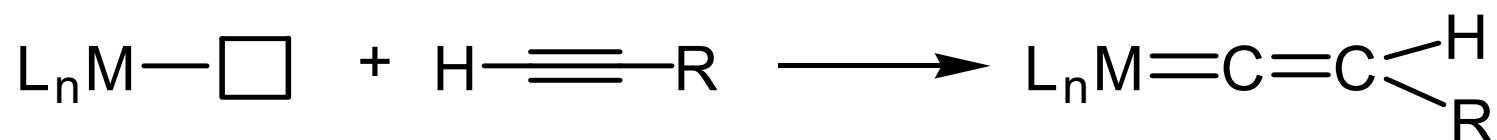
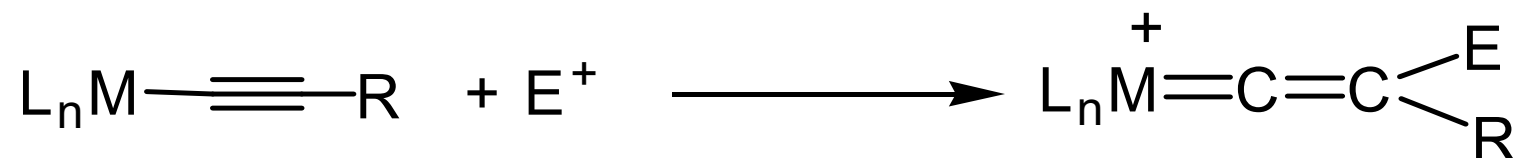
(7) From other sources of carbenes



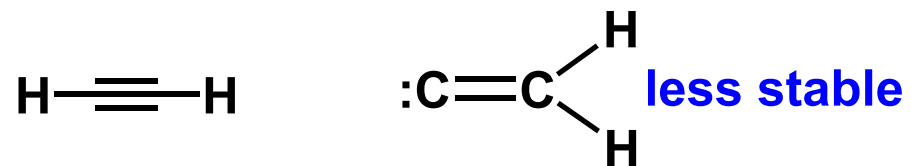
Stable carbenes: N-heterocyclic carbene (NHC) developed by Arduengo (*Acc. Chem. Res.* **1999**, 32, 913) can be used as neutral 2e ligands in place of CO or phosphines. The carbene can either be preformed or prepared *in situ*.



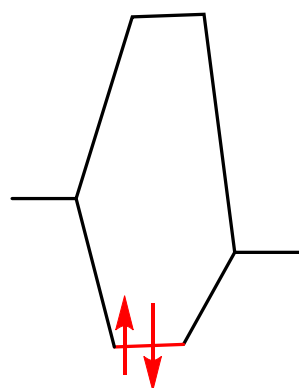
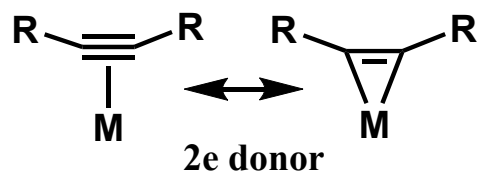
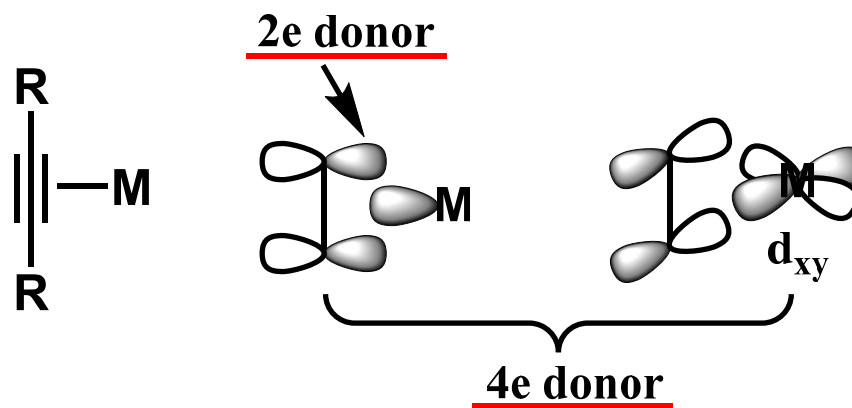
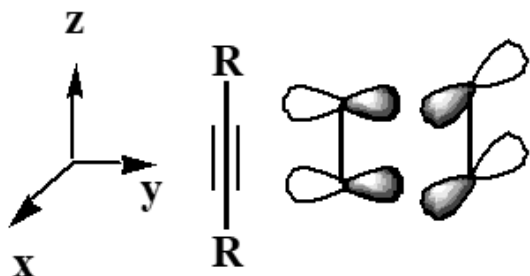
(8) How to prepare $M=C=CR_2$ and $M=C=C=CR_2$?



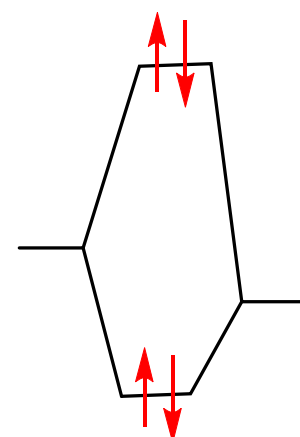
Very common for d^6 metals, e.g. Ru(II), Mn(I)
Common for d^8 square planar complexes



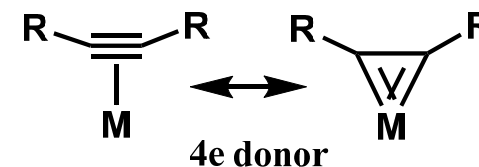
Why?



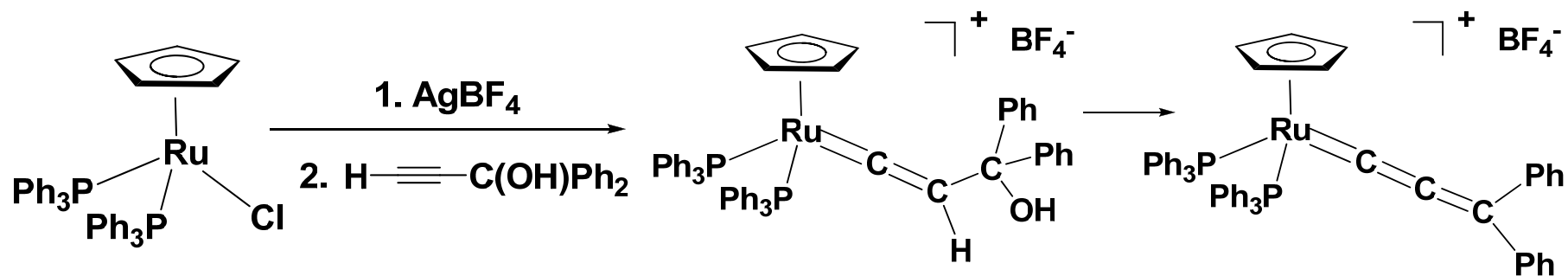
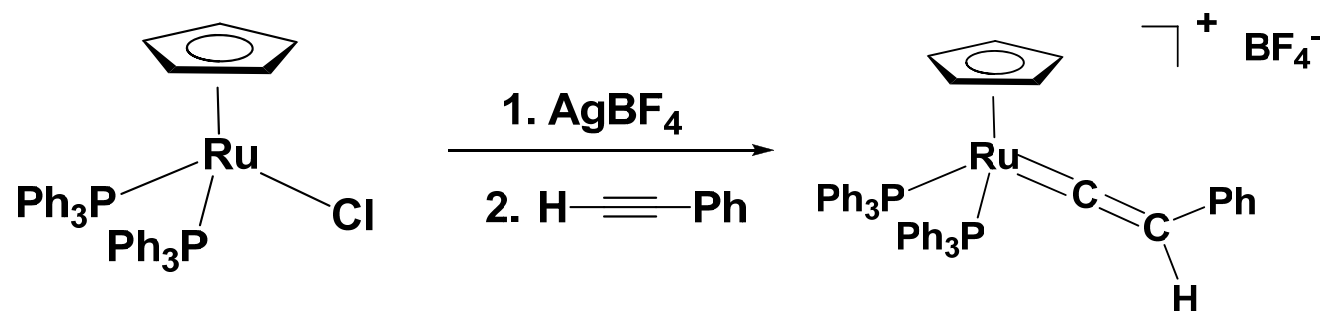
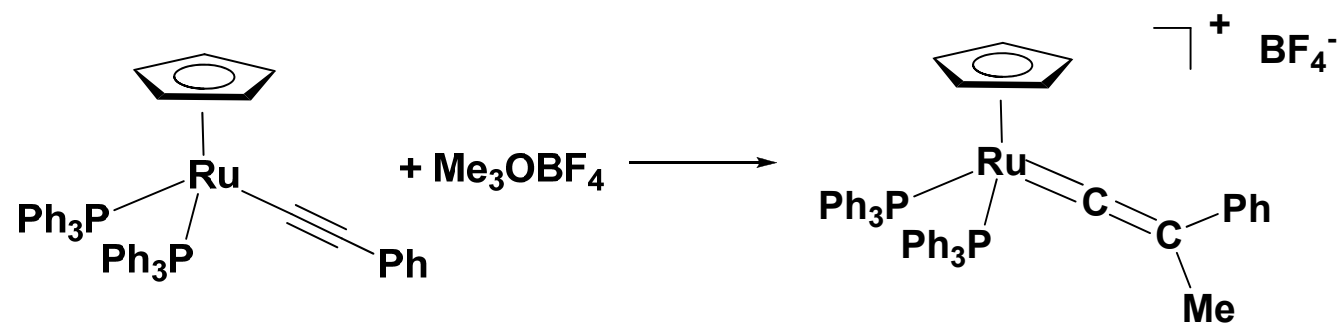
2c-2e, stabilizing



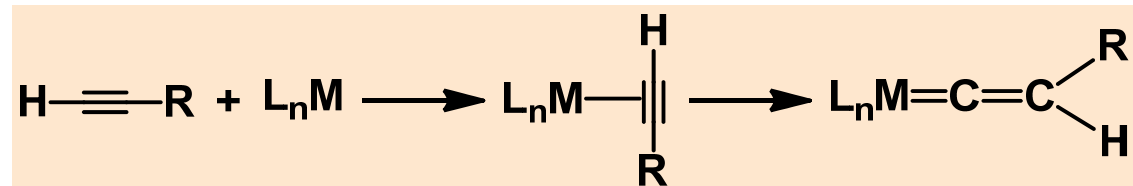
2c-4e, destabilizing



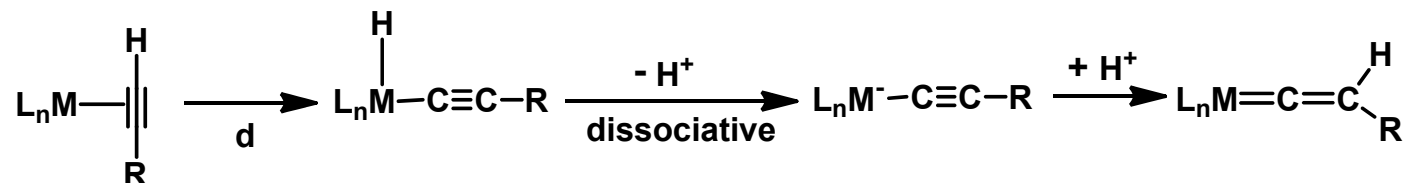
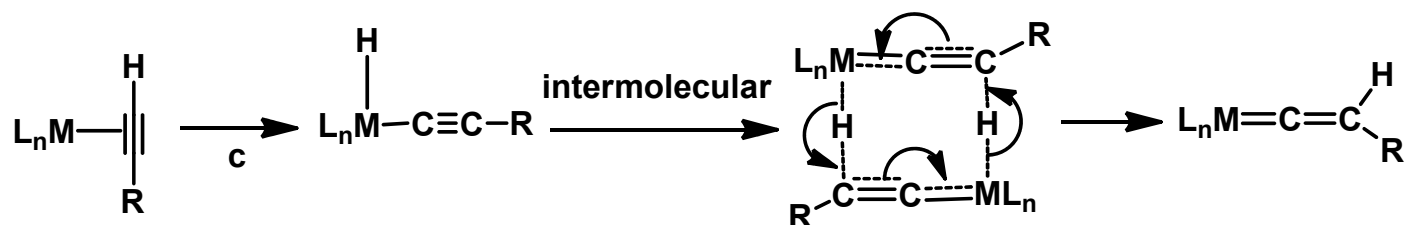
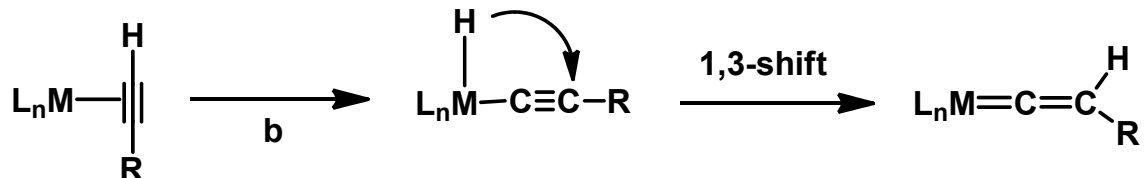
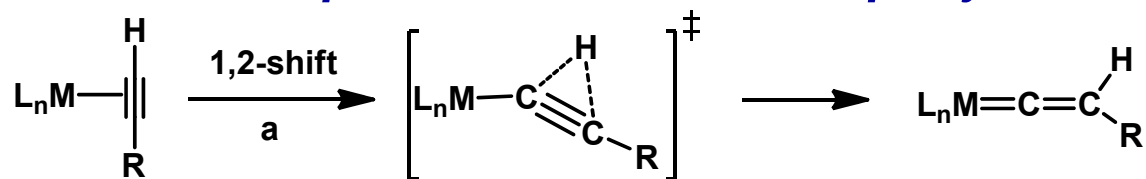
Practice Problems

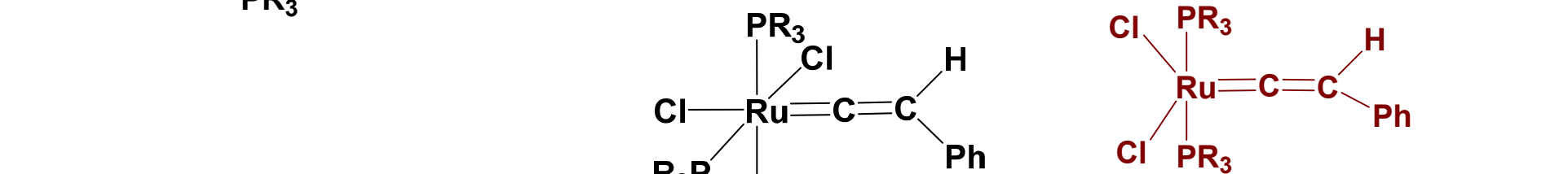
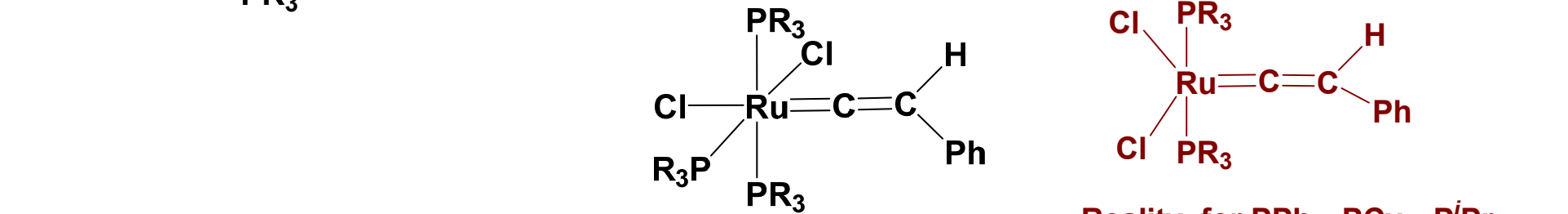
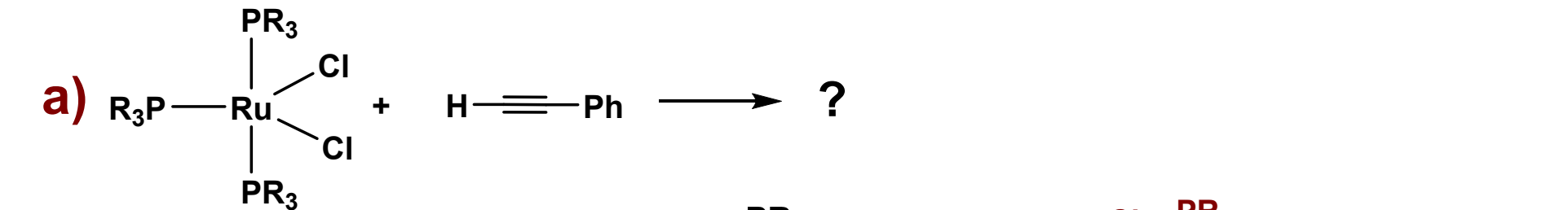


Vinylidene Complexes

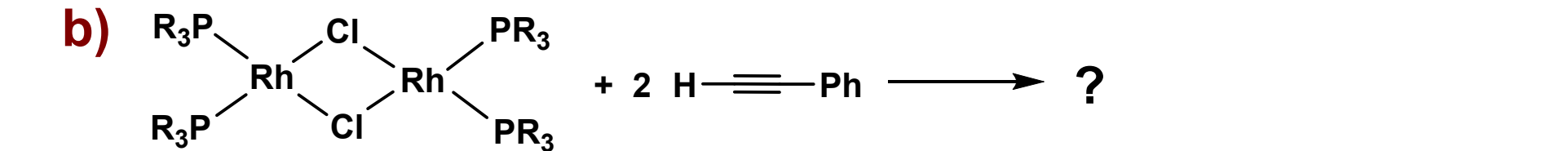


Plausible mechanisms for the tautomerization of alkyne to vinylidene

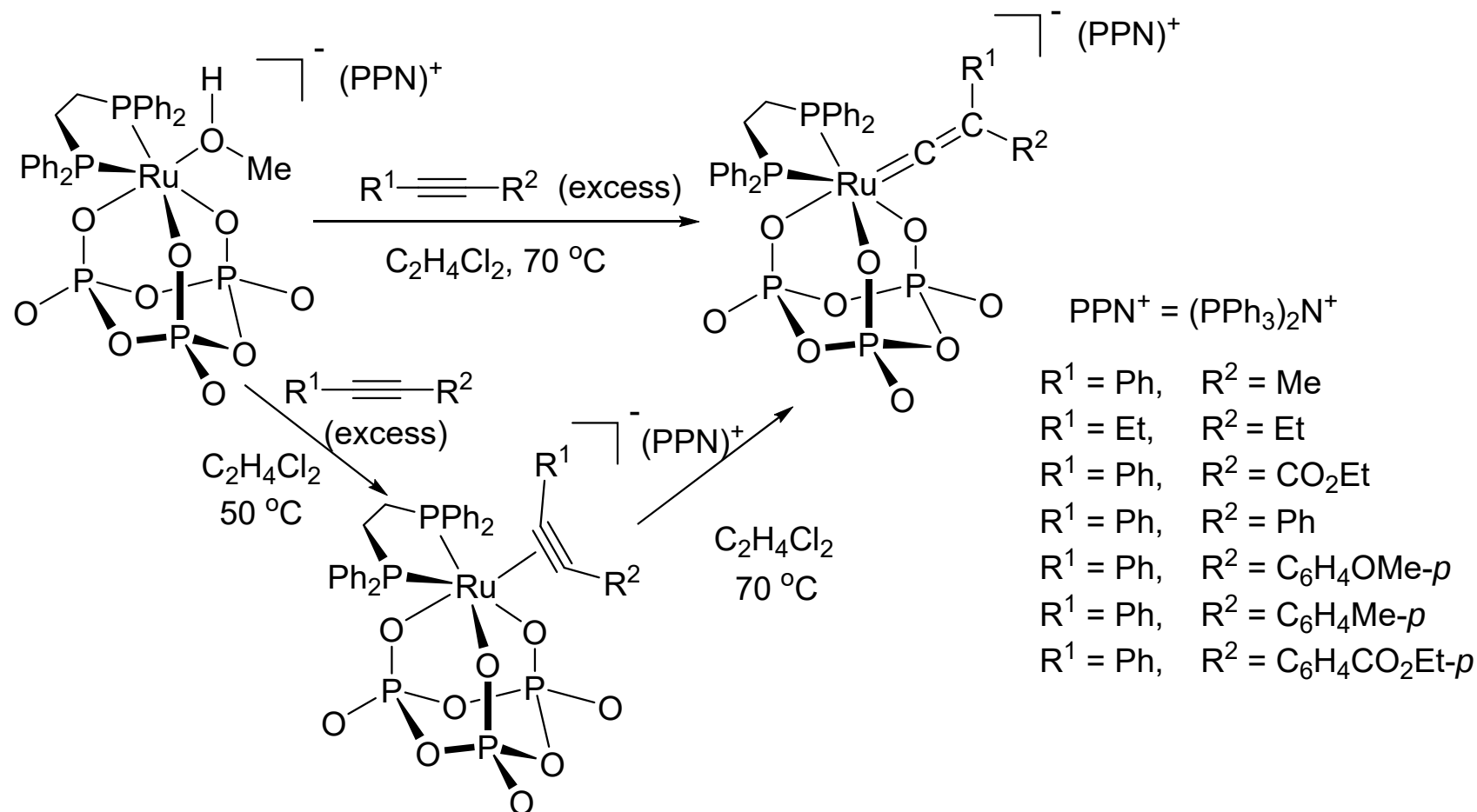


[illegible]

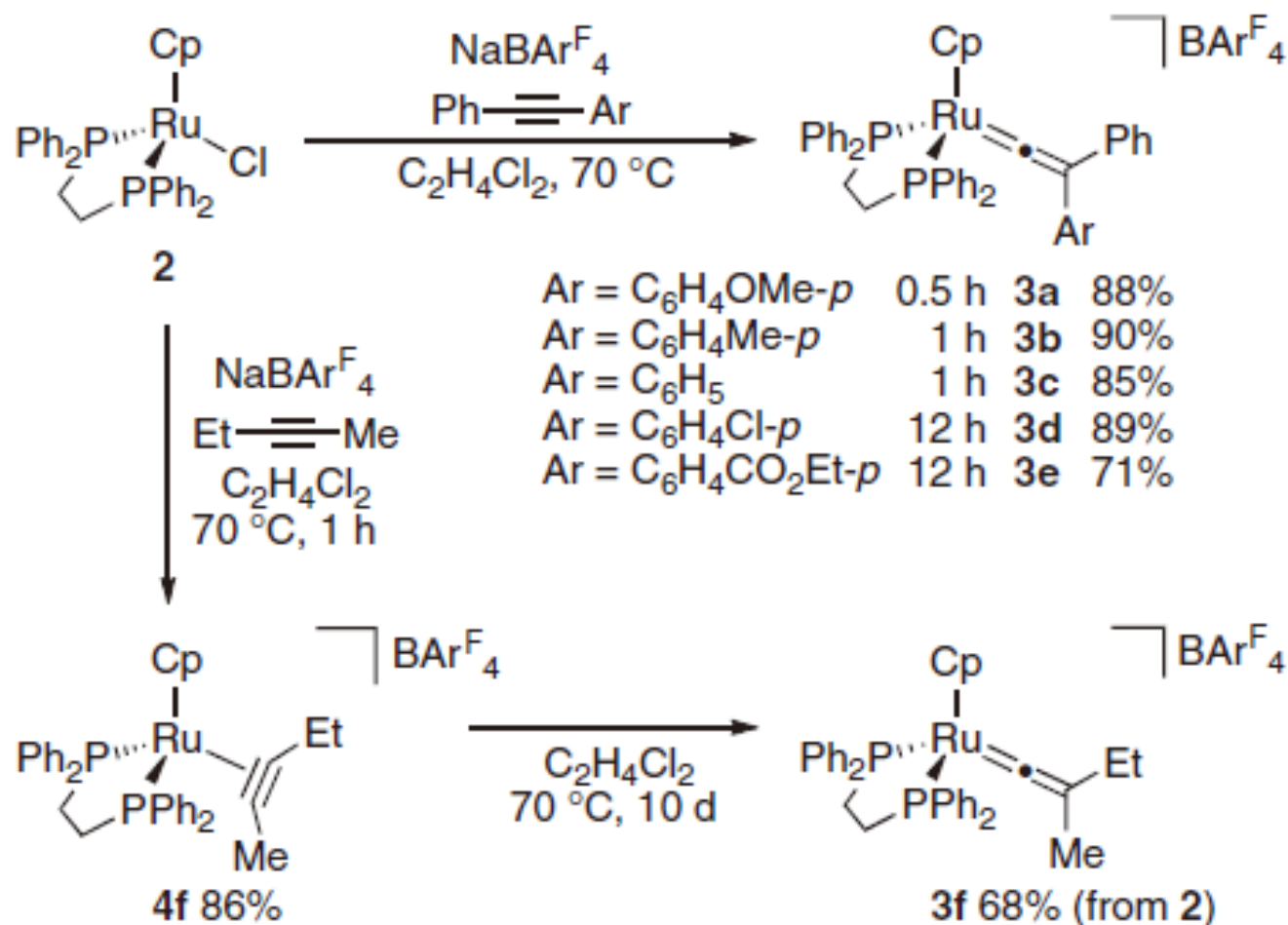
Reality, for PPh₃, PCy₃, P^{*i*}Pr₃



Internal Alkyne-to-vinylidene Isomerization via 1,2-Carbon Migration



Internal Alkyne-to-vinylidene Isomerization via 1,2-Carbon Migration



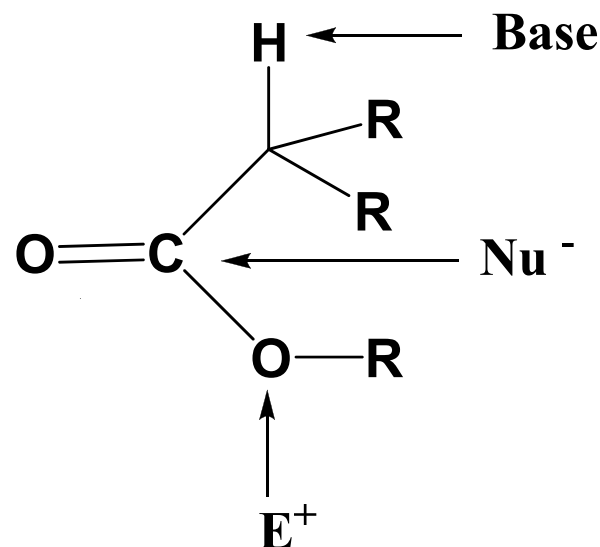
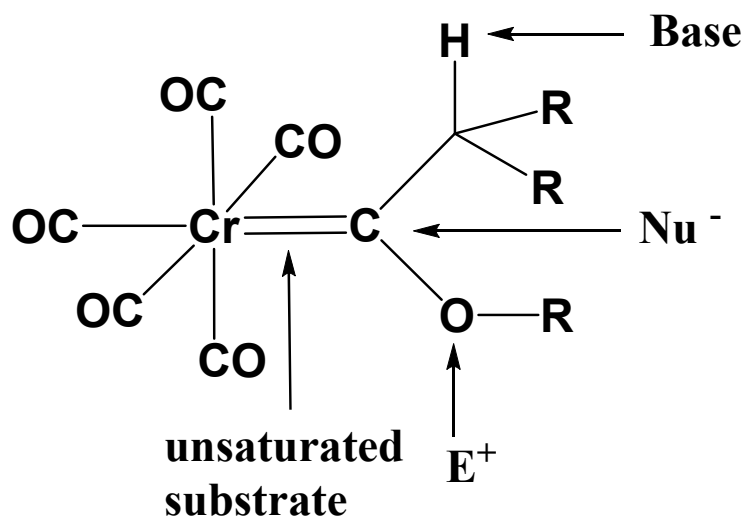
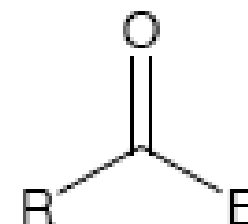
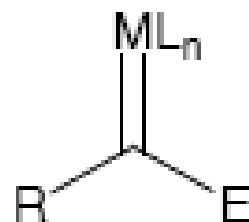
Y. Ishii, et. al. *Chem. Lett.* **2009**, 38, 534.

N. Tsuchida & K. Takano, et. al. *J. Am. Chem. Soc.* **2012**, 134, 17746.

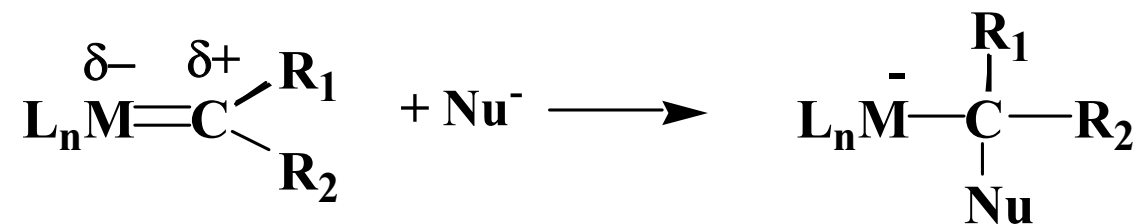
C. Reactivity of carbene complexes

(1) Reactions of Fischer carbene with nucleophiles

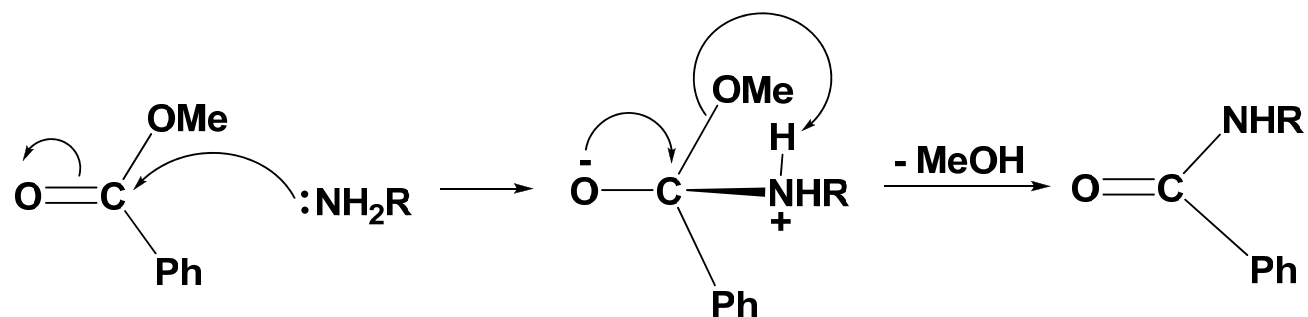
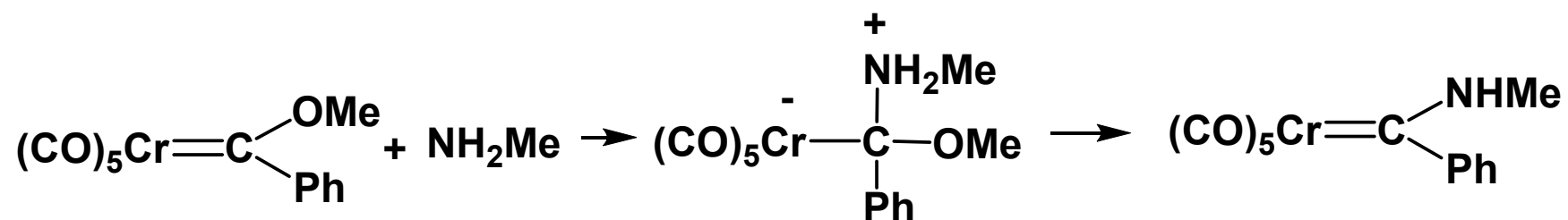
Much of the reactivity of Fischer carbenes is directly analogous to the reactivity of carboxylic acid derivatives.

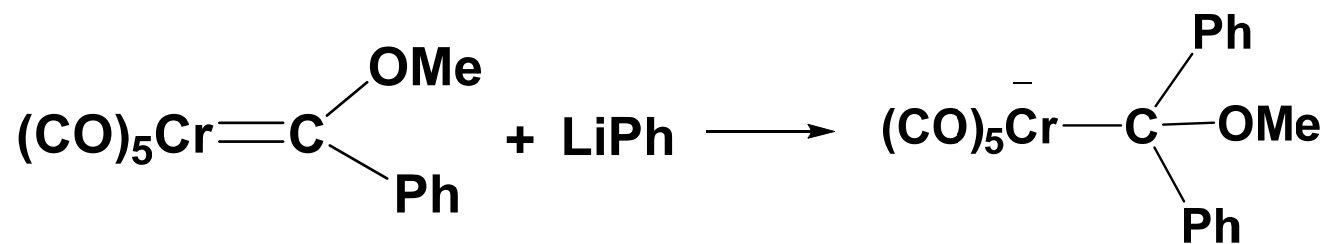
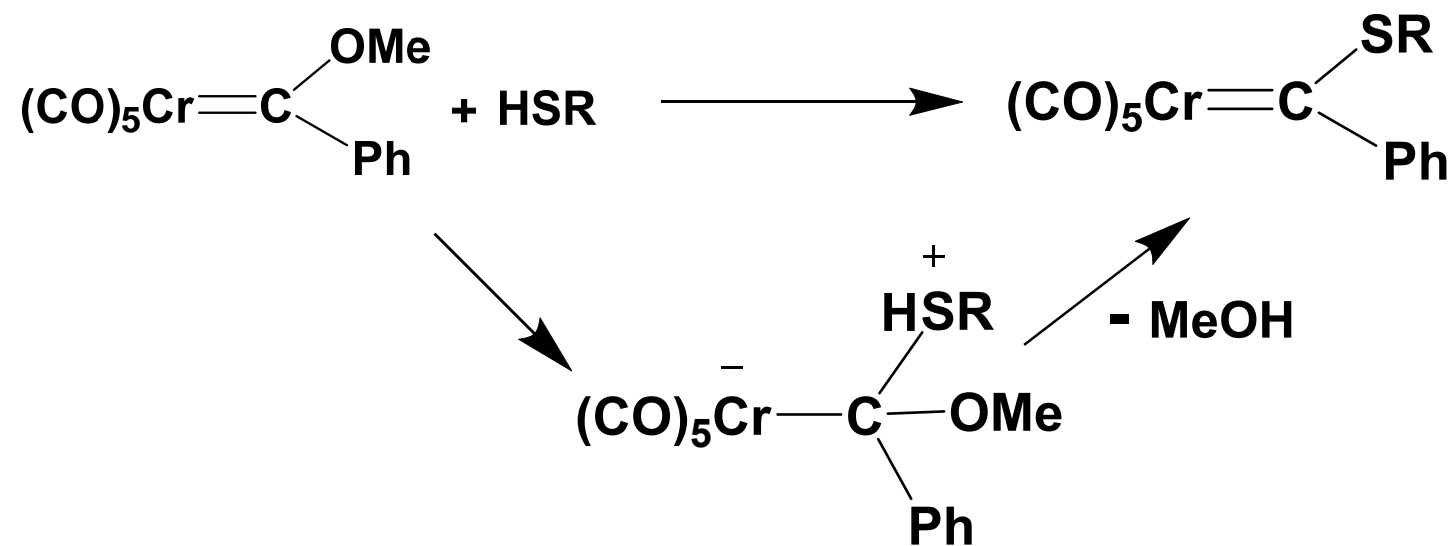


Examples of Reactions of Fisher carbene with nucleophiles

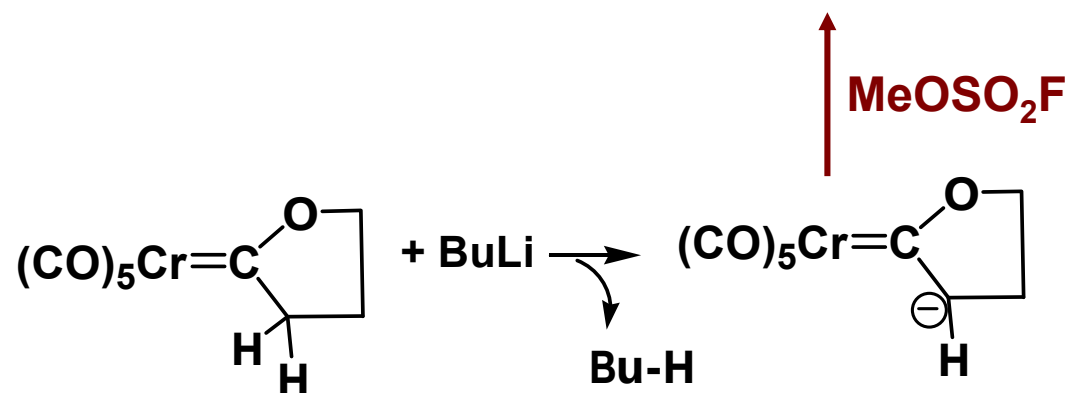
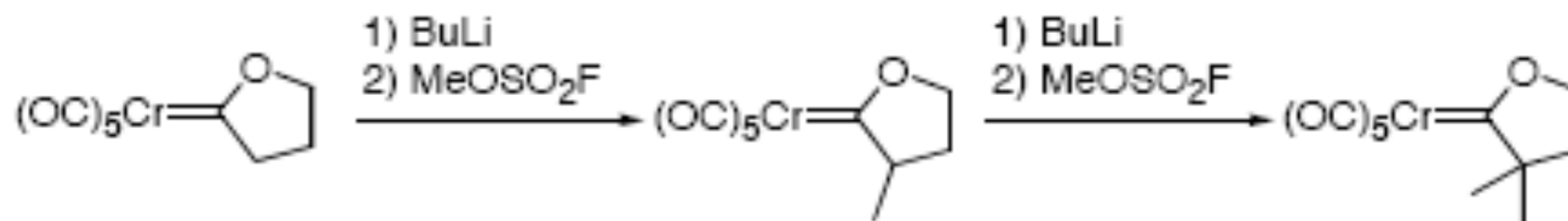
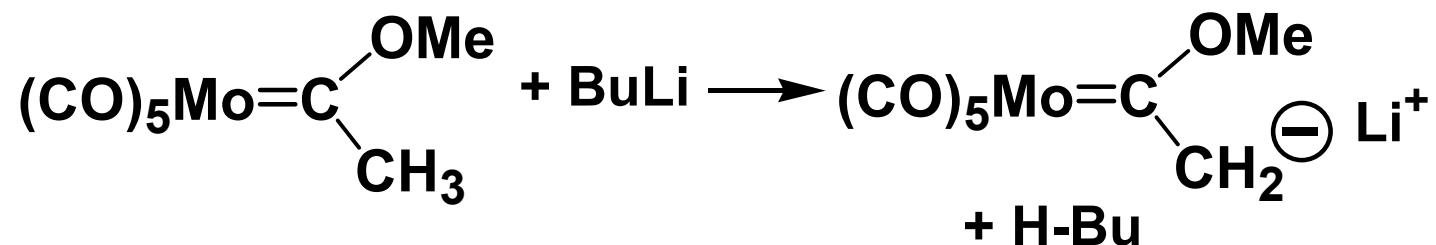


e.g.

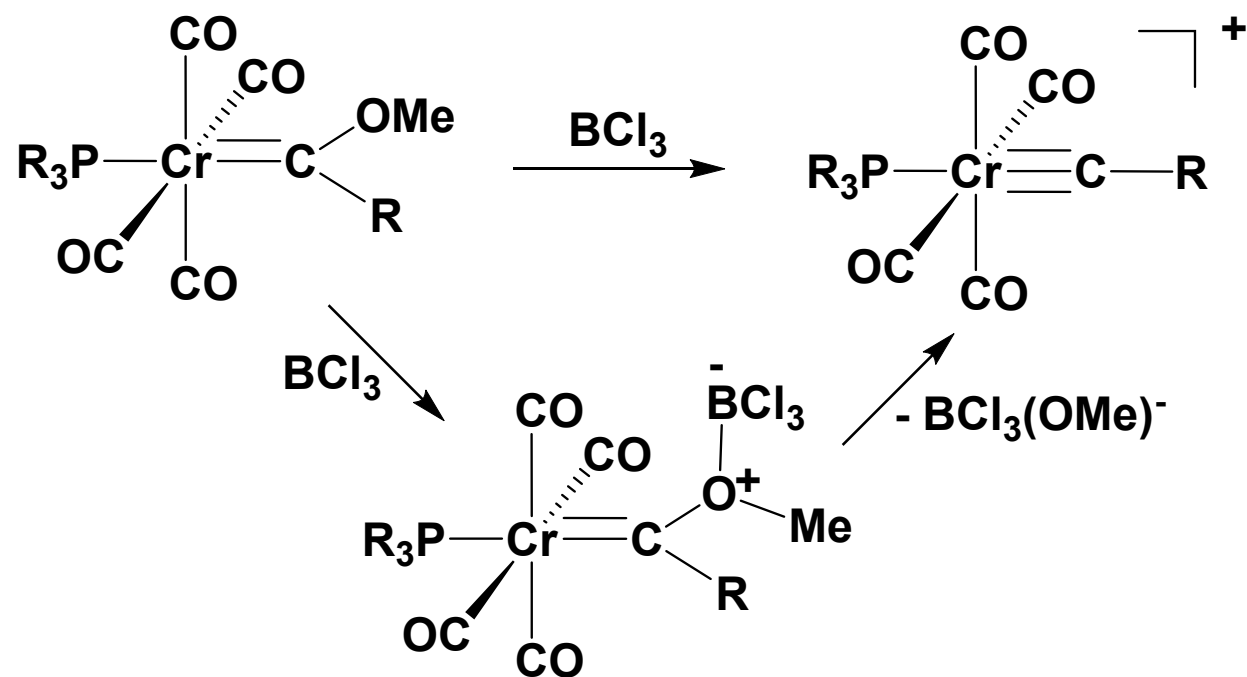
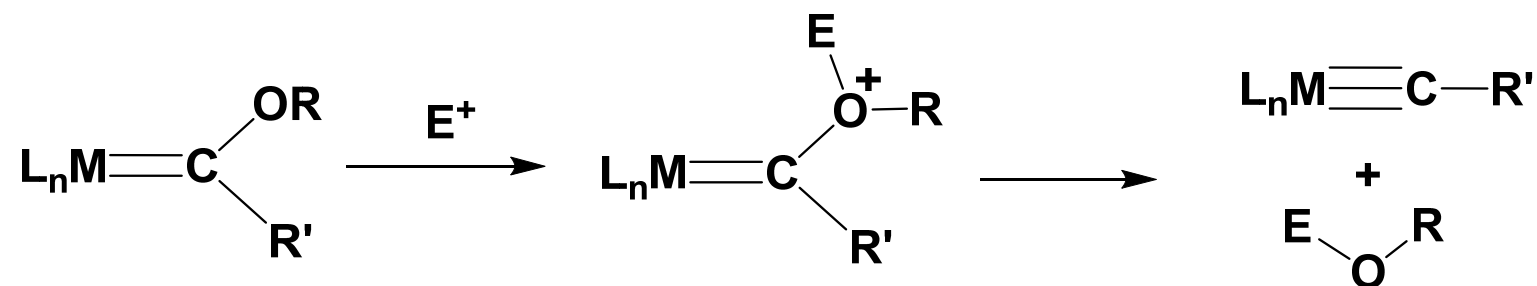


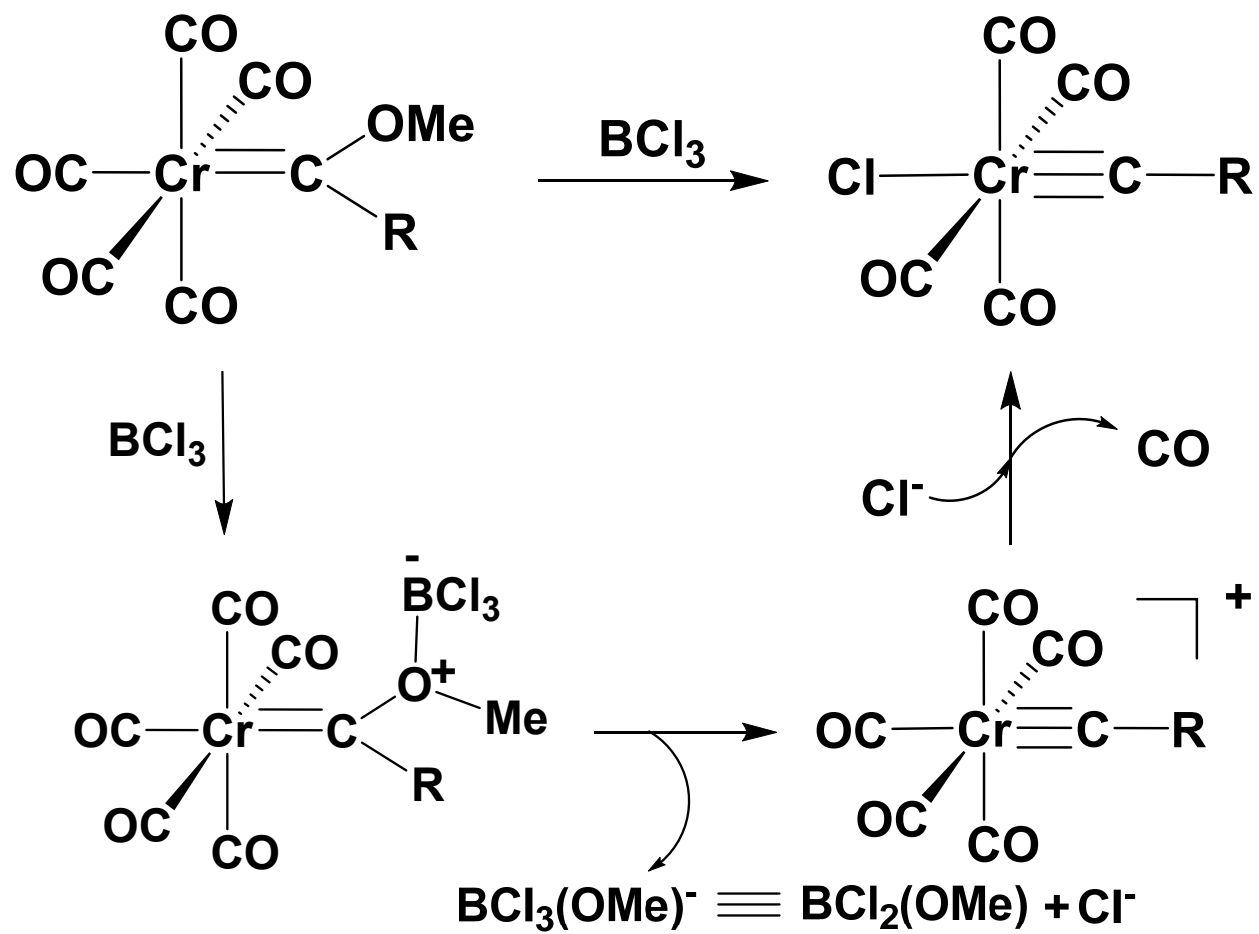


Note: In the presence of hydrogen β to metal, deprotonation may occur with strong bases. e.g.



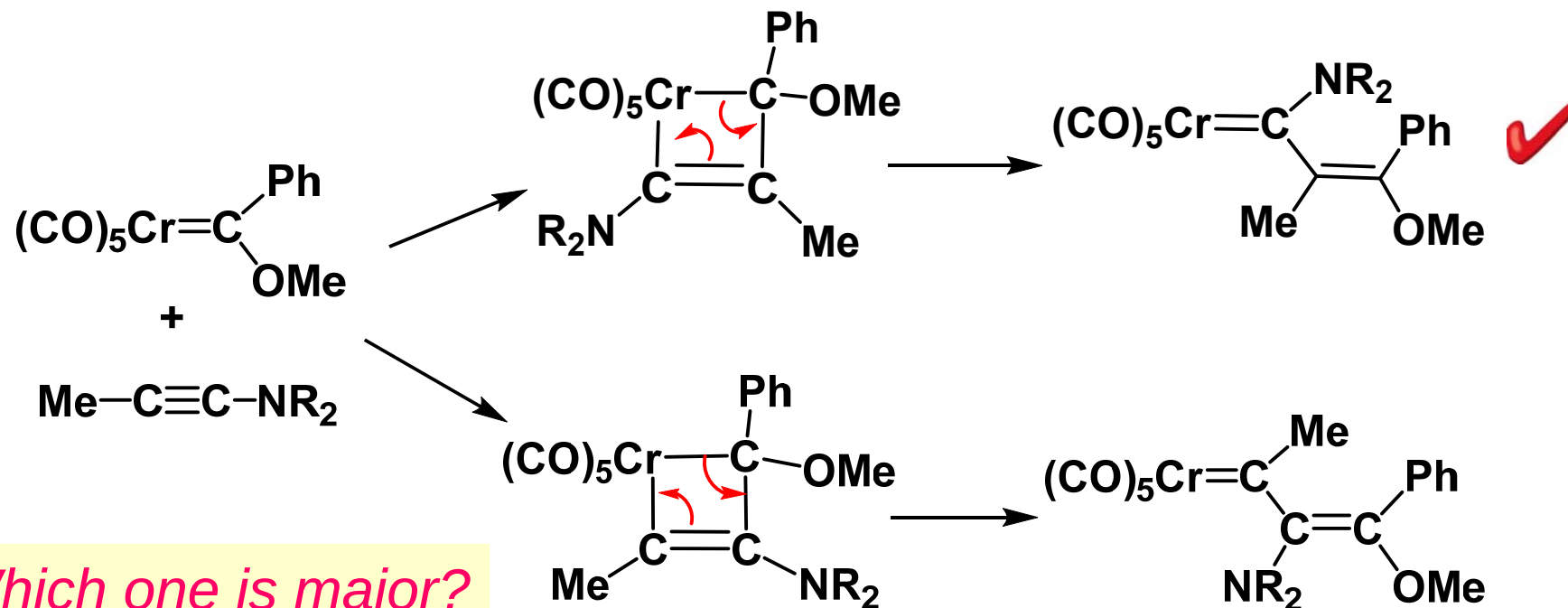
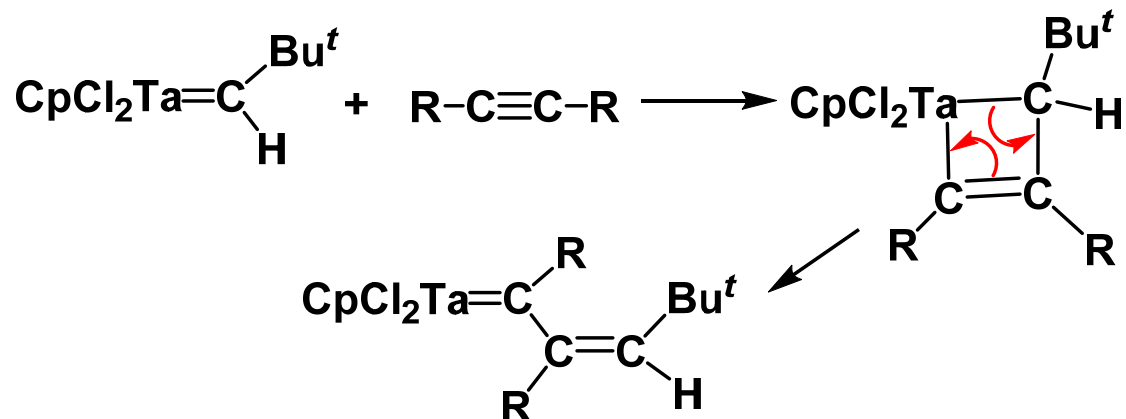
(2) *Reactions with electrophiles*





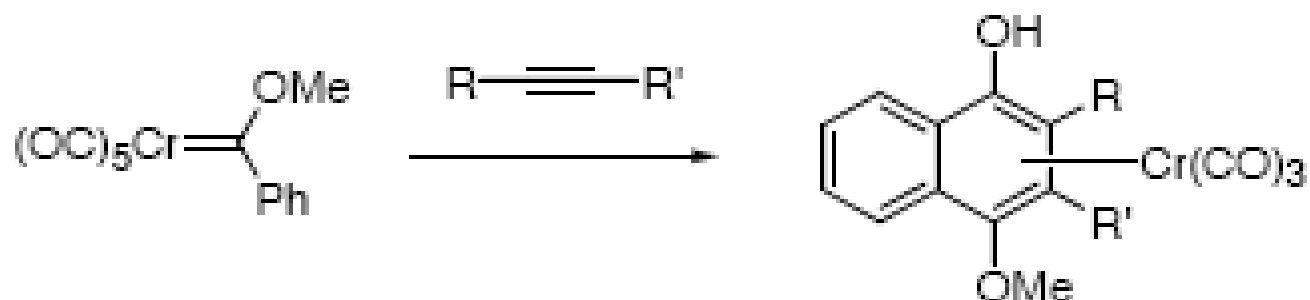
(3) Reactions with acetylenes

Both Fisher and Schrock carbene can react with acetylenes.

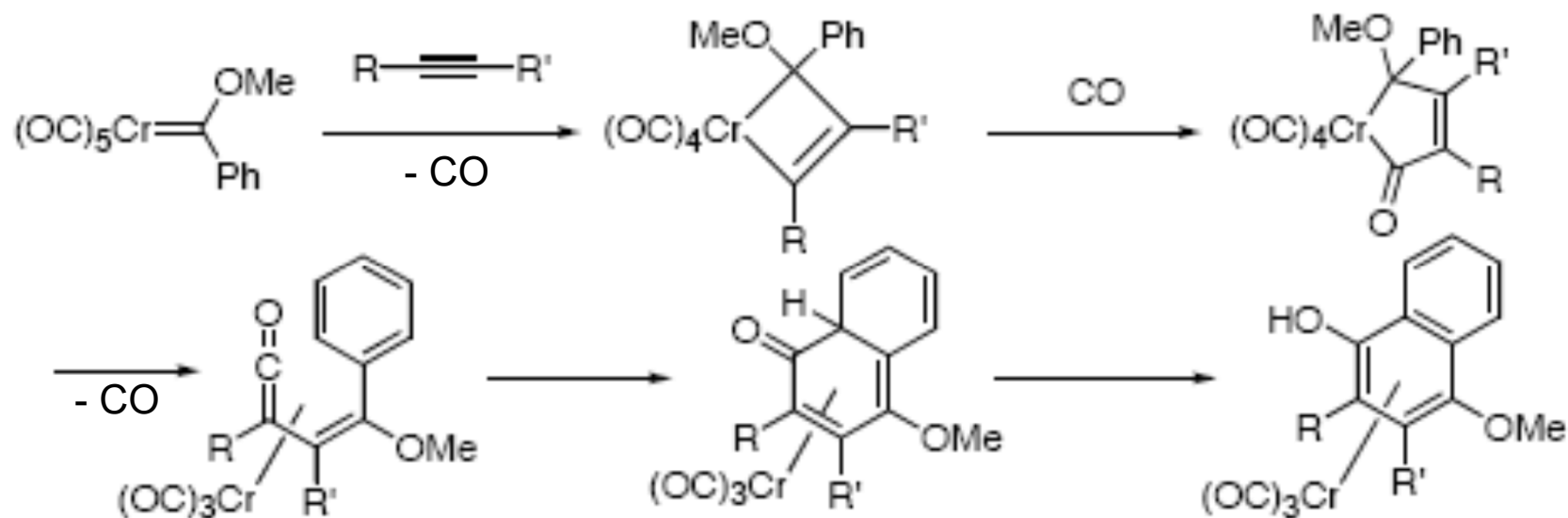


Which one is major?

The Dötz reaction. (*Angew. Chem. Int. Ed. Engl*, 1975, 14, 644)

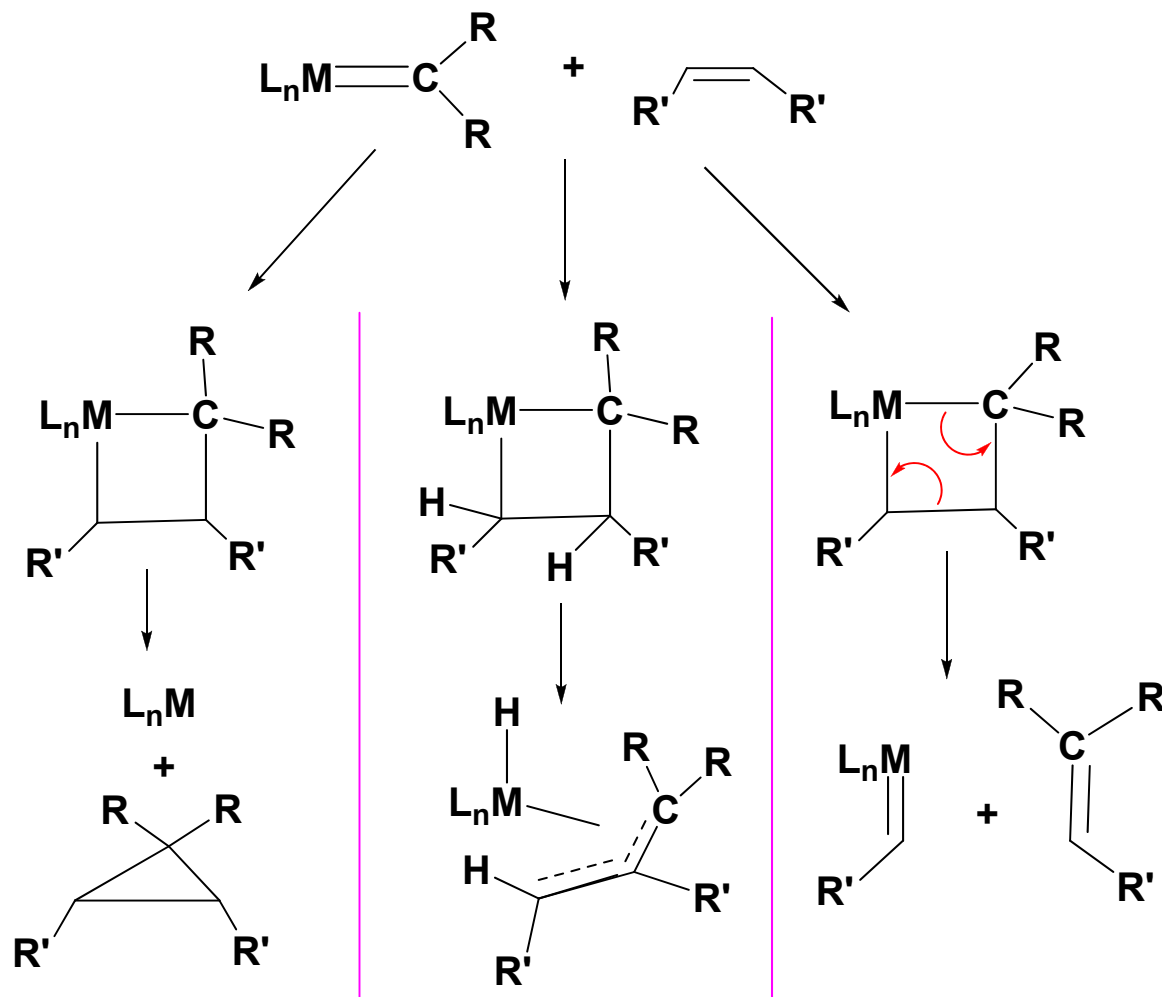


Dötz and Fischer have proposed the following mechanism: (*Chem. Ber*, 1982, 115, 1355).

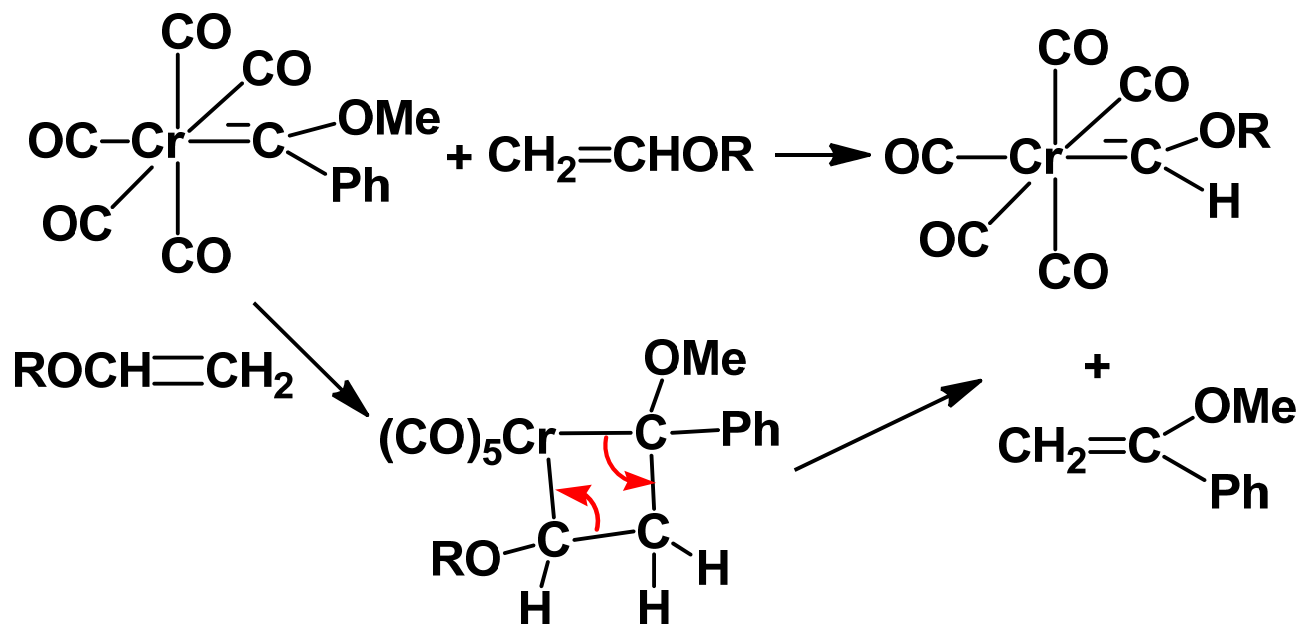


(4) Reactions with olefins

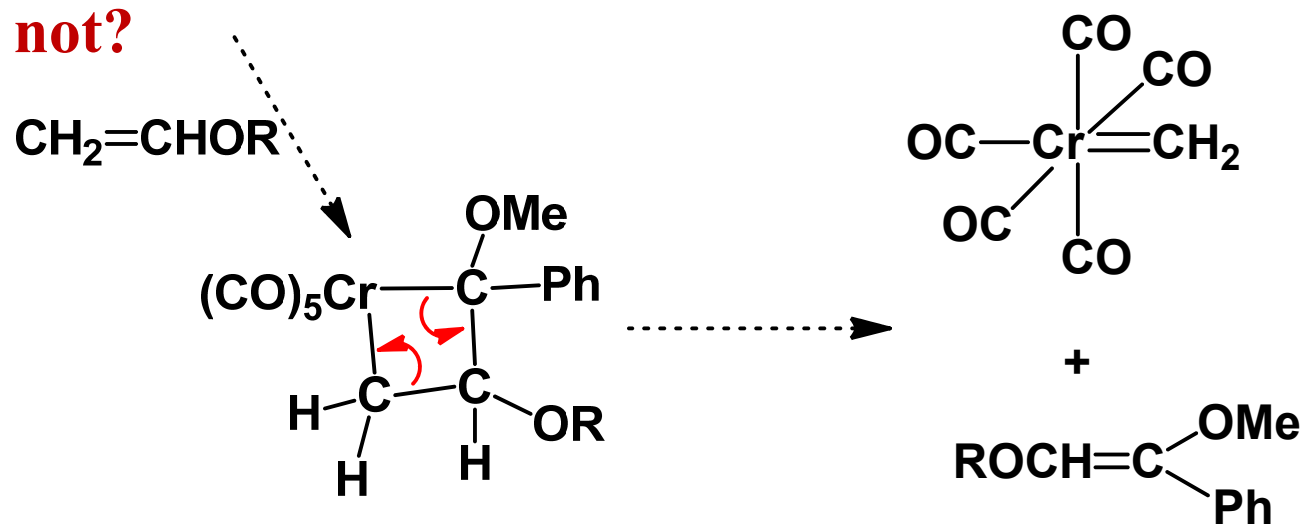
There are at least three possible products for the reactions, depending on metal complexes and olefins used.



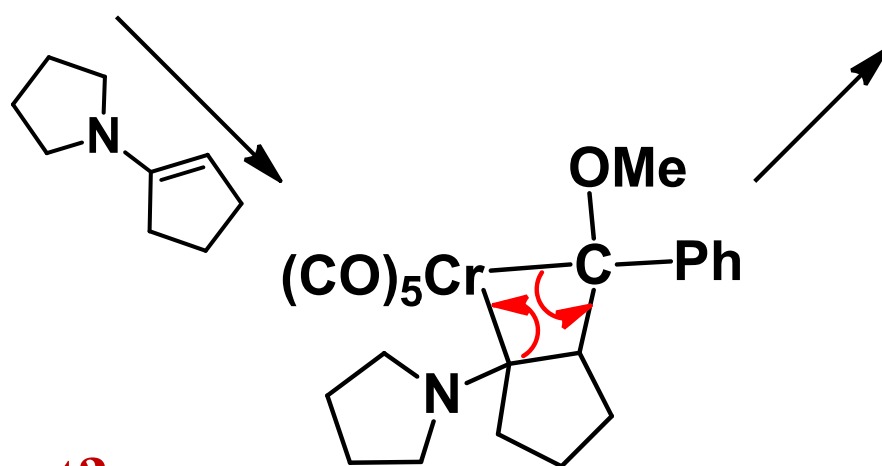
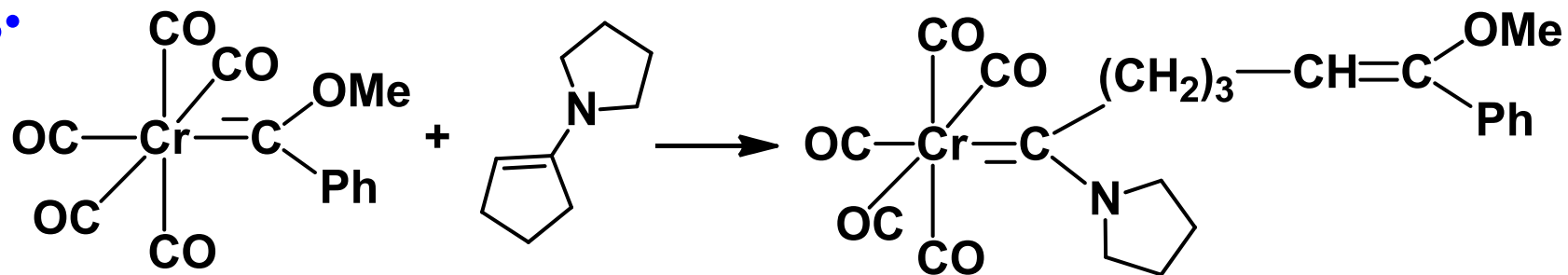
e.g.



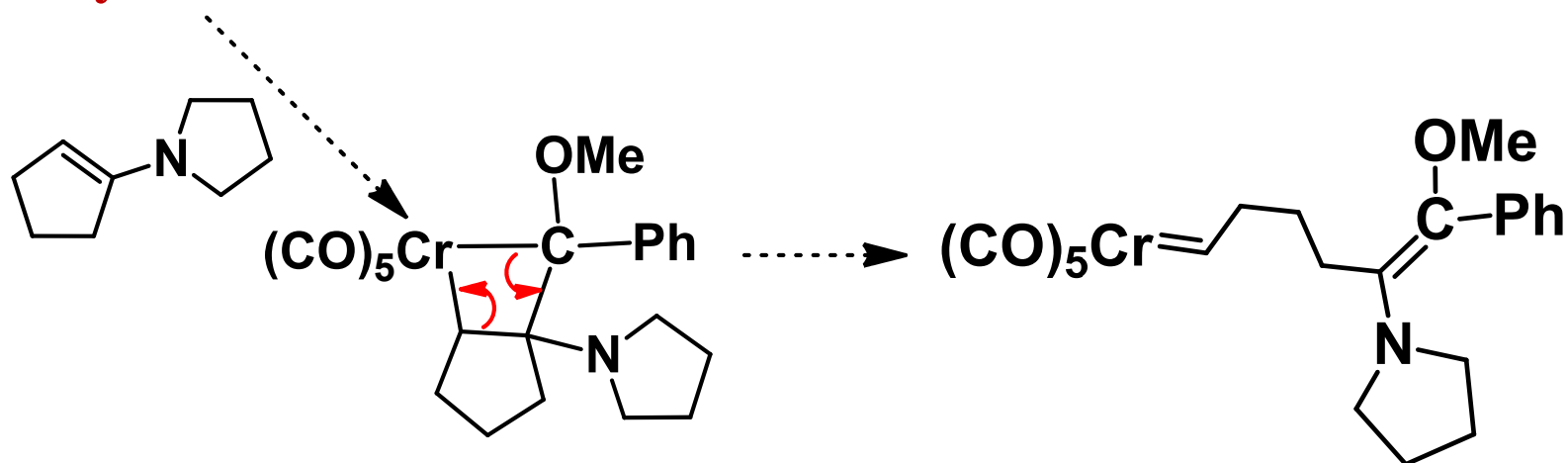
Why not?

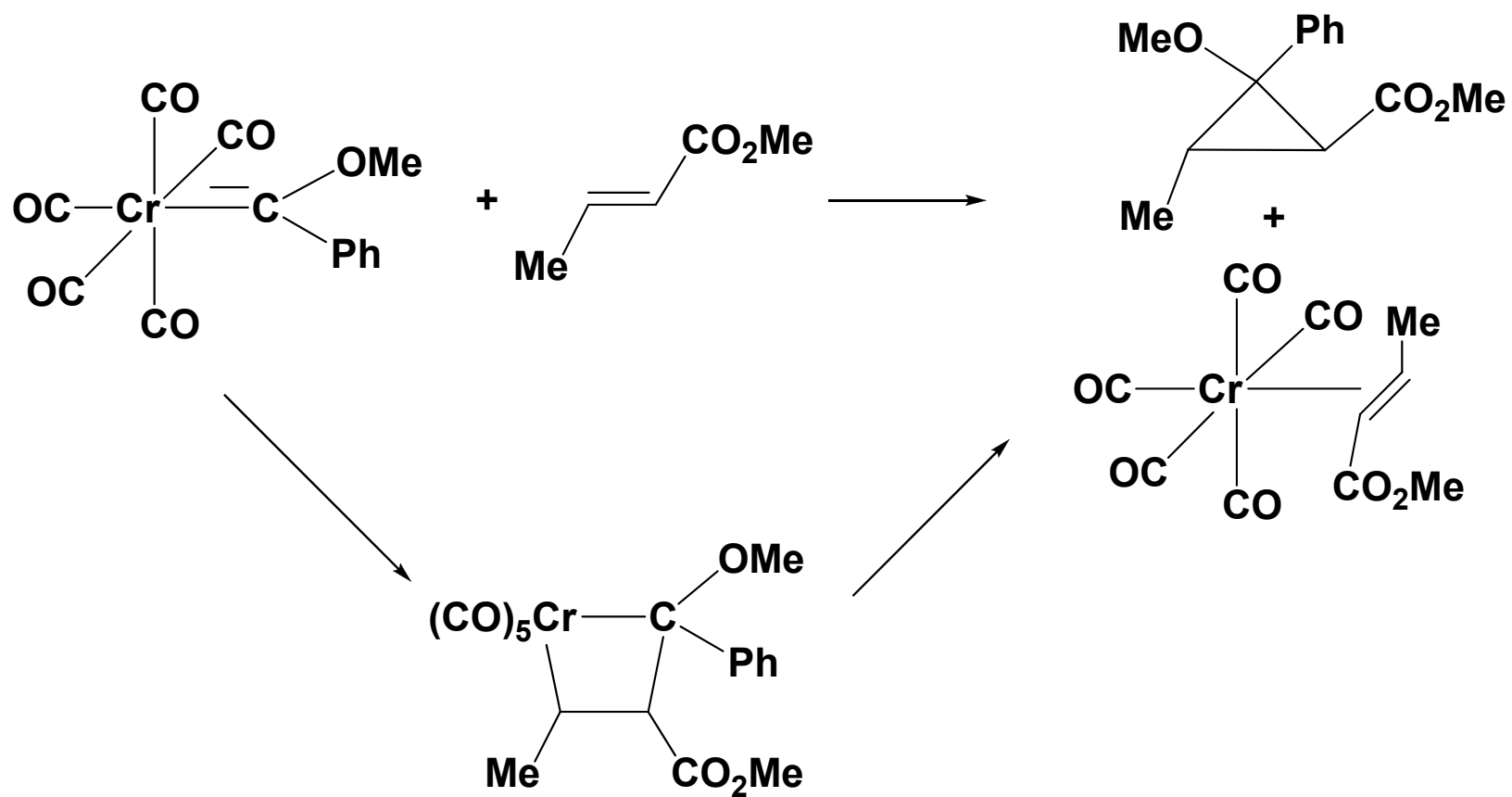


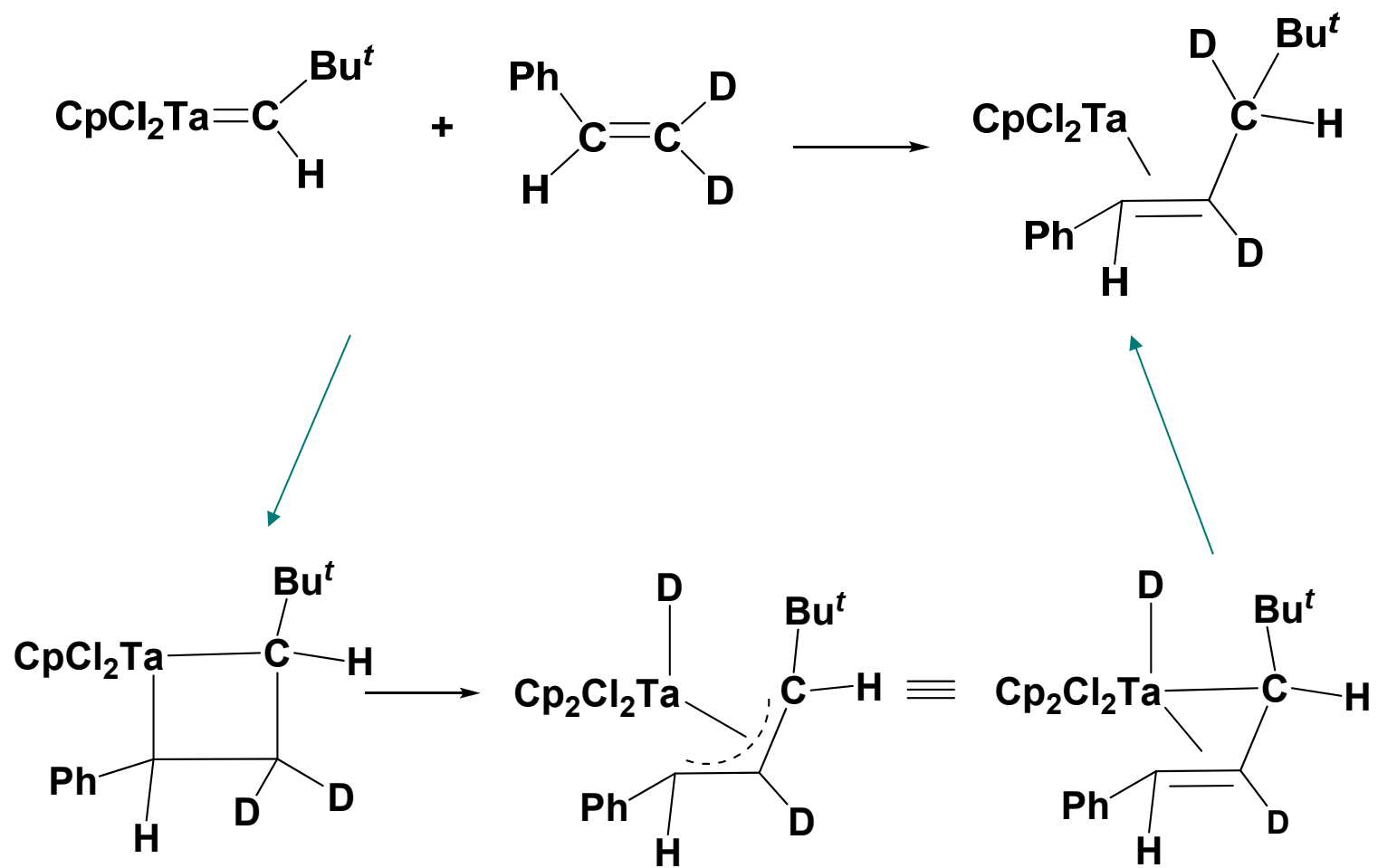
e.g.



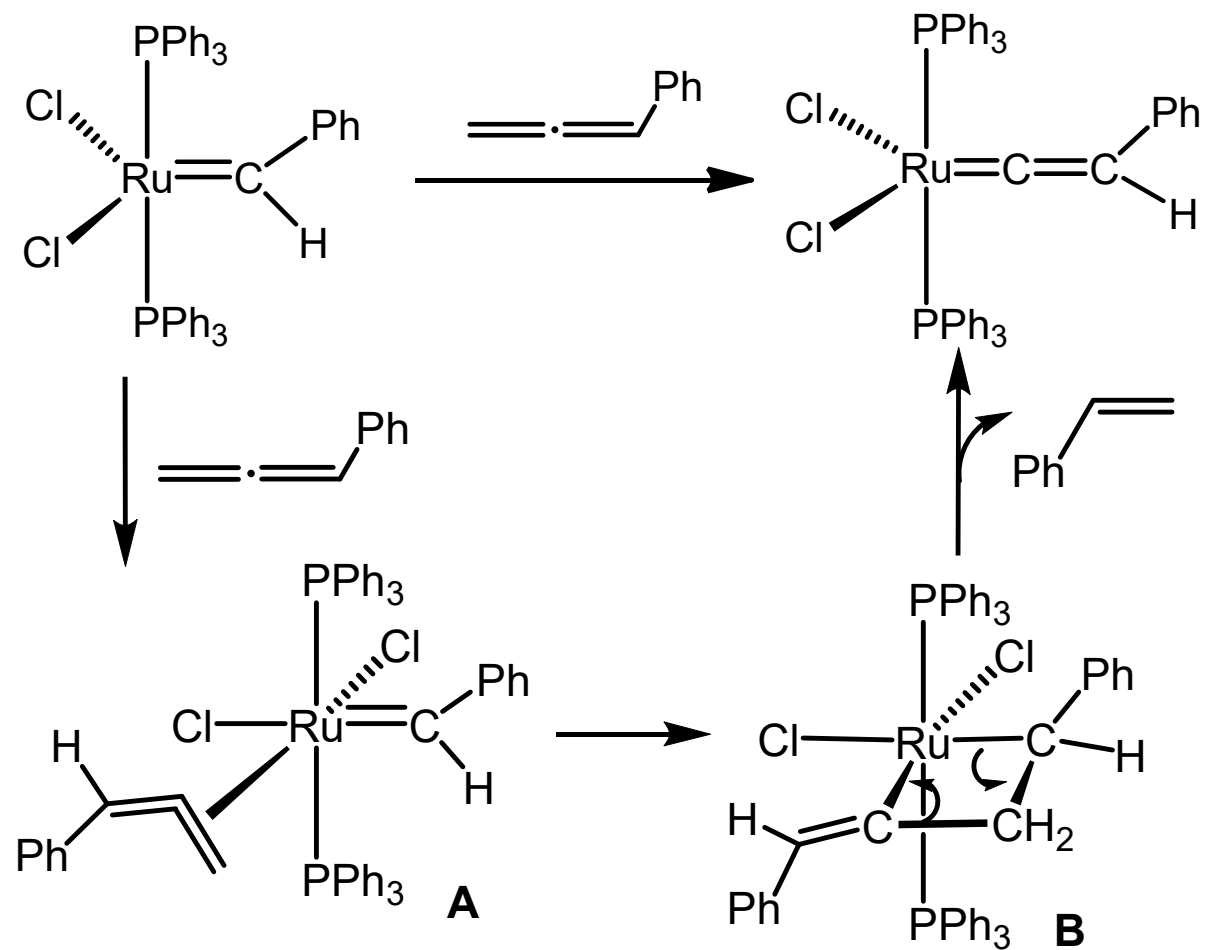
Why not?







e.g.

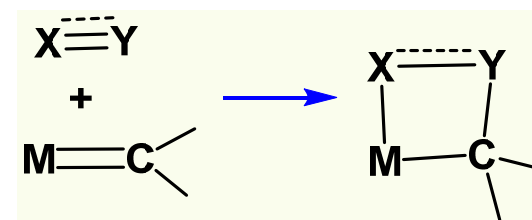
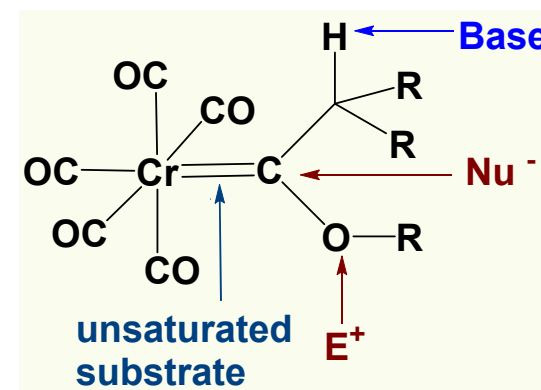


Summary of last lecture

- Introduction to M-C multiple bonds
 - Fischer carbene/carbyne
 - Schrock carbene/carbyne
- Common synthetic routes to carbene complexes
 - M-CO, M-CNR, M-CR=CHR', M-CXRR'
 - HC≡CR, N₂CRR', R₃P=CRR', stable carbenes

- Reactivity of carbene complexes

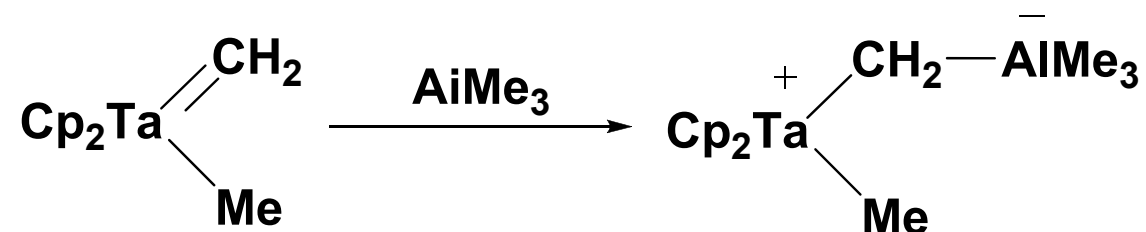
- Fischer carbene
 - ✓ Nucleophilic addition/abstraction to carbene.
 - ✓ Electrophilic abstraction of OR
- Reactions of carbenes with unsaturated substrates.
- Other reactions
 - Schrock carbene: Wittig-type reaction.



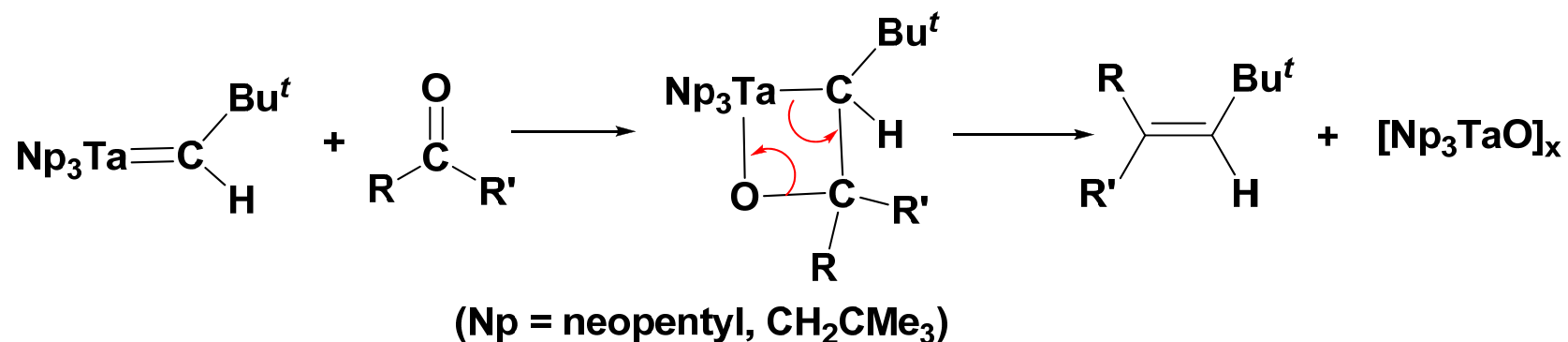
(5) Other reactions

Schrock carbene: nucleophilic character

◆ They form adducts with Lewis acid AlMe_3 .



◆ They can react with ketones in the same way as a Wittig reagent.



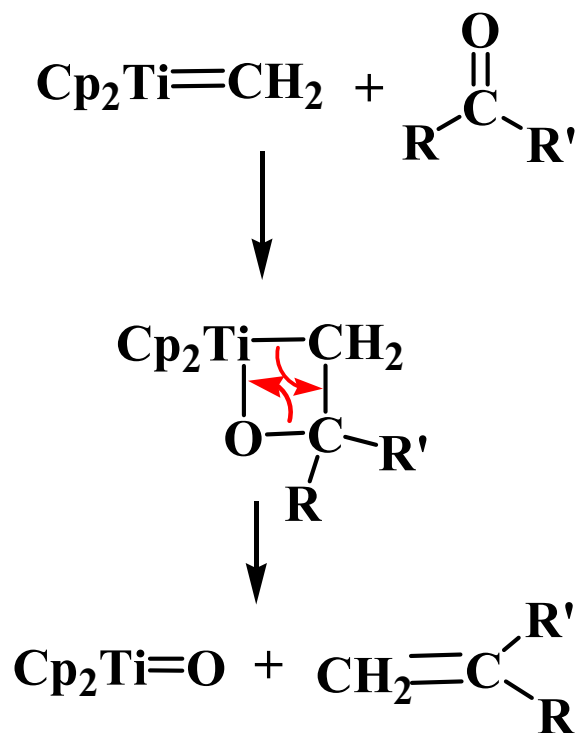
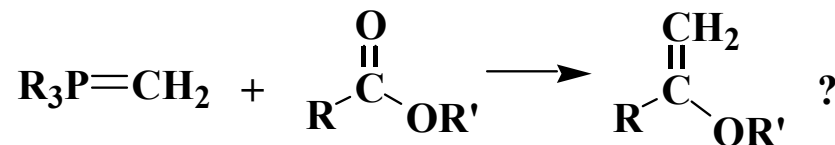
Tebbe's reagent:

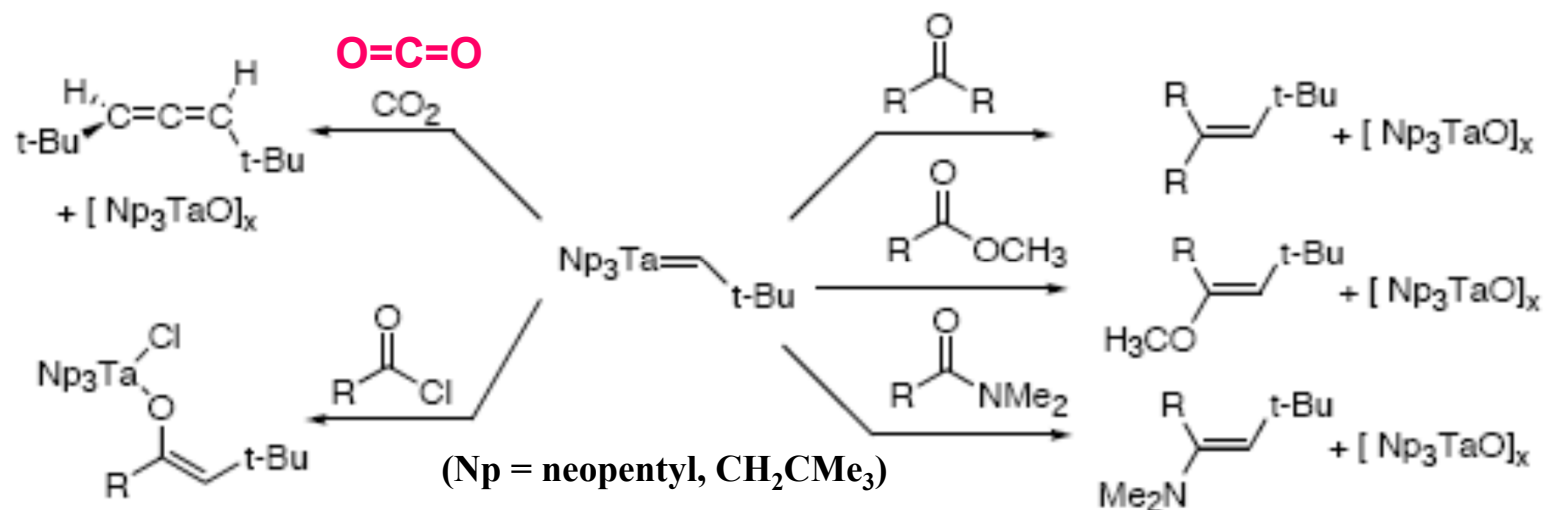
$$\text{Cp}_2\text{TiCl}_2 \xrightarrow{\text{AlMe}_3} \text{Cp}_2\text{TiCl}(\text{CH}_3)\text{AlMe}_2 \xrightarrow{-\text{CH}_4} \text{Cp}_2\text{TiCl}(\text{CH}_2)\text{AlMe}_2$$

bridging carbene

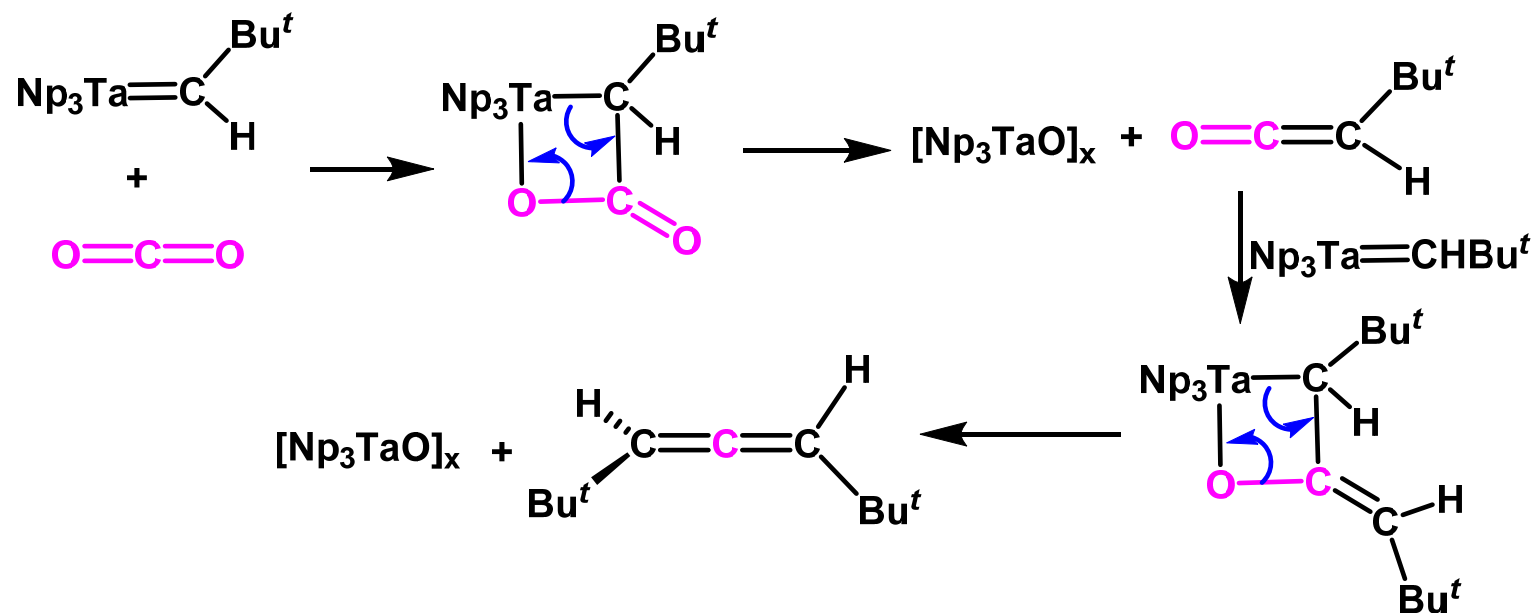


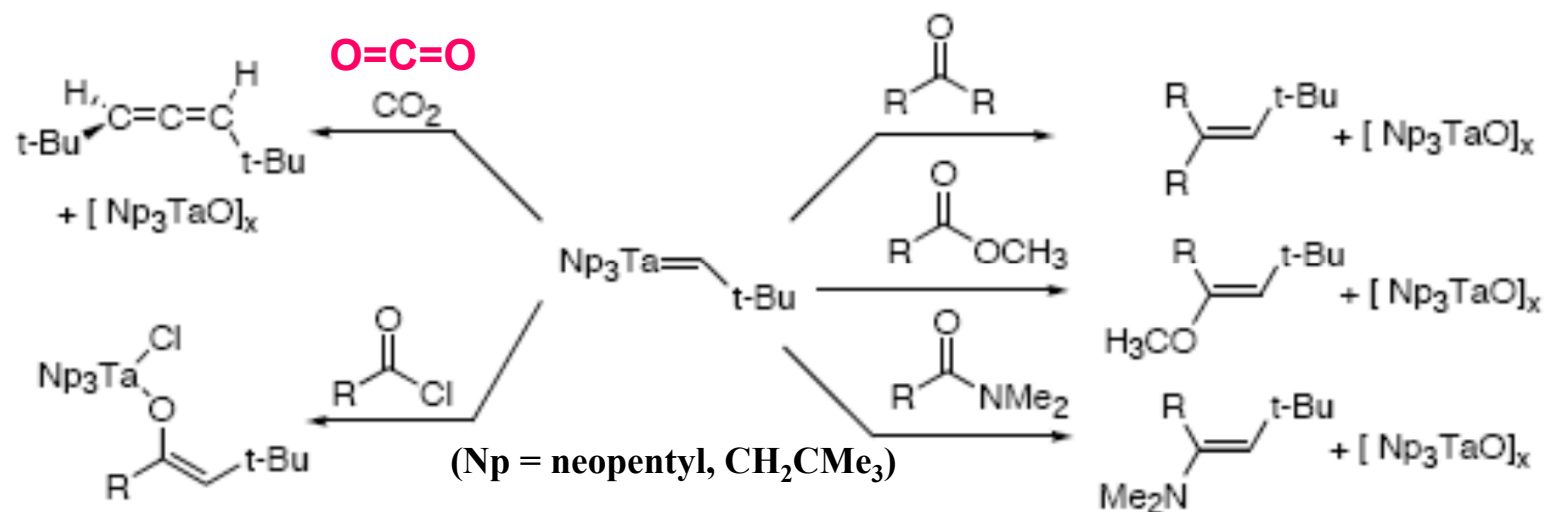
Even give Wittig-type product with esters



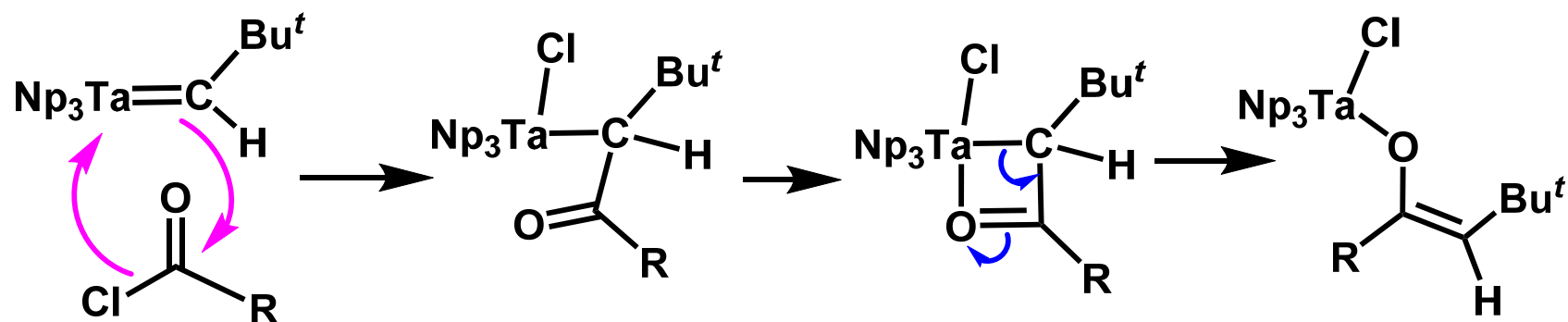


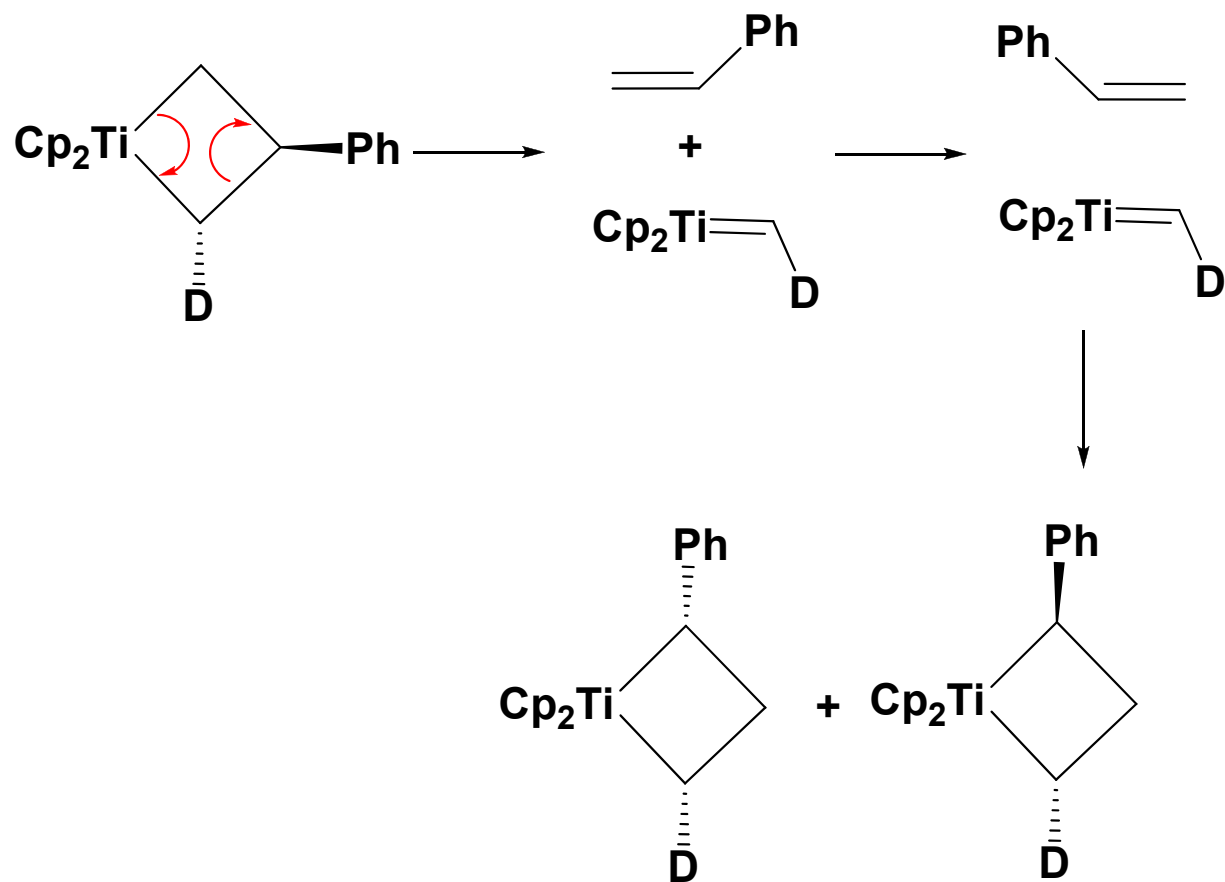
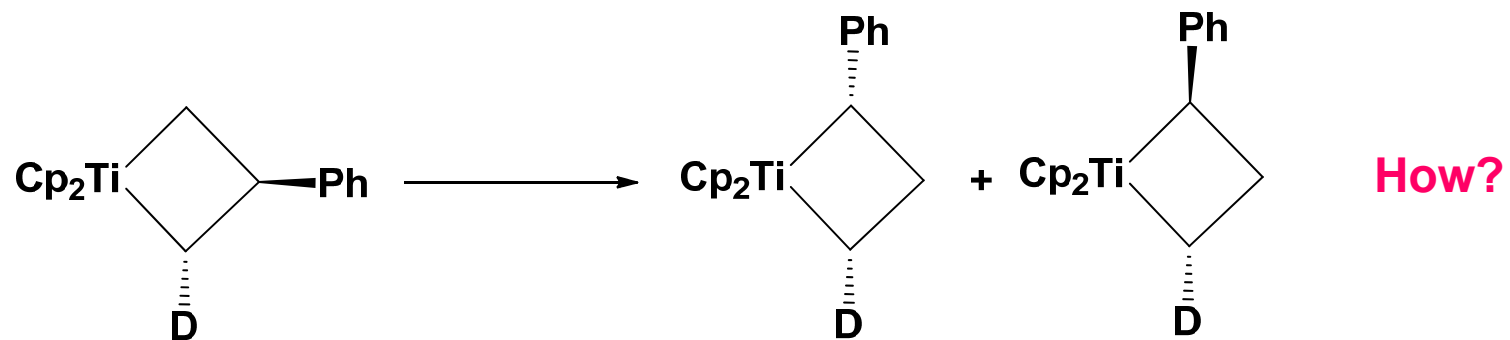
How?

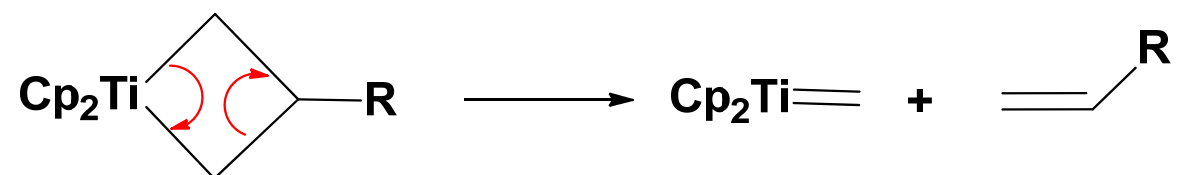
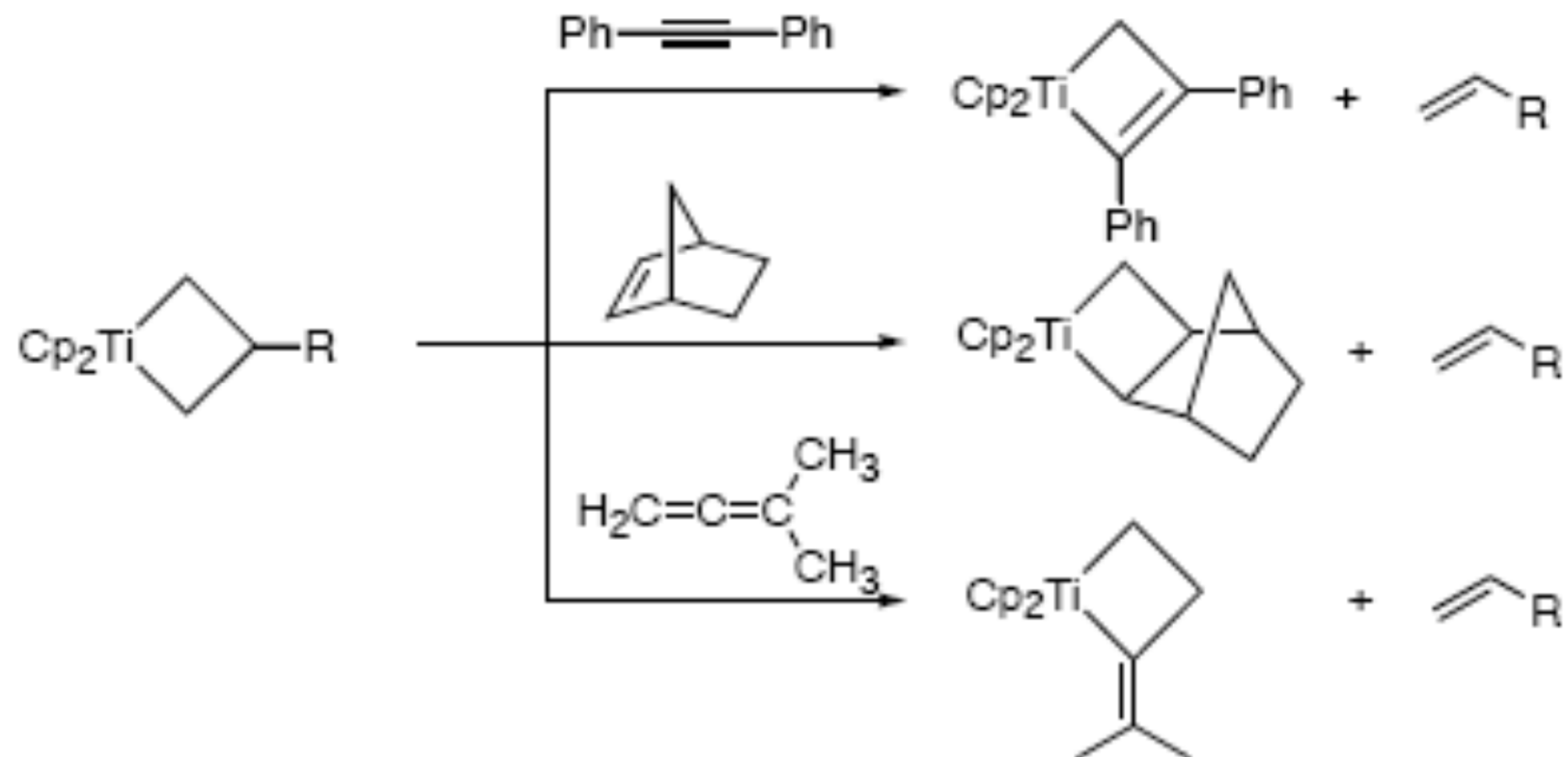




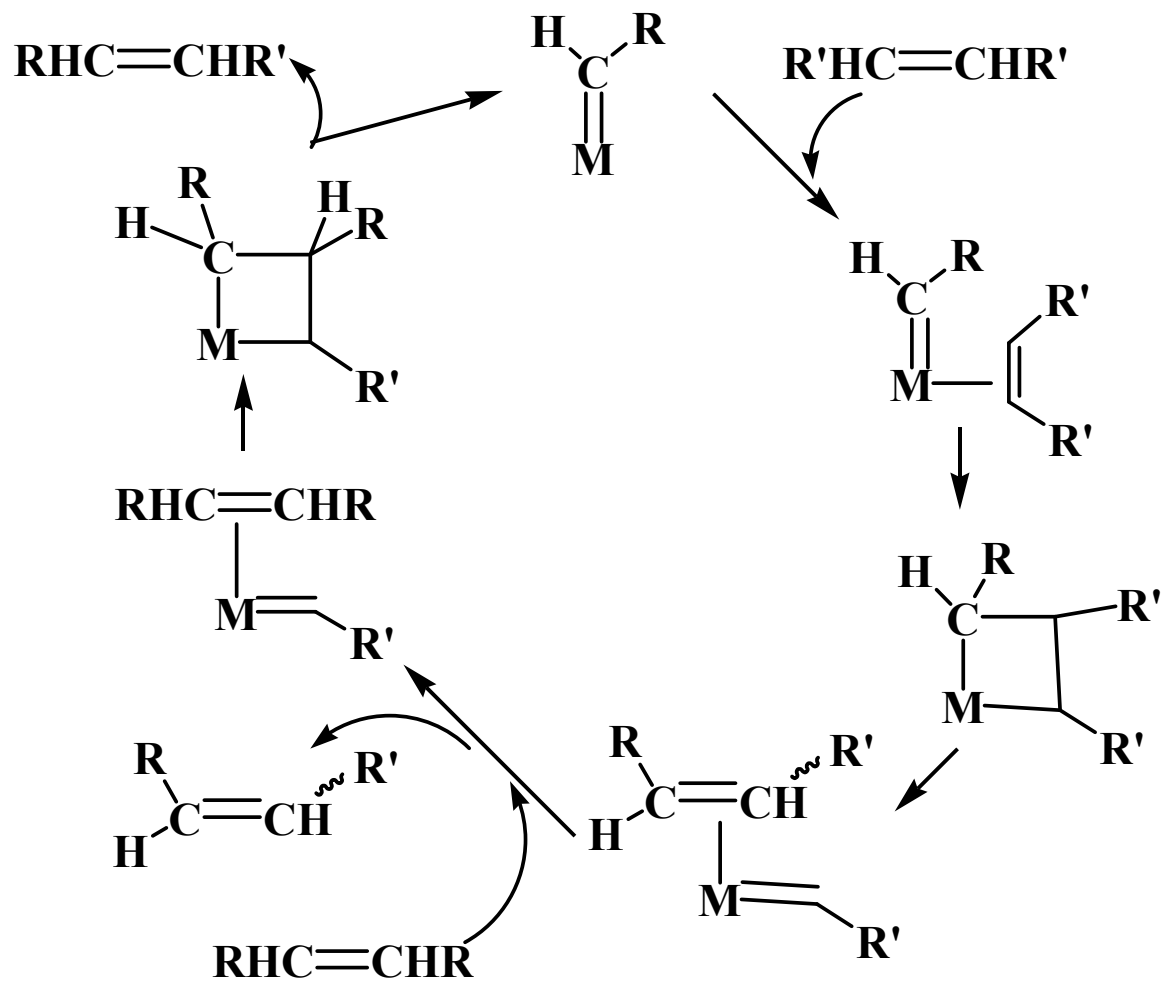
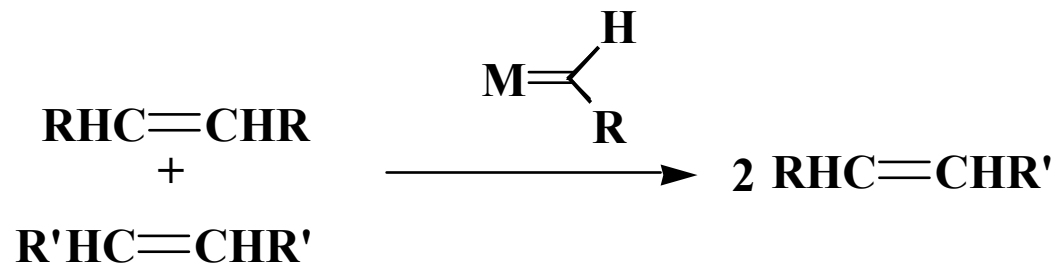
Acyl halides do not methylenate, but give an enolate



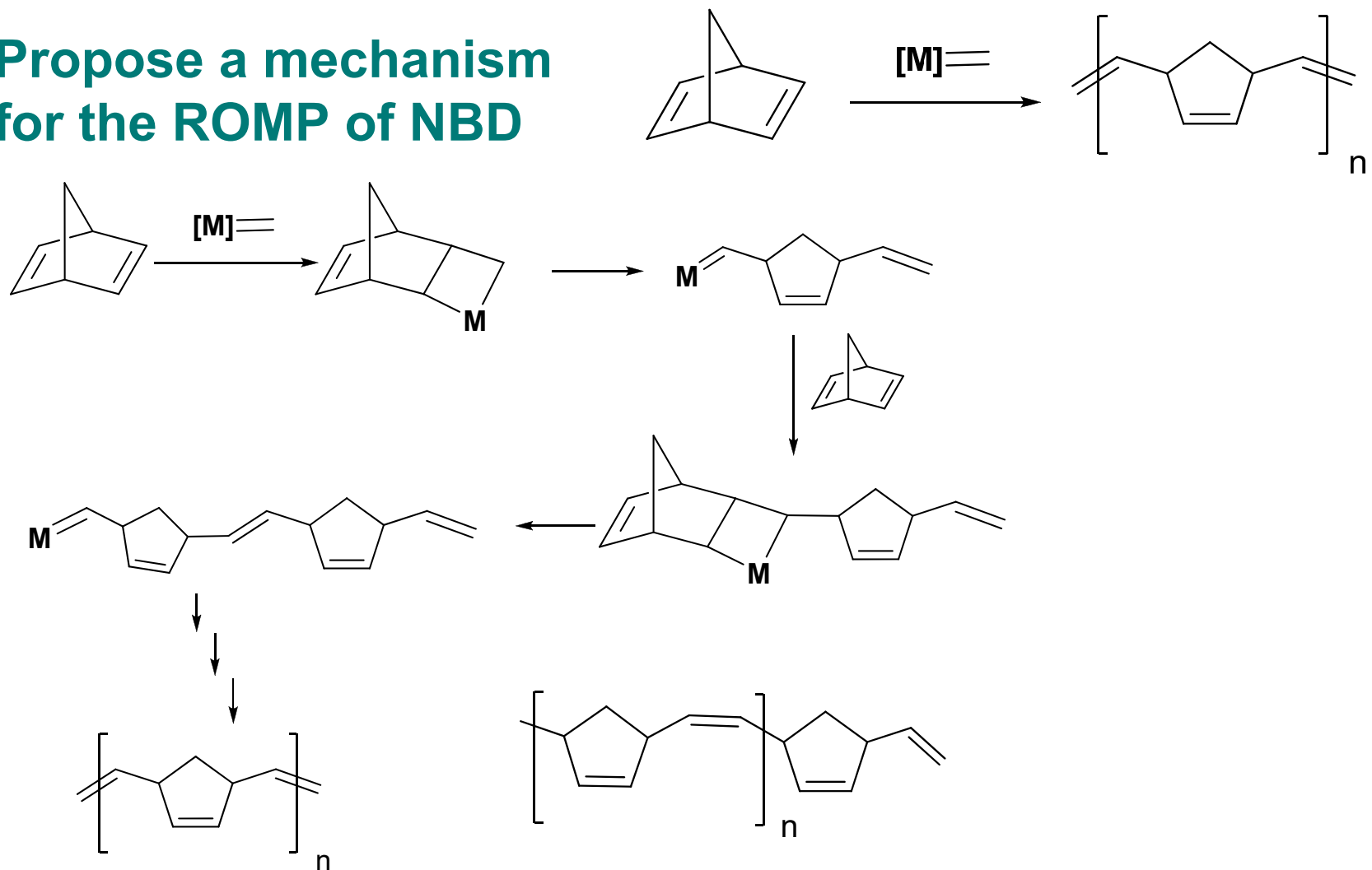




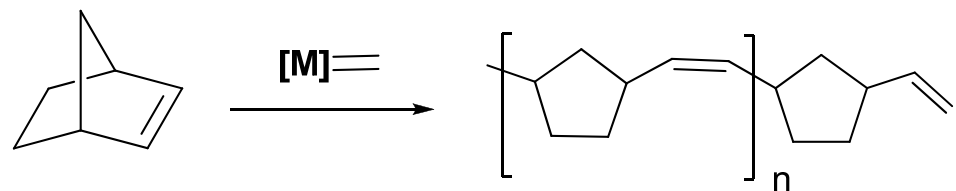
Olefin metathesis



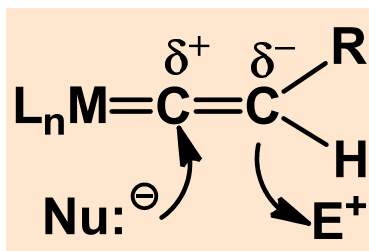
Propose a mechanism for the ROMP of NBD



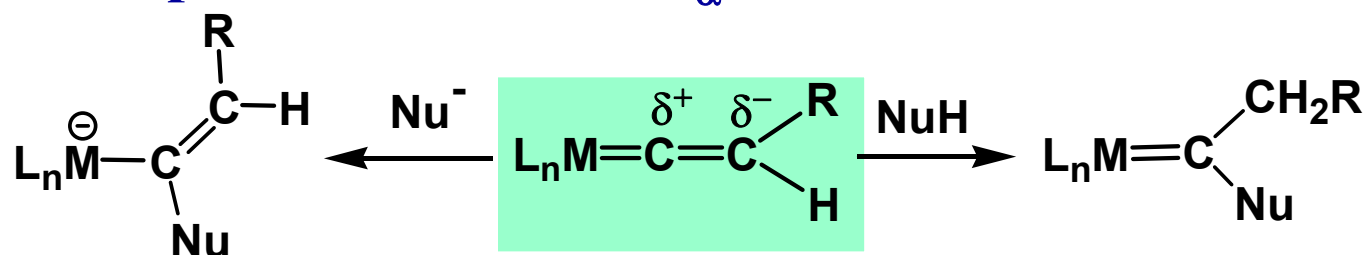
Propose a mechanism for the ROMP of NBE



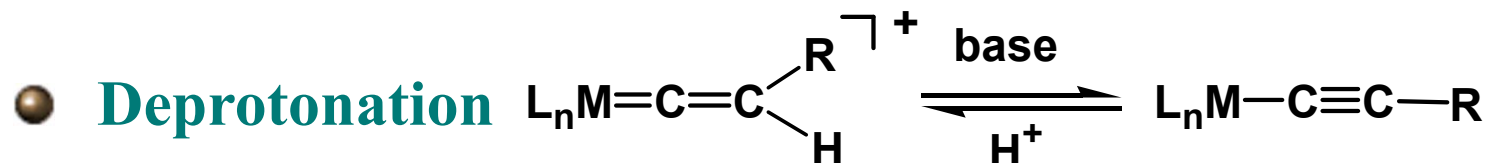
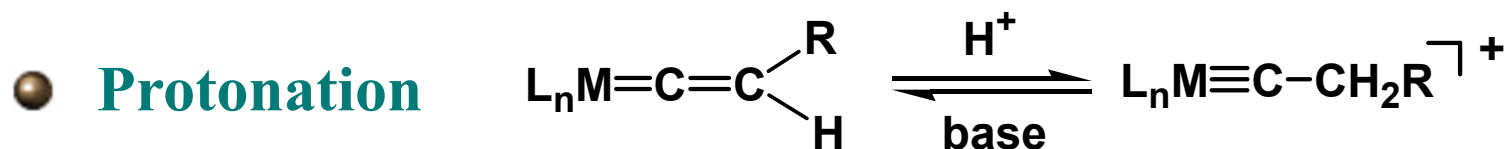
(6) Reactivity of Metal Vinylidene Complexes



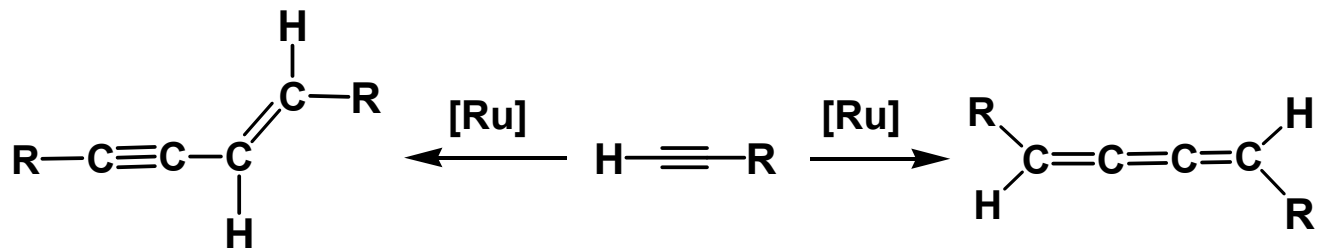
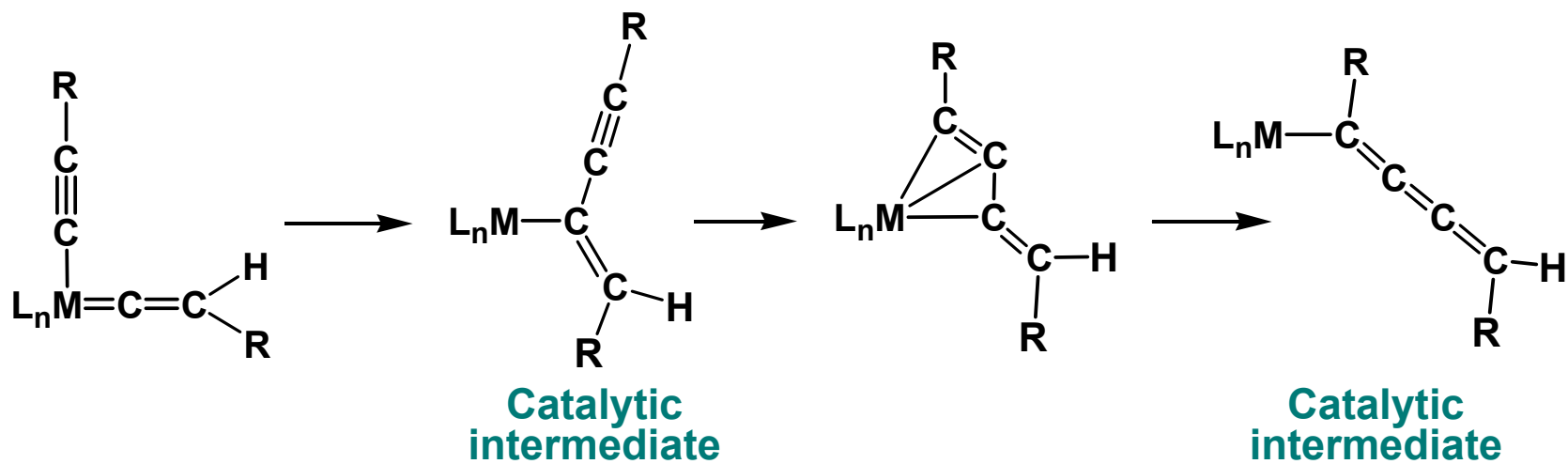
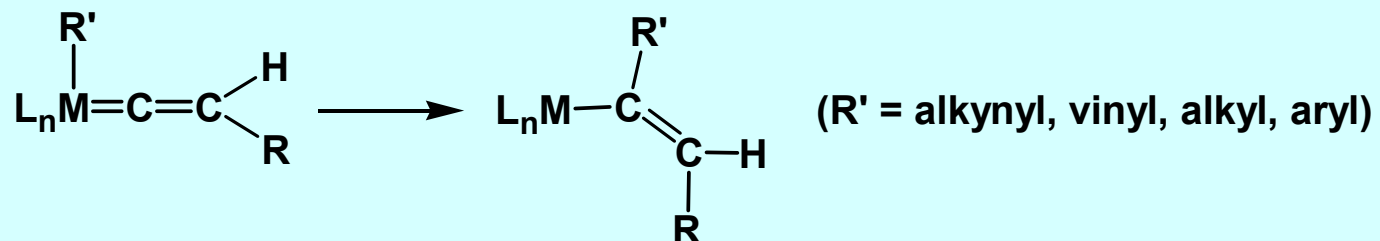
● Nucleophilic addition to C_α



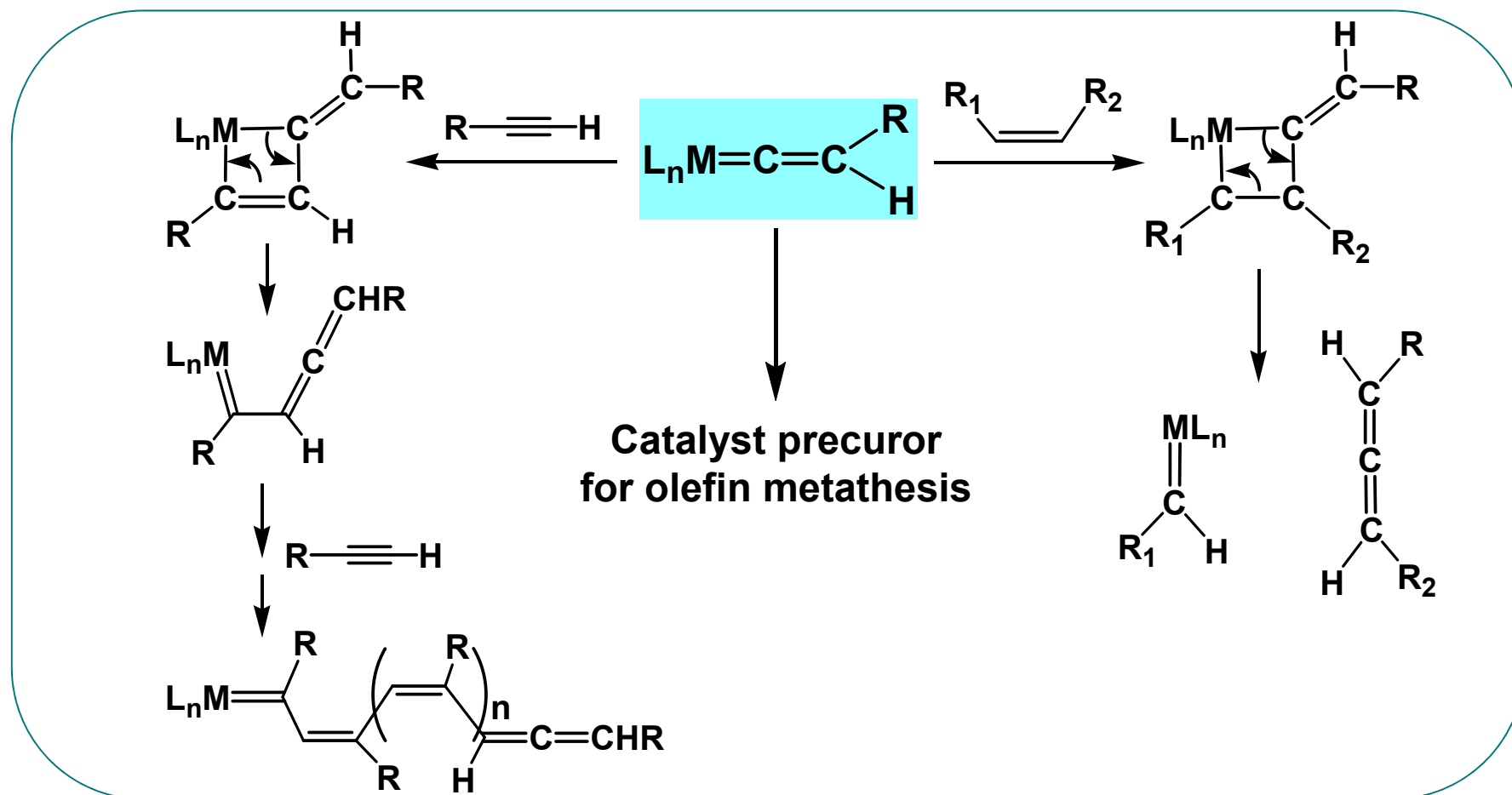
(Nu = PR'_3 , NR_3 ; Nu^- = halide; NuH = H_2O , $\text{R}'\text{OH}$, RSH)



● Promote C-C bond formation

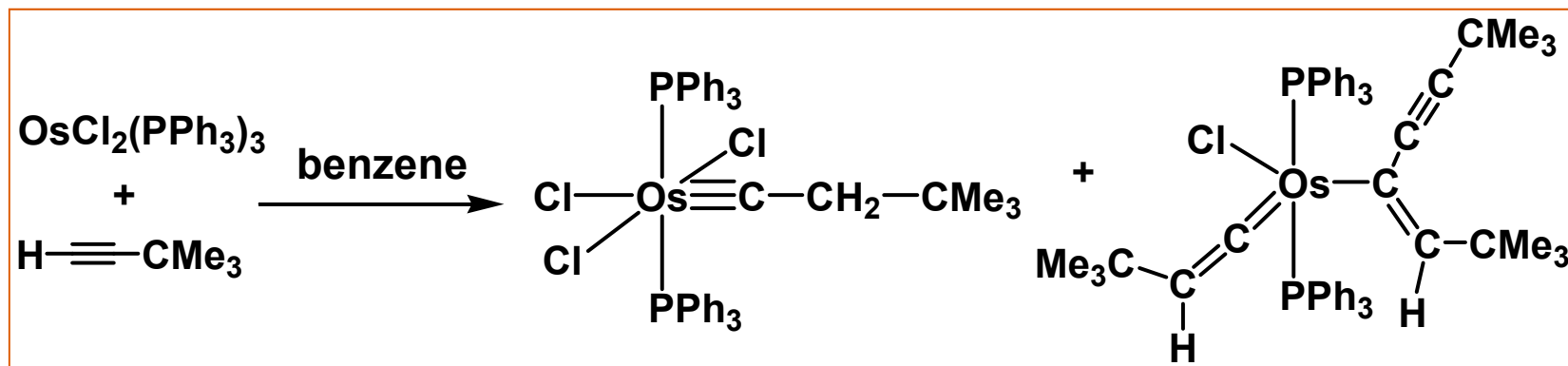


● [2+2] Cycloaddition

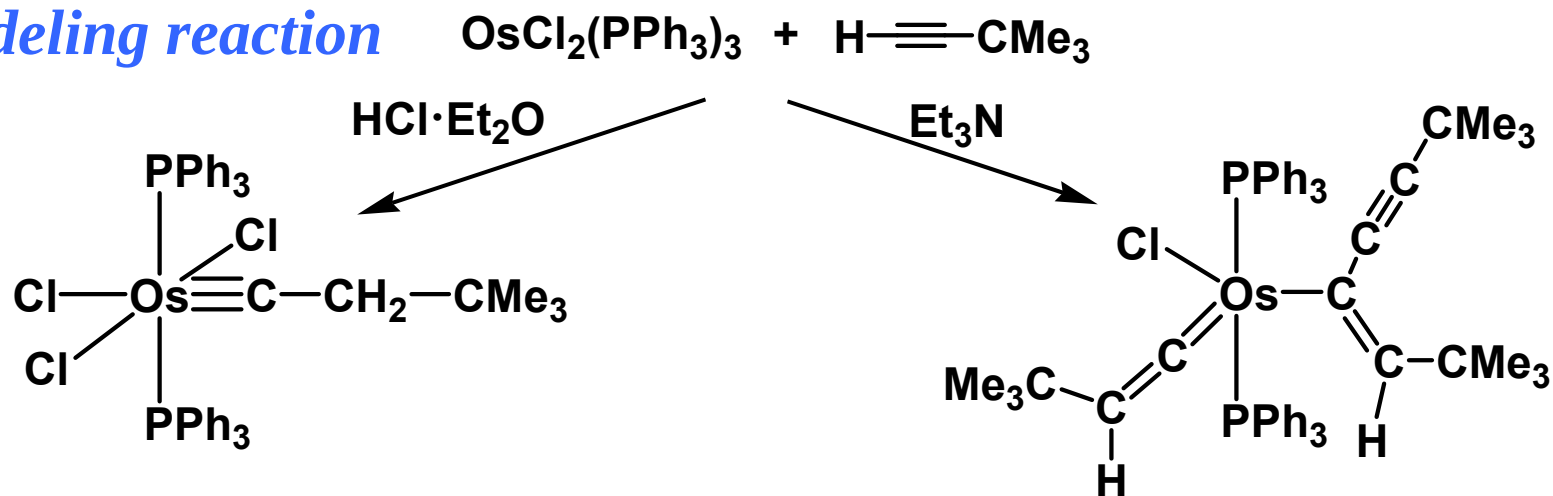


Organometallic vinylidenes are at the heart of selective transformations of terminal alkynes with atom economy.

Reaction of $\text{OsCl}_2(\text{PPh}_3)_3$ with $\text{HC}\equiv\text{CCMe}_3$

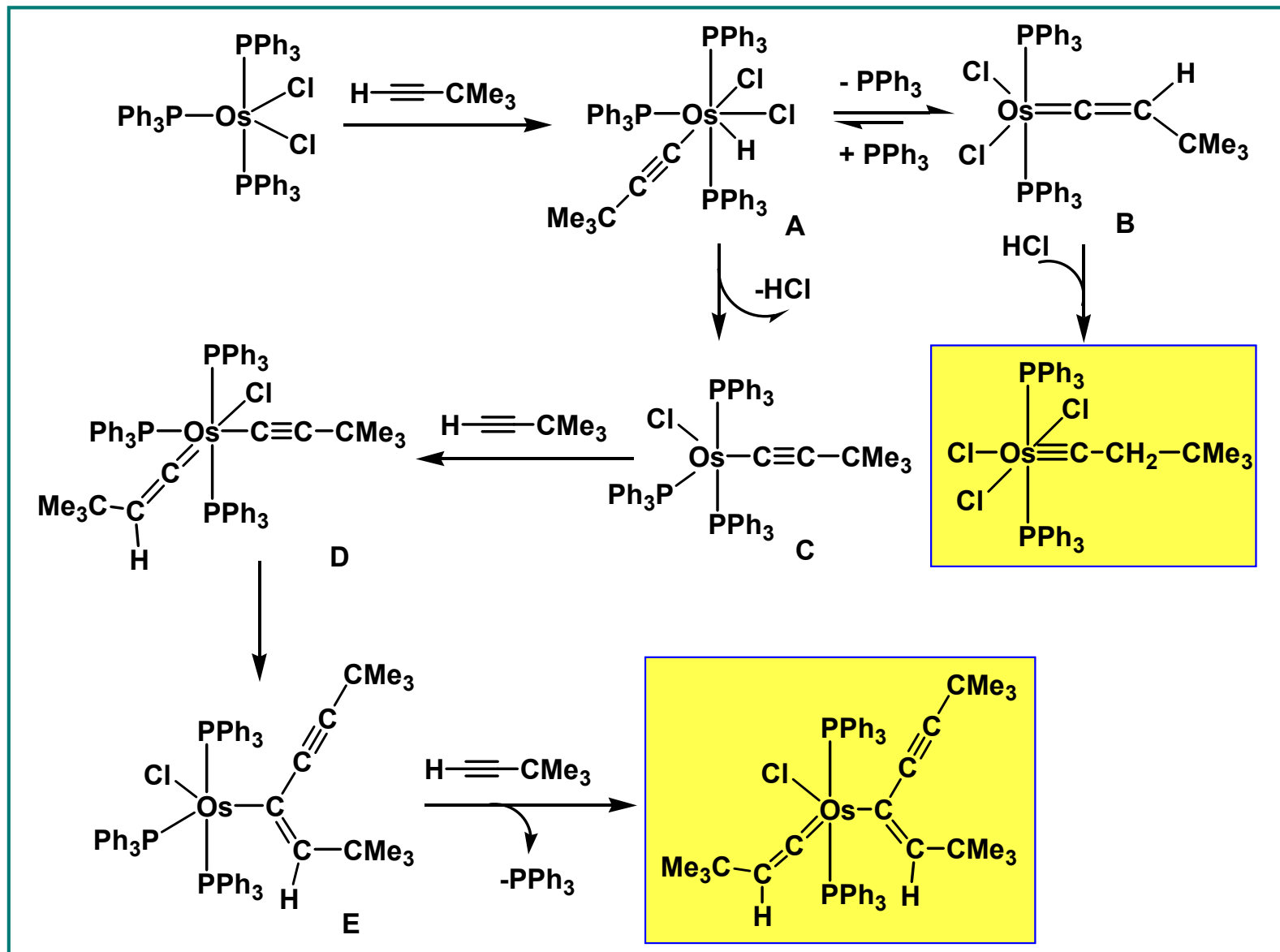


Modeling reaction

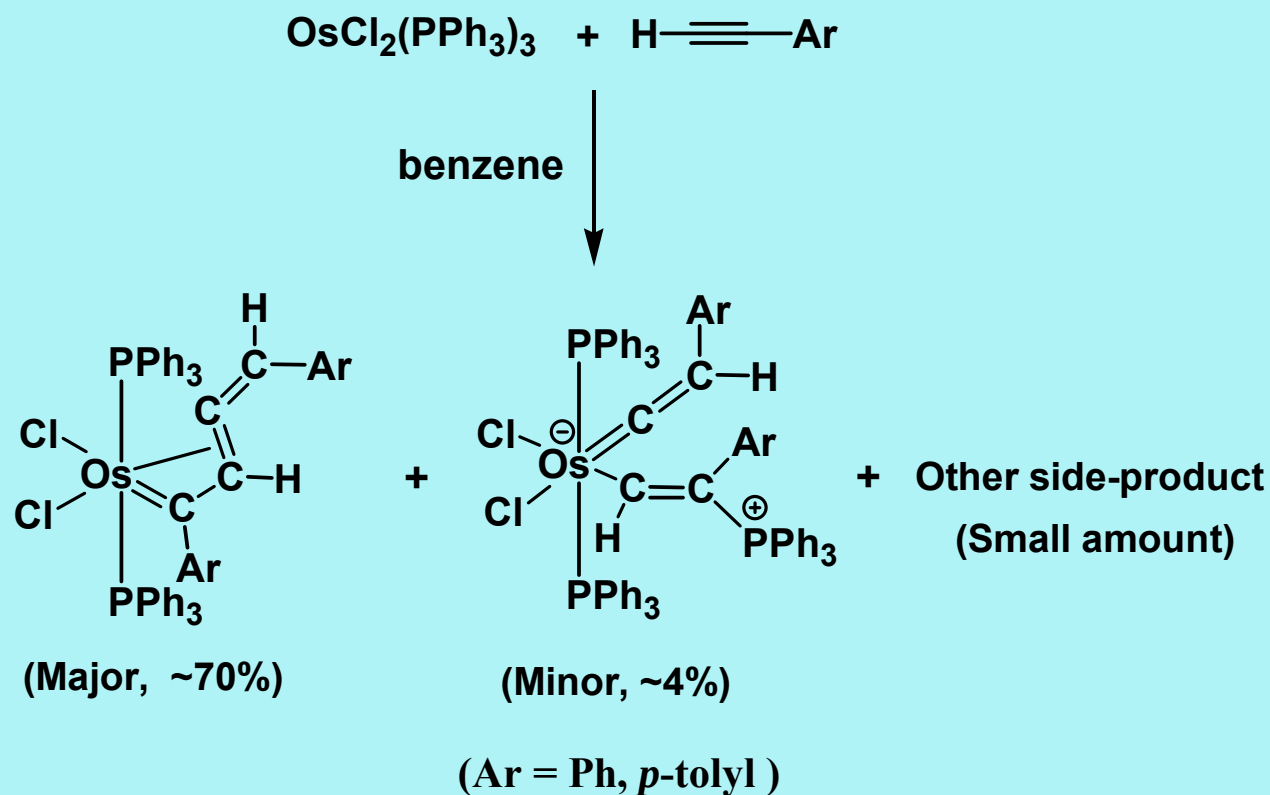


(Wen, T. B.; Yang, S. Y.; Zhou, Z. Y.; Lin, Z.; Lau, C. P.; Jia, G.
Organometallics 2000, 19, 3757.)

Plausible mechanism for reaction of $\text{OsCl}_2(\text{PPh}_3)_3$ with $\text{HC}\equiv\text{CCMe}_3$

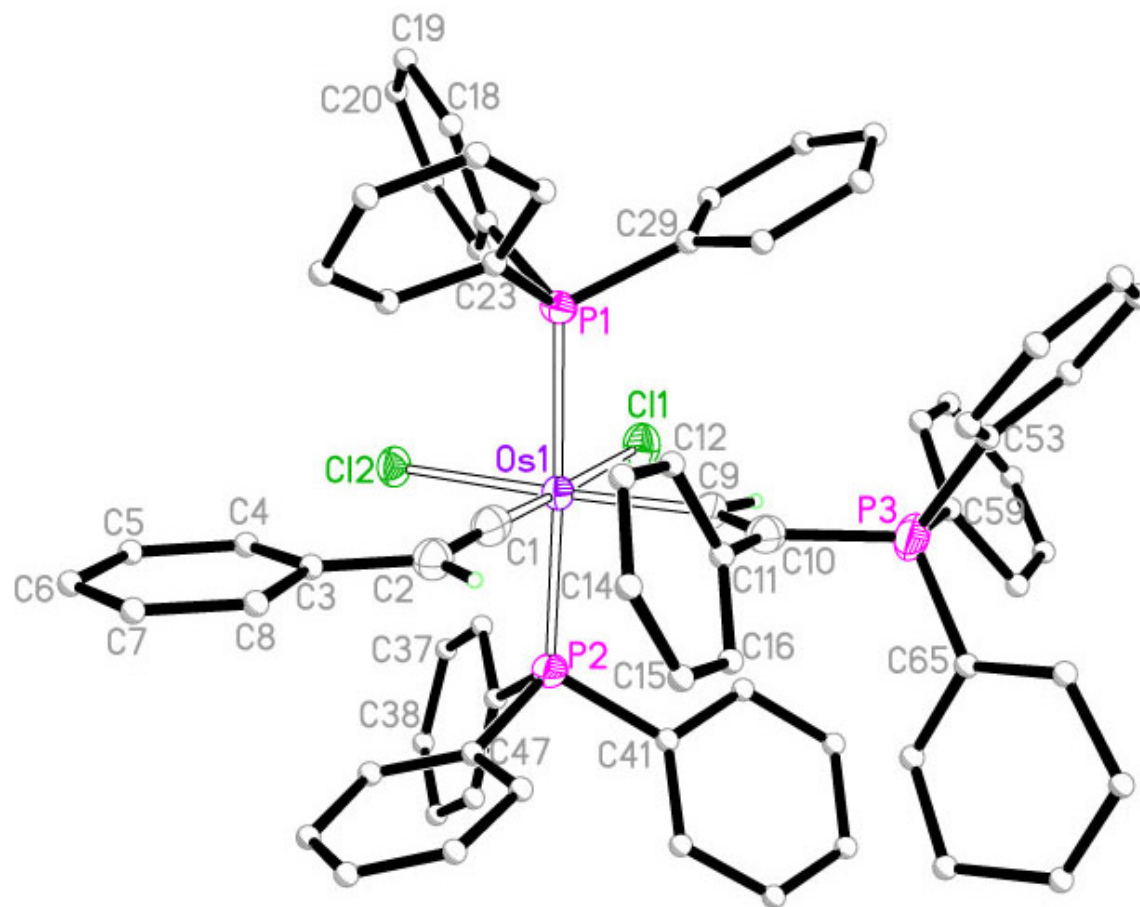
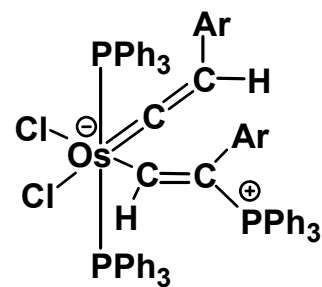


Reaction of $\text{OsCl}_2(\text{PPh}_3)_3$ with $\text{HC}\equiv\text{CAr}$ *(Ar = Ph, *p*-tolyl)*

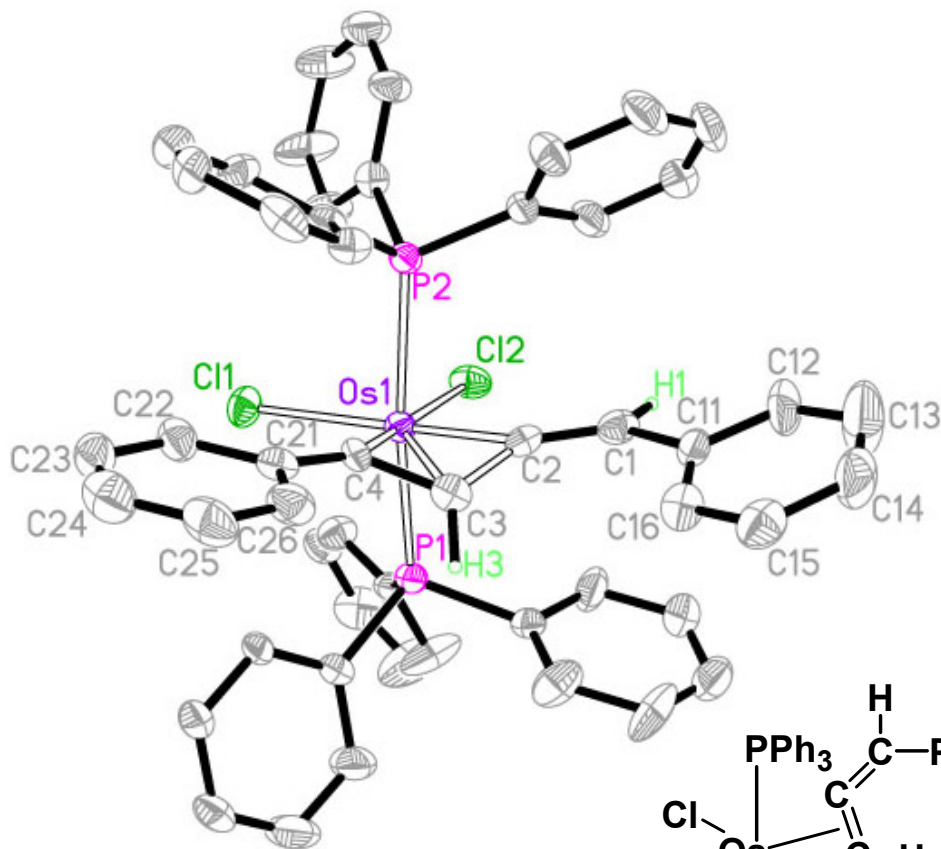
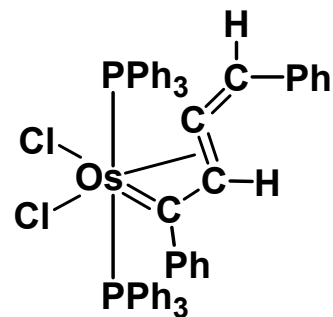


(Wen, T. B.; Jia, G. et. al, *J. Am. Chem. Soc.*, 2005, 127, 2856.)

Crystal structure of



X-ray Structure of



Bond Distances (Å)

Os(1)-C(2) 2.069(11)

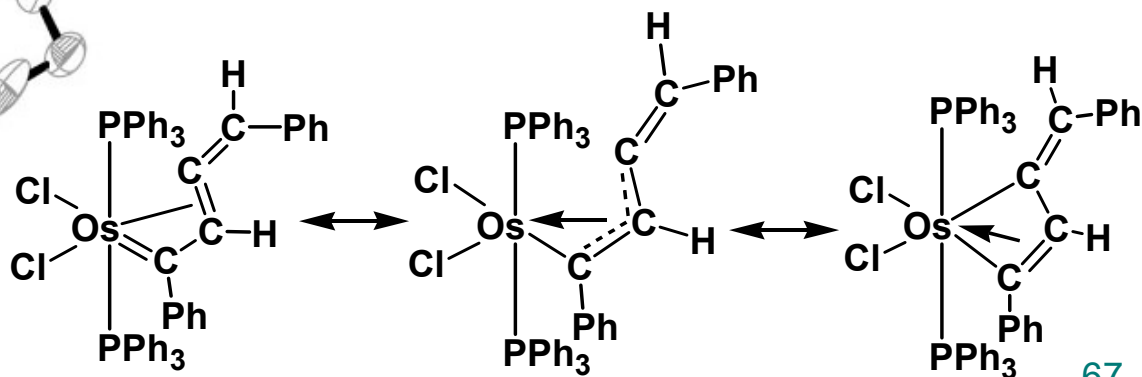
Os(1)-C(3) 2.166(11)

Os(1)-C(4) 1.895(9)

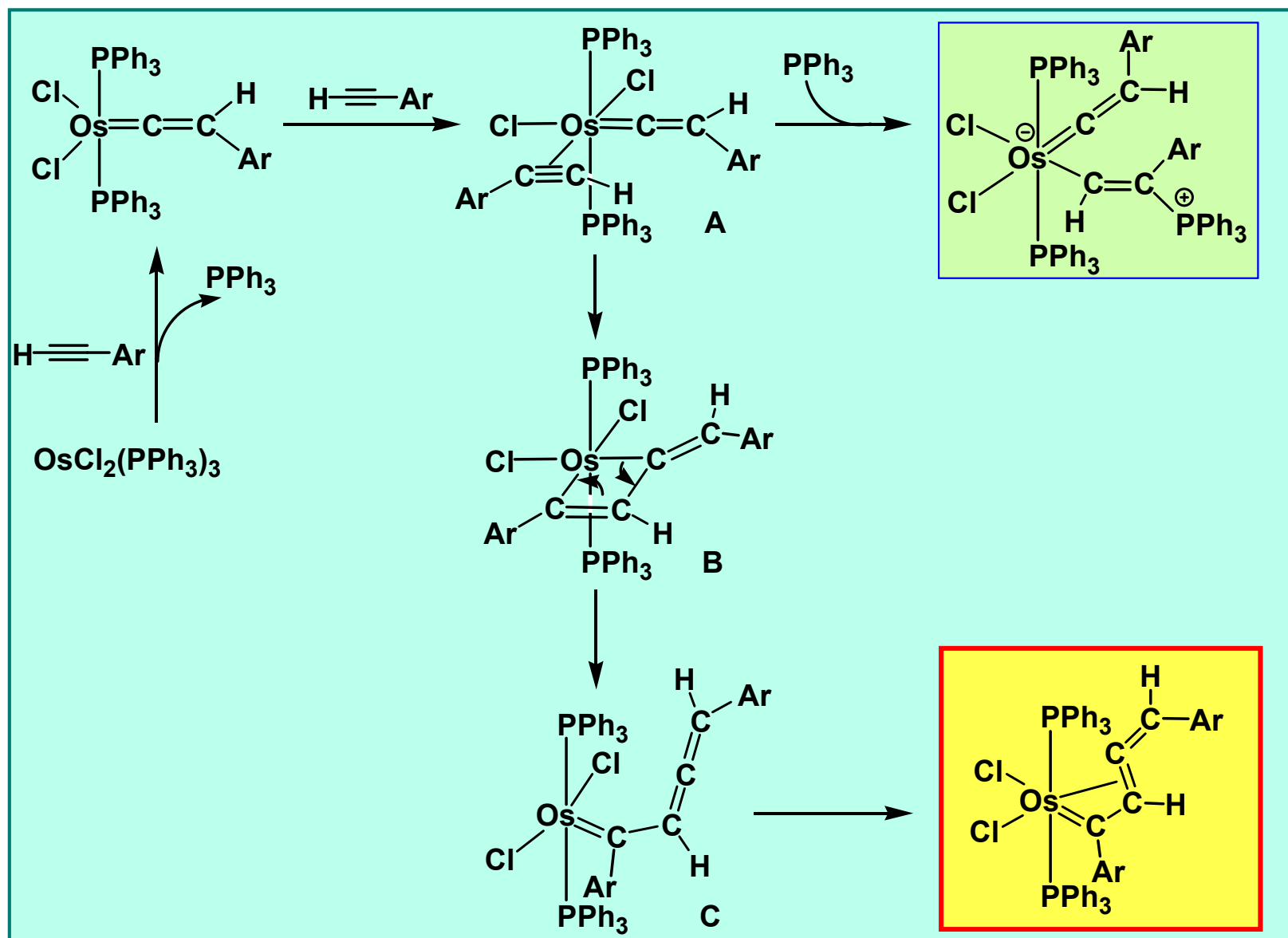
C(1)-C(2) 1.319(15)

C(2)-C(3) 1.445(15)

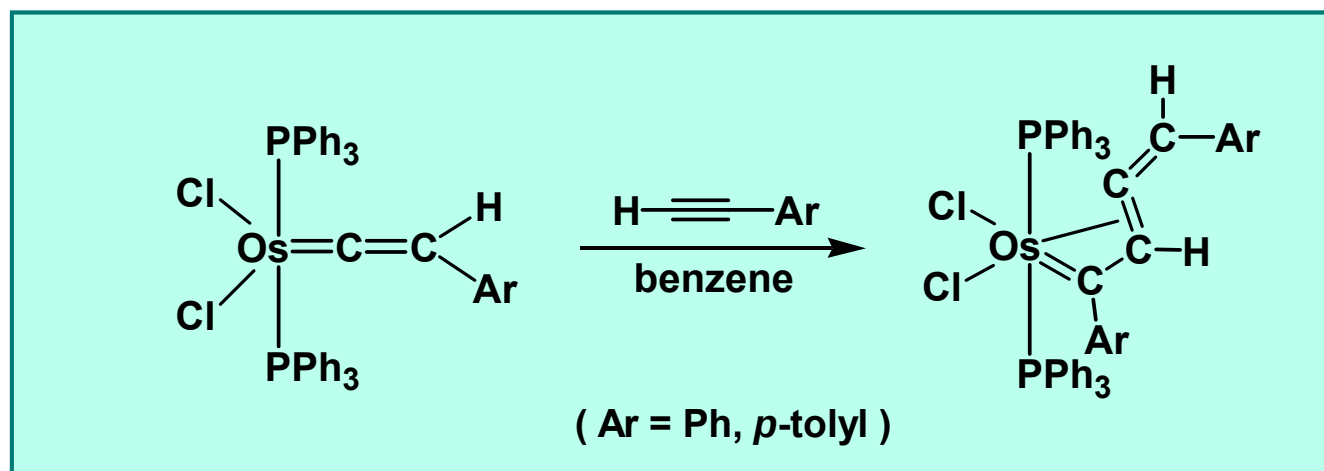
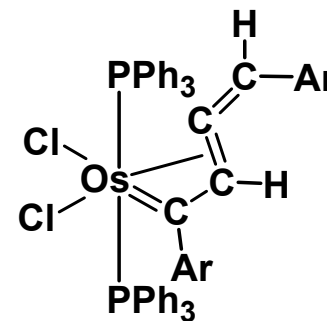
C(3)-C(4) 1.432(15)



Plausible mechanism for reaction of $\text{OsCl}_2(\text{PPh}_3)_3$ with $\text{HC}\equiv\text{CCAr}$



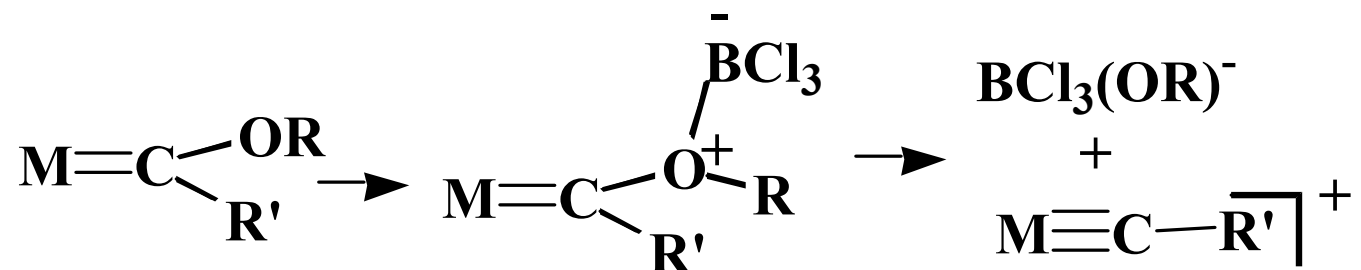
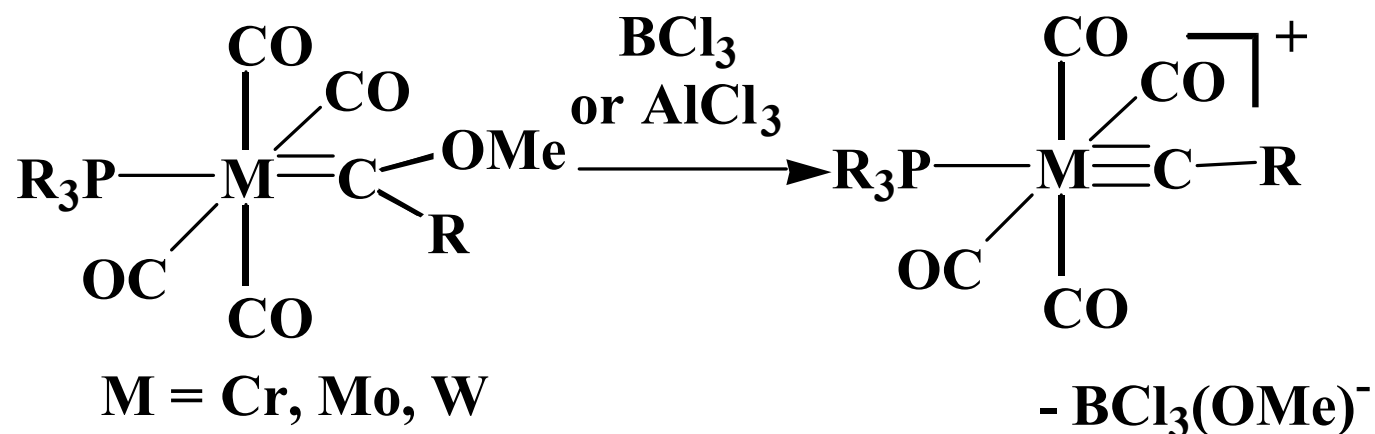
Modeling reaction for the formation of

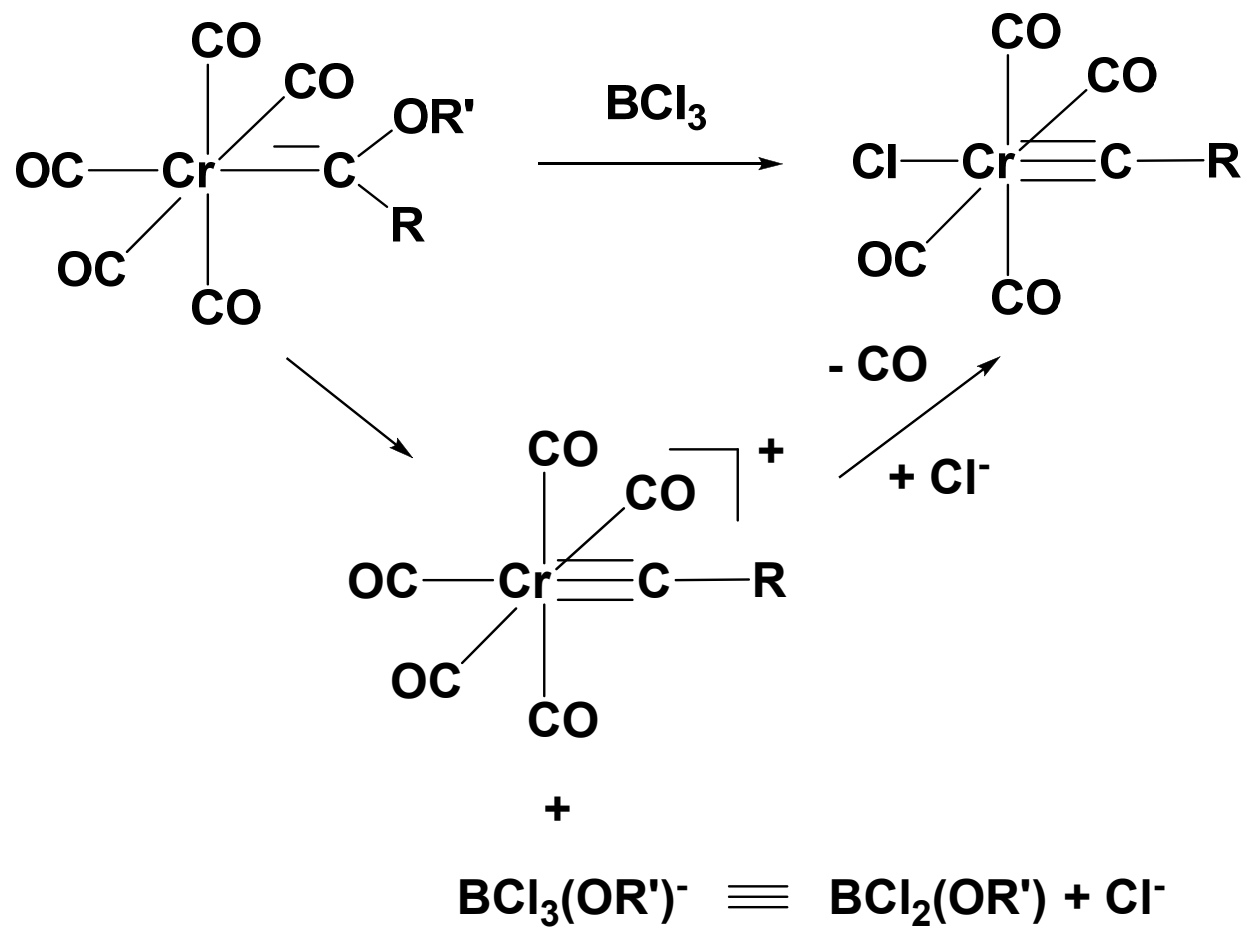


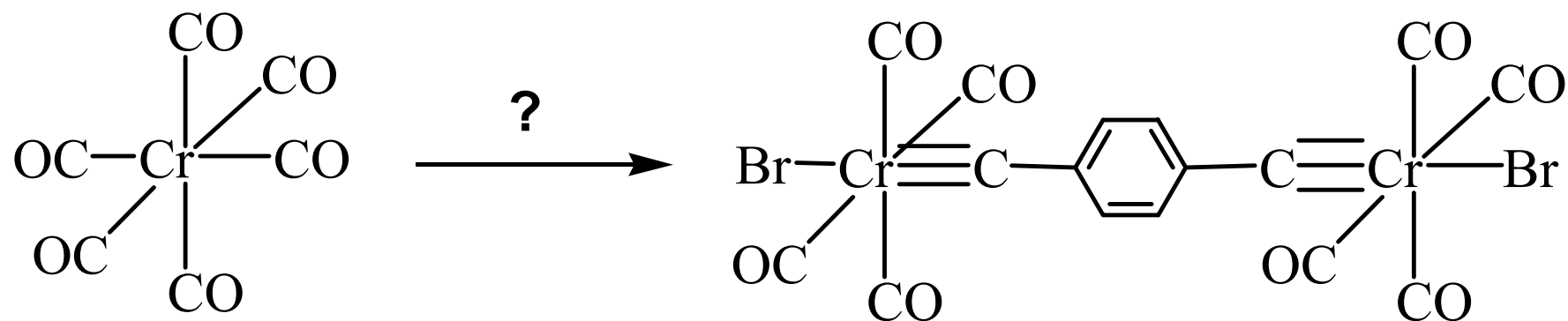
(Wen, T. B.; Jia, G. et al. *J. Am. Chem. Soc.* **2005**, 127, 2856; OM, **2016**, 35, 1514.)

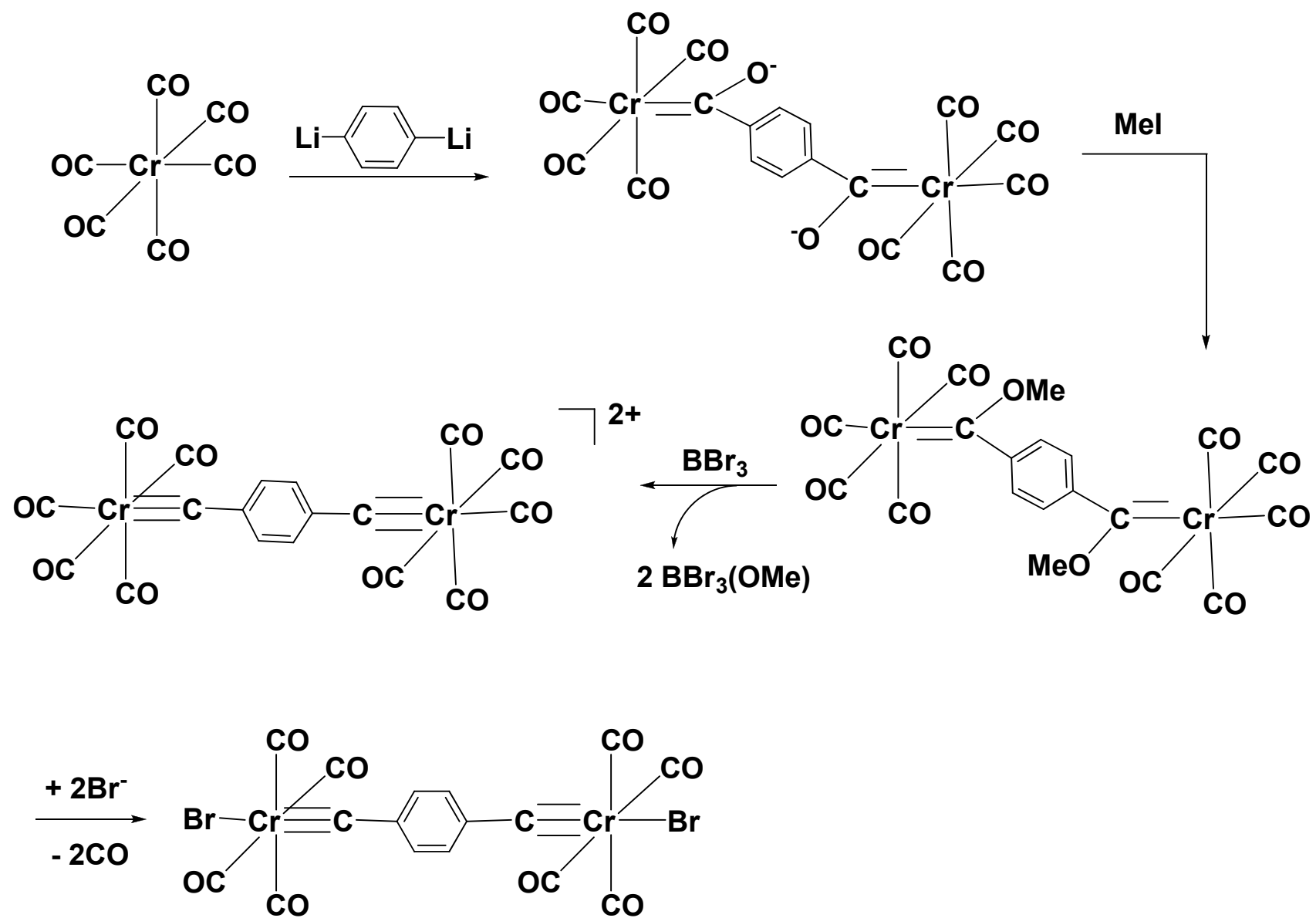
D. Common synthetic routes to carbyne complexes

(1) Lewis-acid-assisted abstraction of OR.

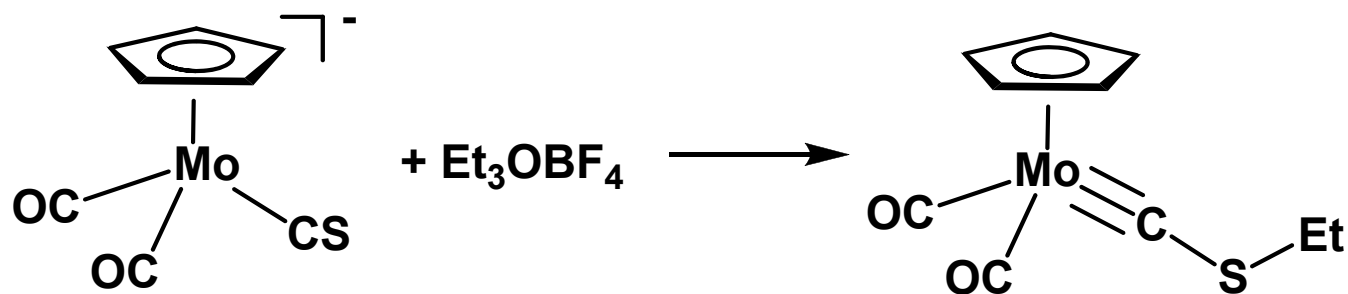
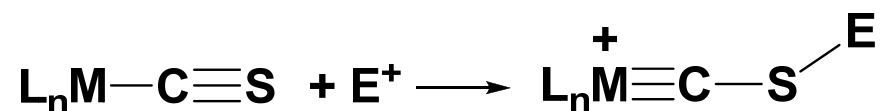
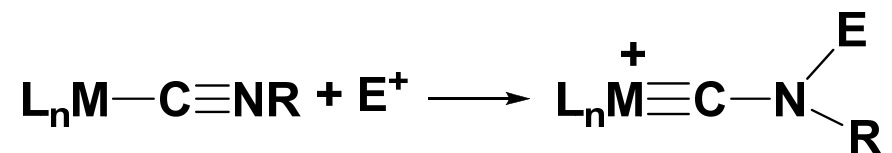


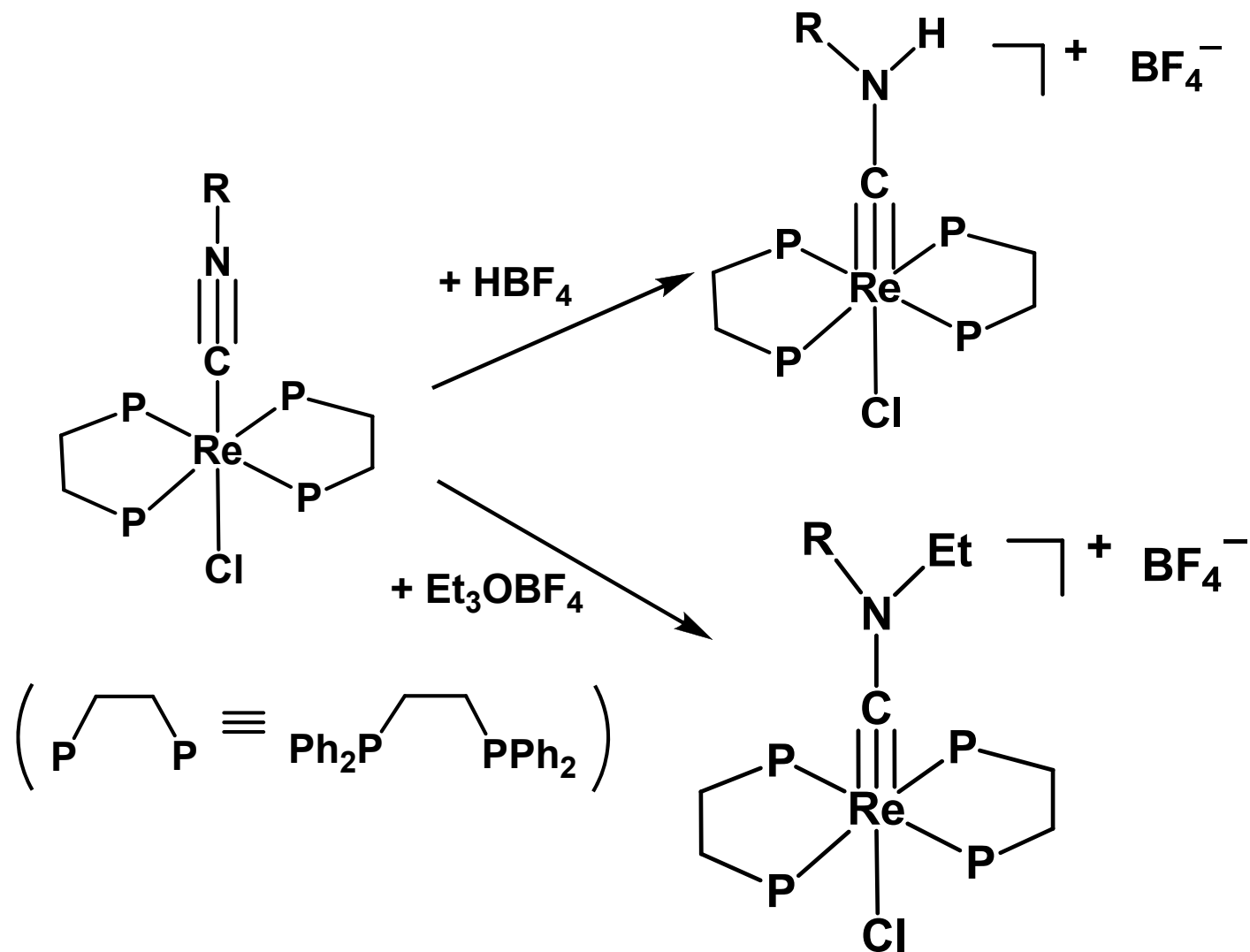




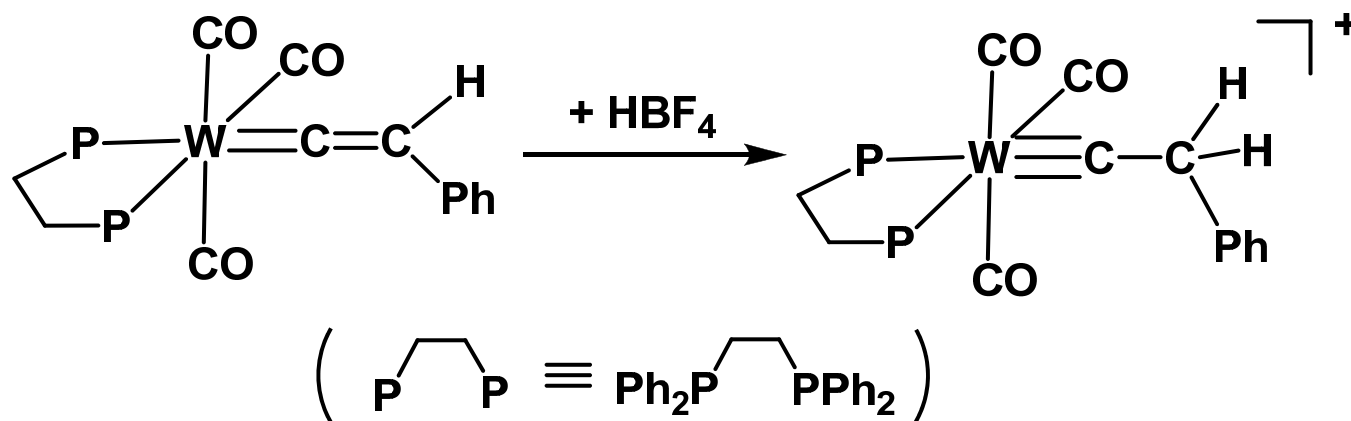
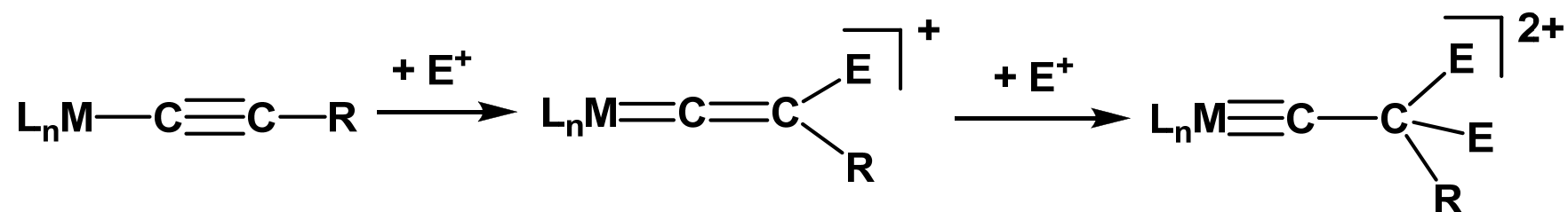


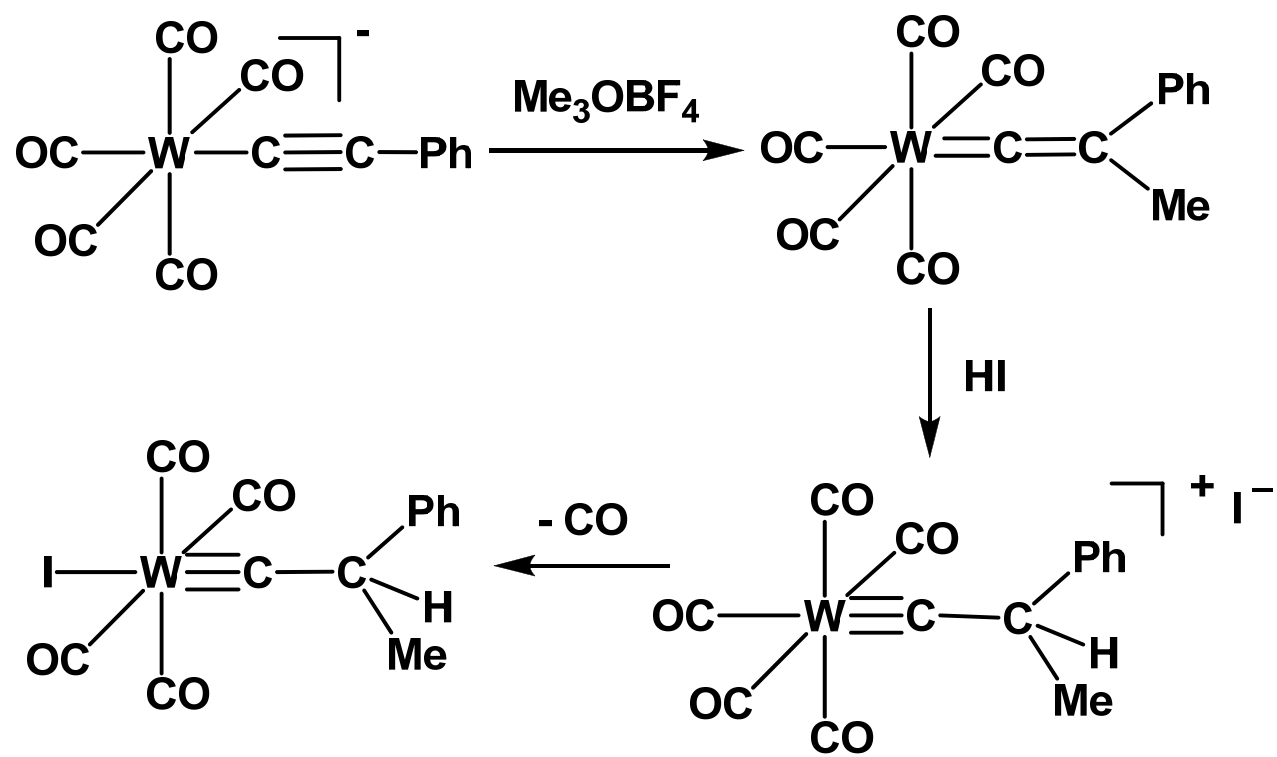
(2) Electrophilic addition to M-CNR and M-CS





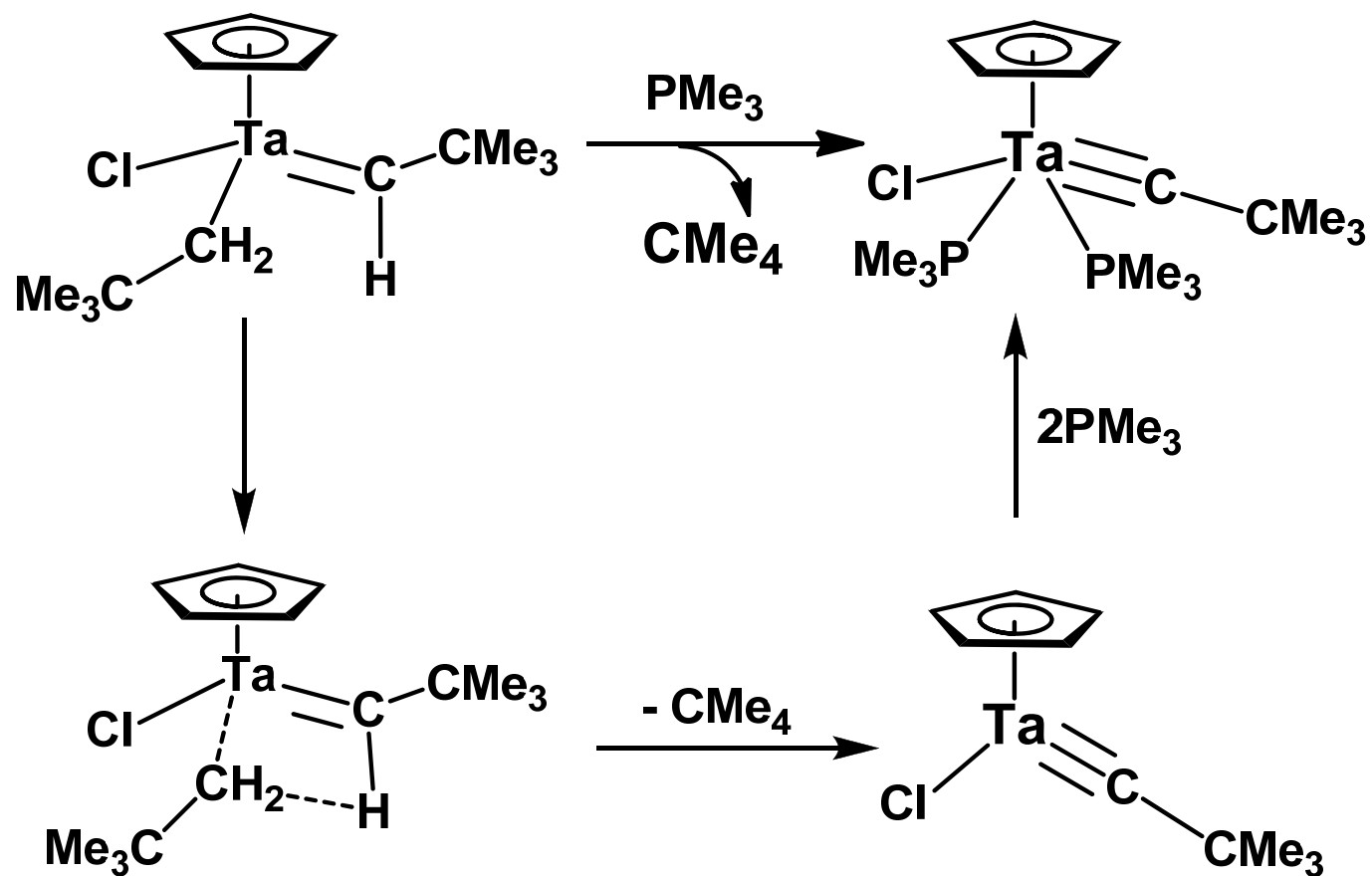
(3) Transformation of $M-C\equiv CR$ and $M=C=CRR'$

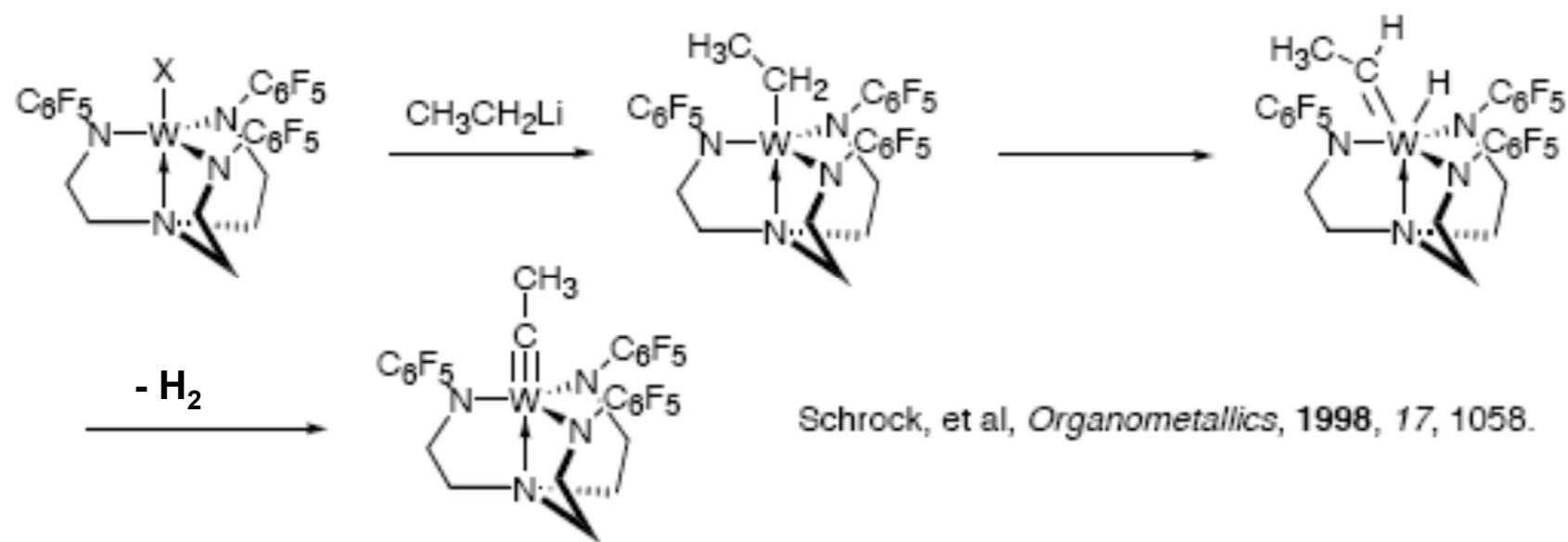




(4) By α -H elimination

e.g.

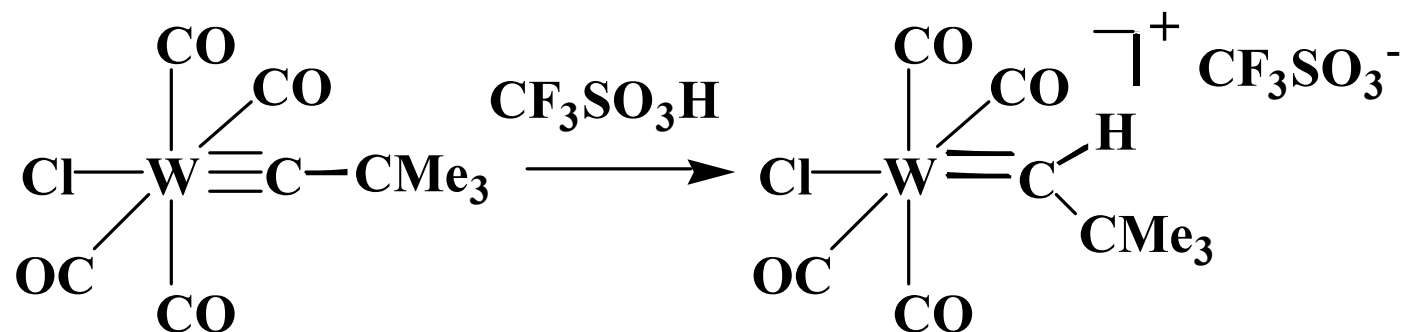
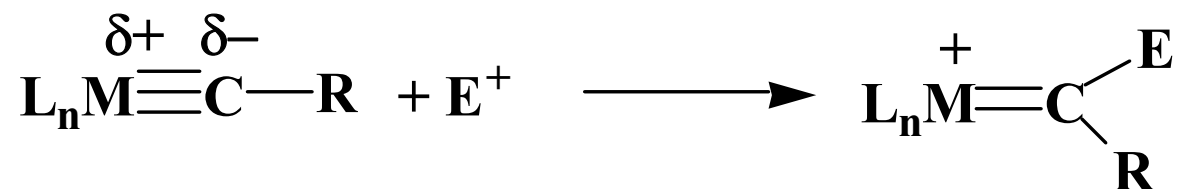


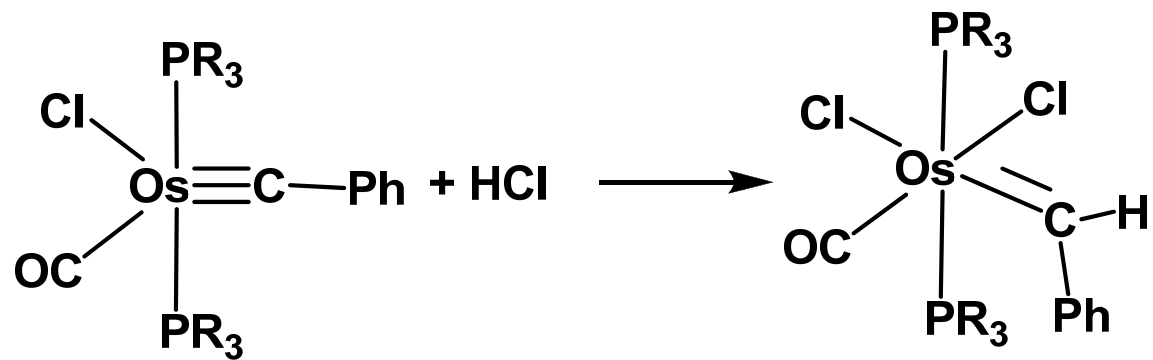
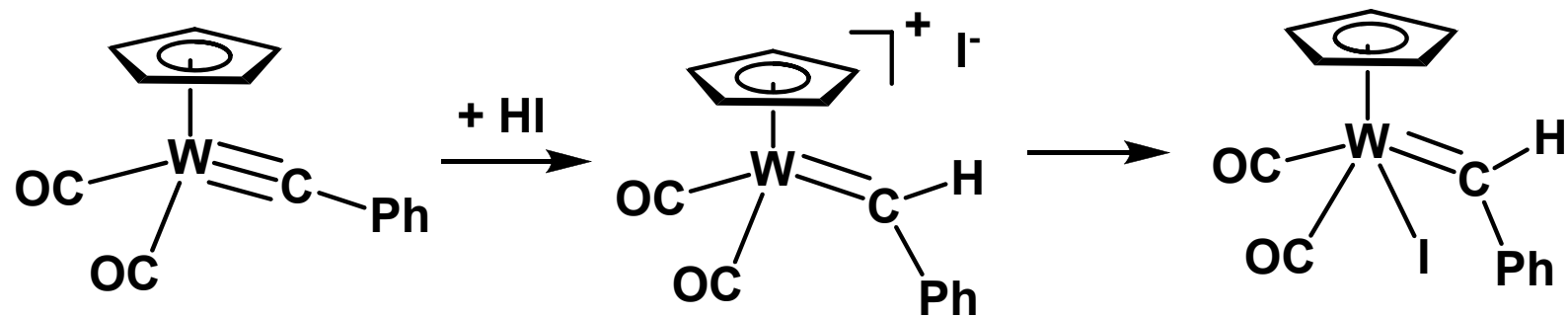


Schrock, et al, *Organometallics*, 1998, 17, 1058.

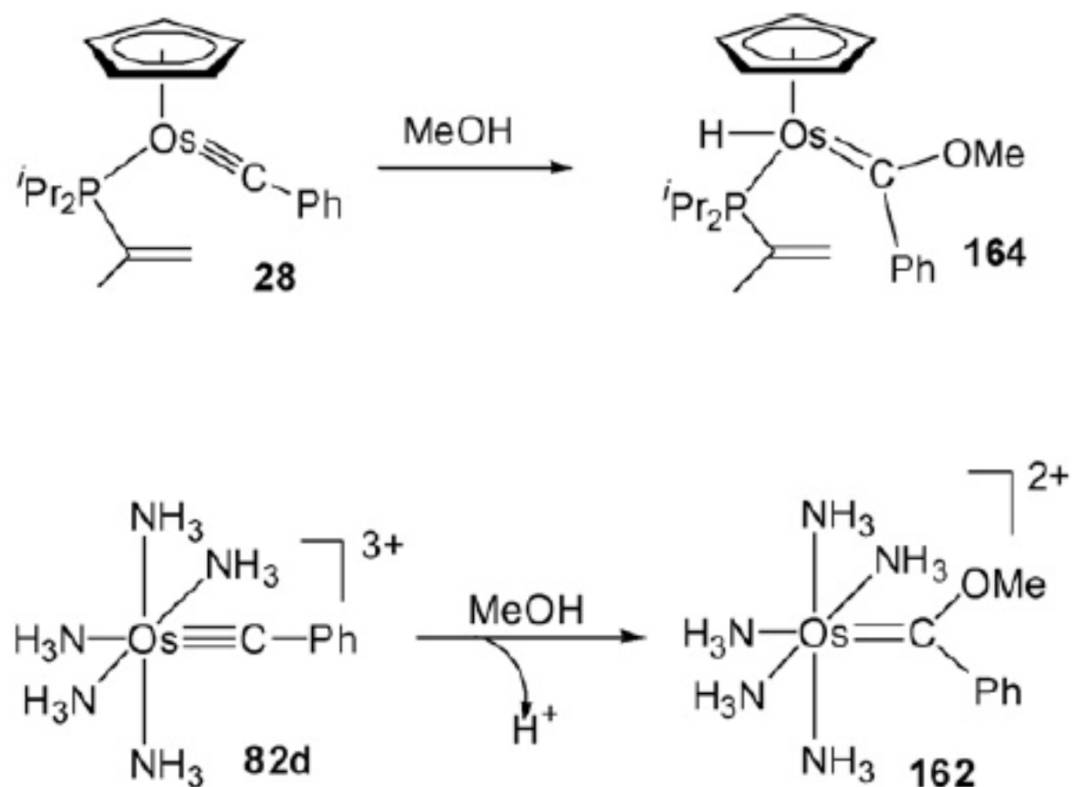
E. Reactivity of $M\equiv CR$

(1) *Reactions with electrophiles*

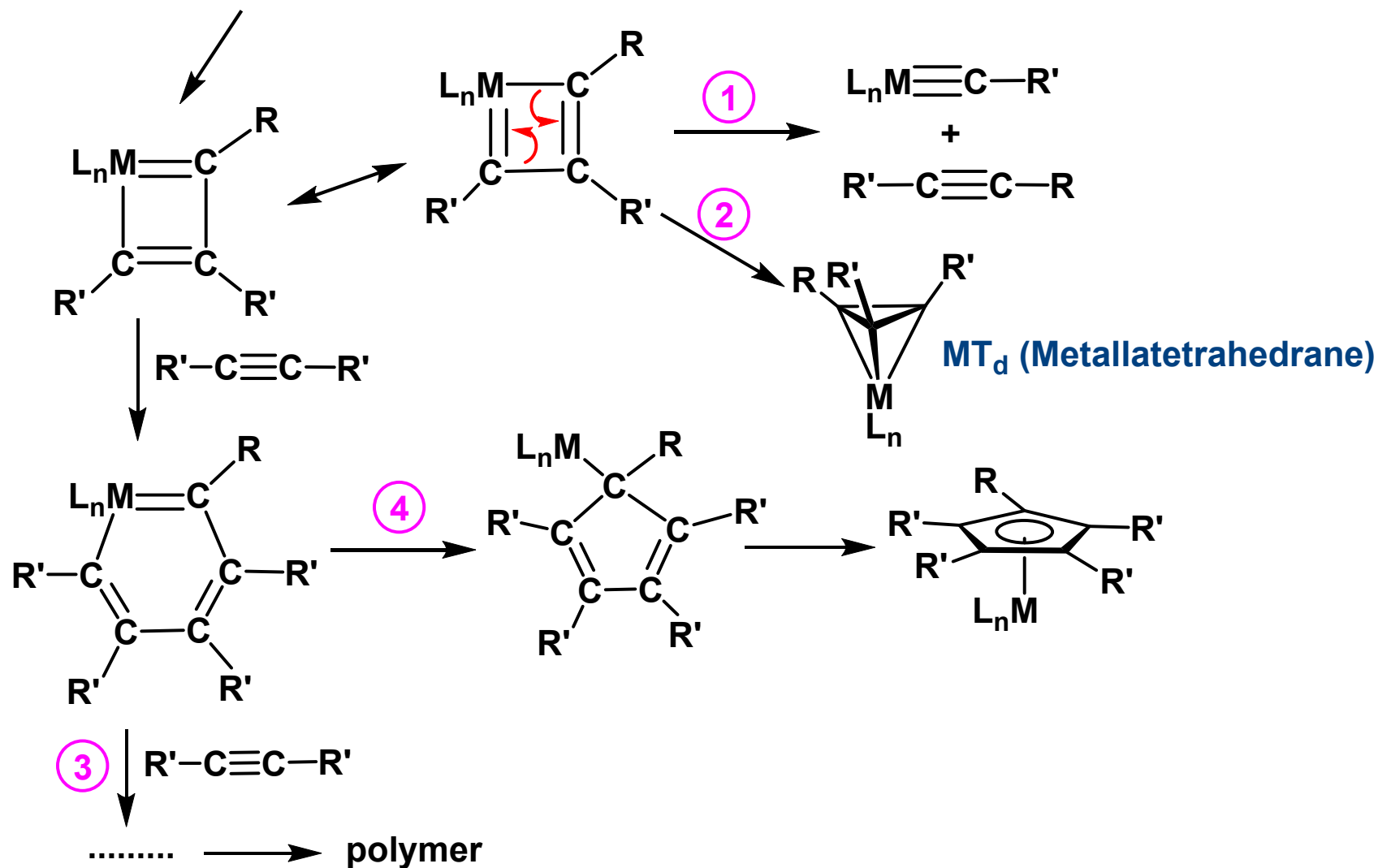
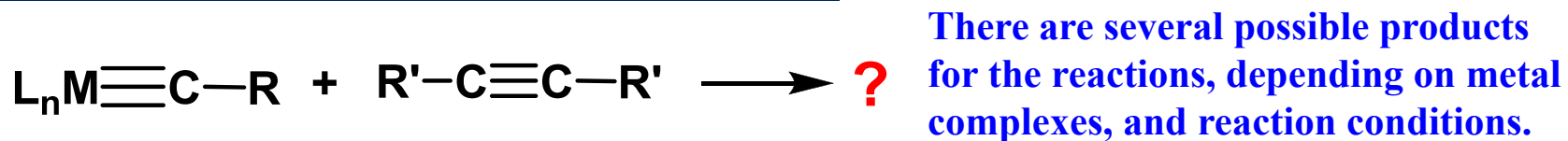




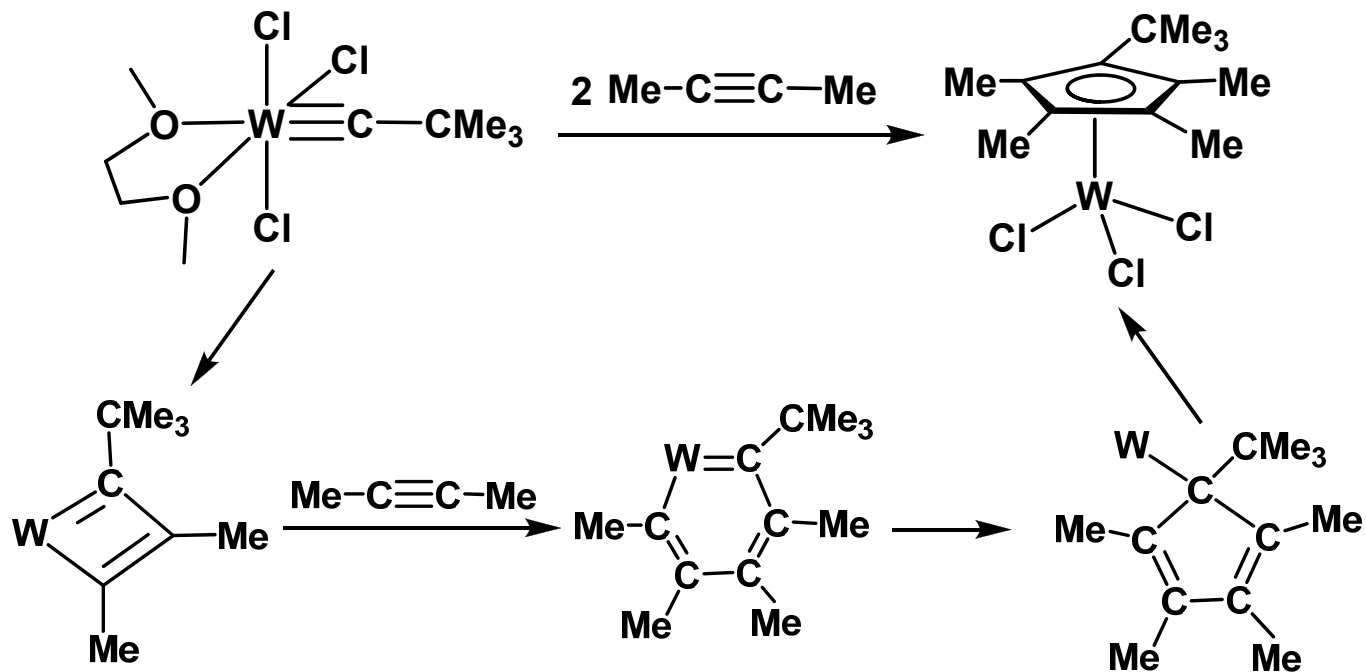
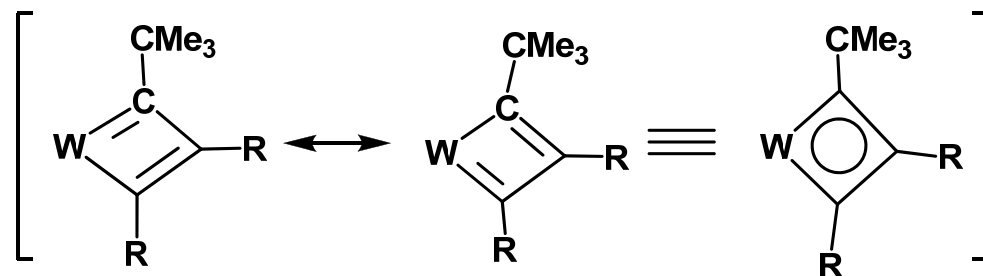
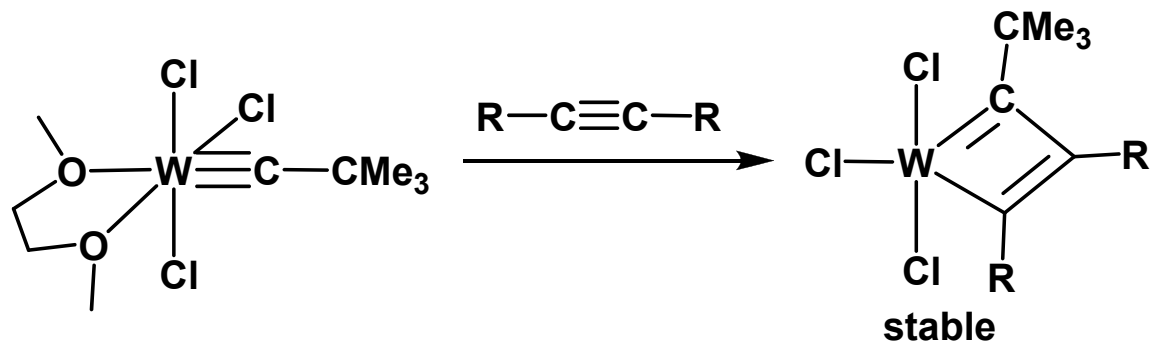
Note: some “high valent” carbyne carbon can be attacked by nucleophiles, e.g.



(2) Reactions with acetylenes

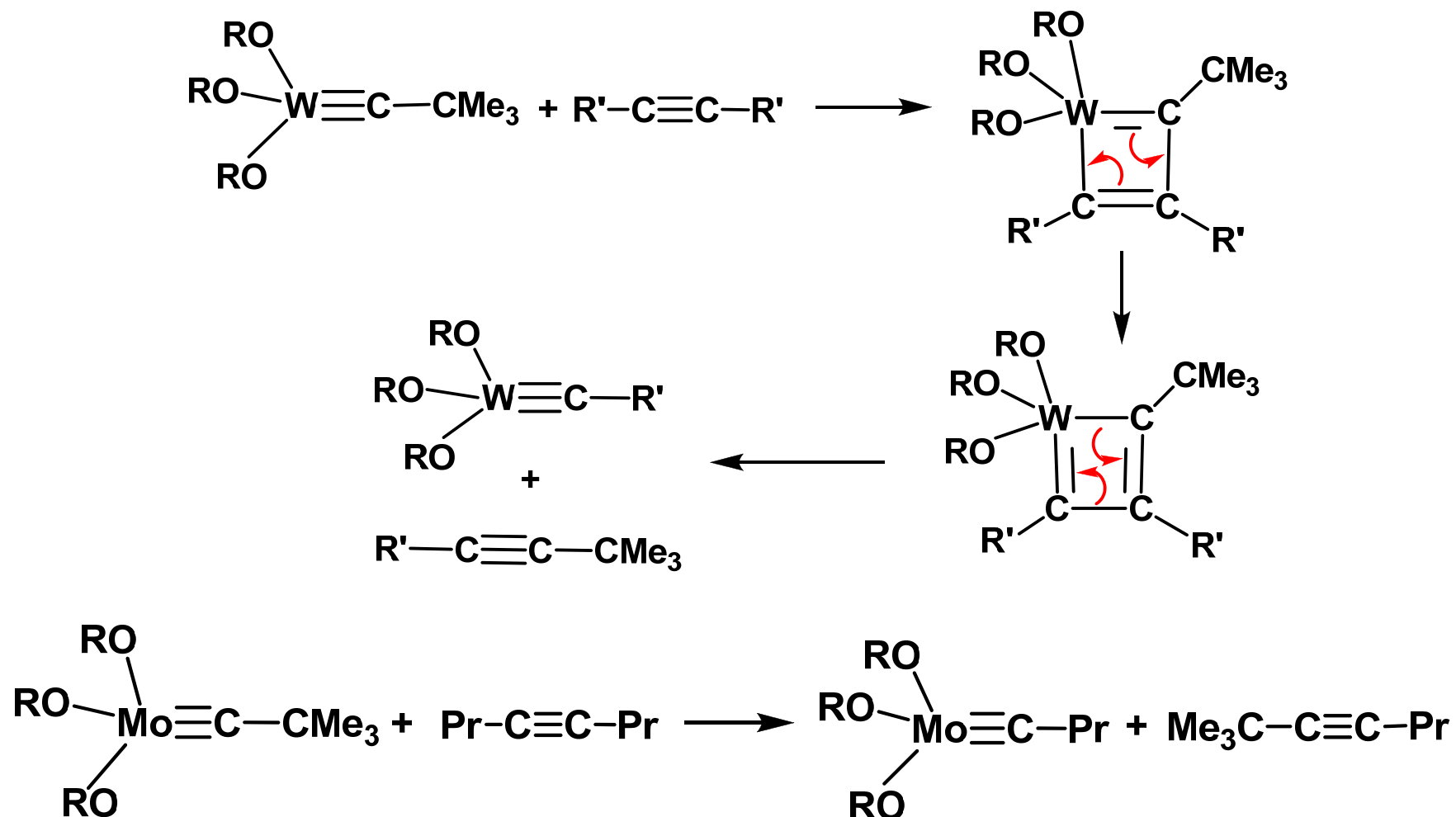


e.g.

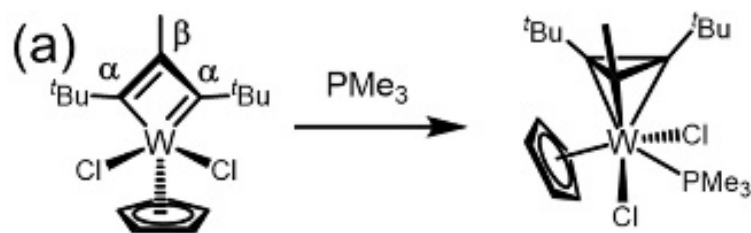


Schrock carbynes undergo metathesis reactions

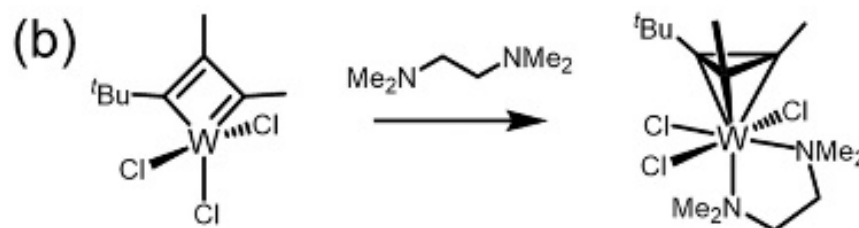
Alkyne Metathesis



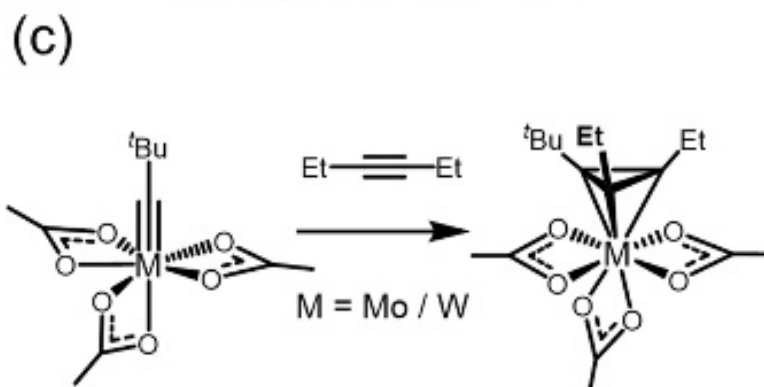
Formation of Mo(VI)/W(VI)-Based Metallatetrahedrane (MT_d)



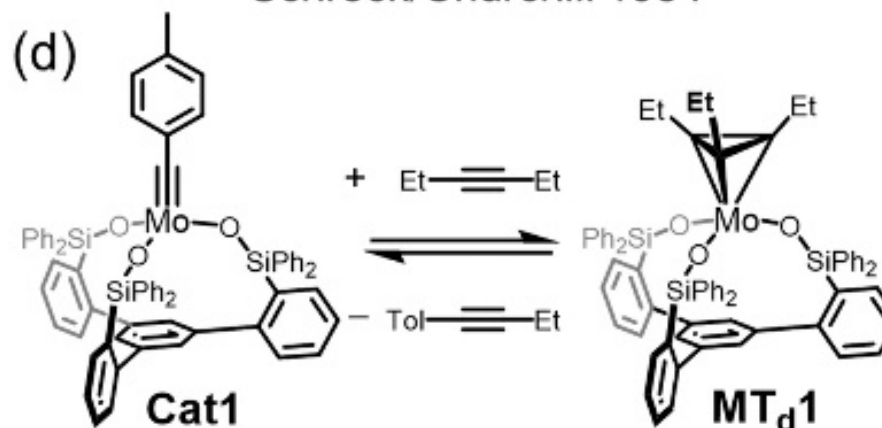
Schrock/Churchill 1984



Schrock/Churchill 1984



Schrock/Churchill 1986



Lee/Thompson 2019

Metallatetrahedrane is considered as an off-cycle decomposition product in alkyne metathesis.

Suggested Reading

Impact of Ligands and Metals on the Formation of Metallacyclic Intermediates and a Nontraditional Mechanism for Group VI Alkyne Metathesis Catalysts

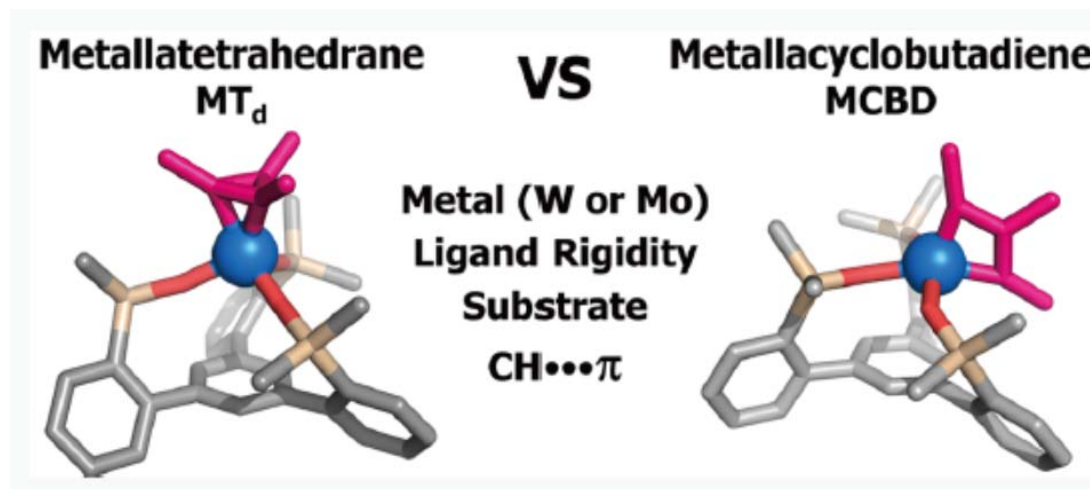
Richard R. Thompson,[§] Madeline E. Rotella,[§] Xin Zhou, Frank R. Fronczek, Osvaldo Gutierrez,^{*} and Semin Lee^{*}



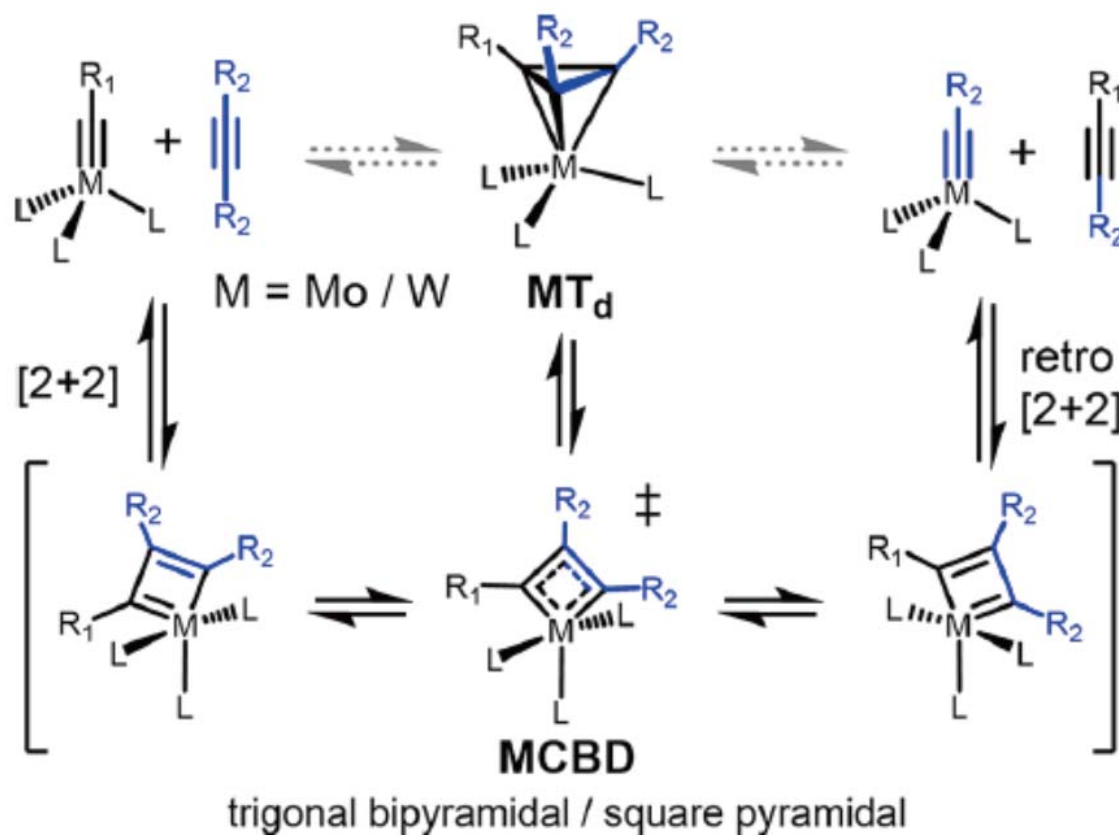
Cite This: *J. Am. Chem. Soc.* 2021, 143, 9026–9039



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Scheme 1. Commonly Accepted Mechanism for Alkyne Metathesis by Way of a MCBBD Intermediate^a



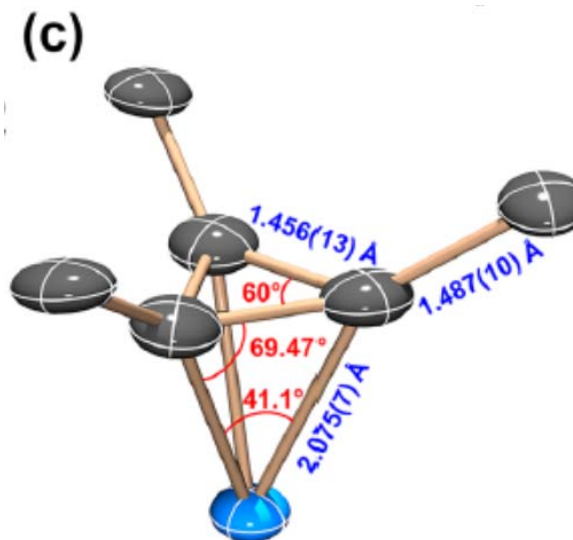
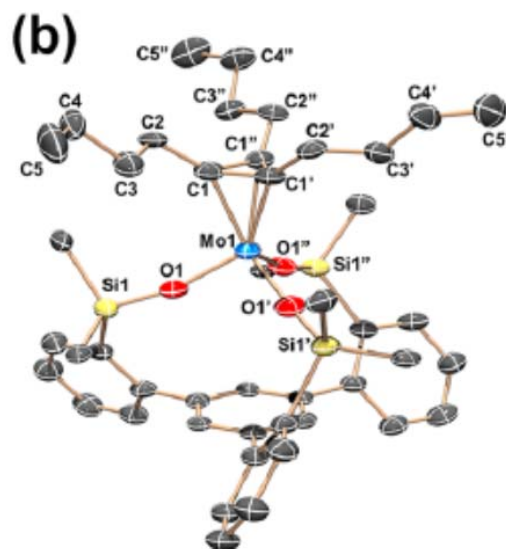
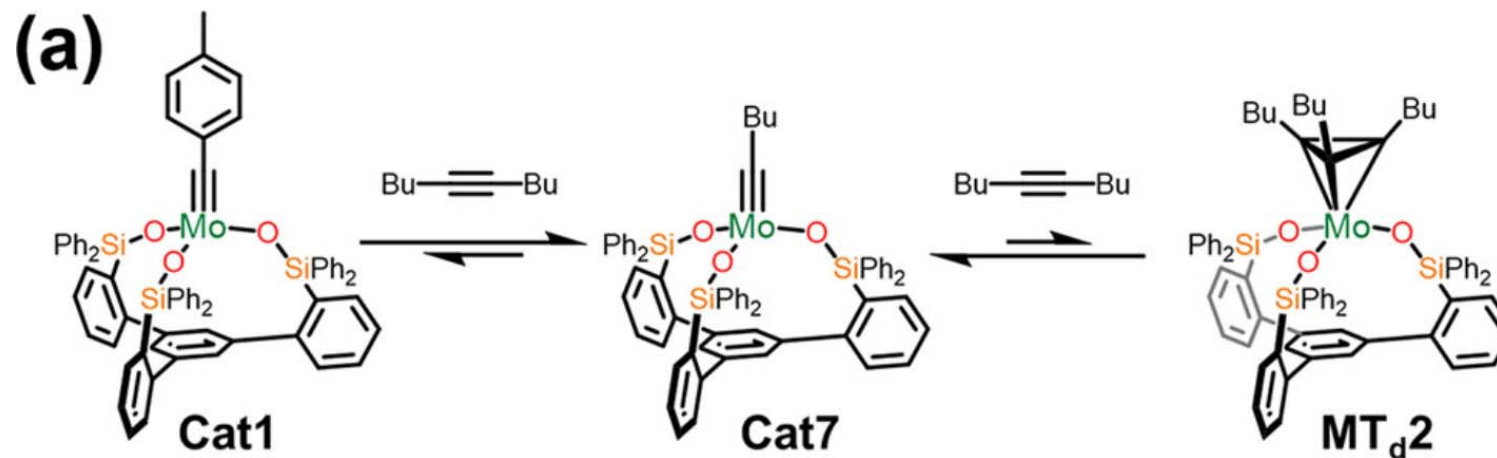
^aMT_d formation is considered as an off-cycle intermediate.

MCBD = Metallacyclobutadiene;
MT_d = Metallatetrahedrane

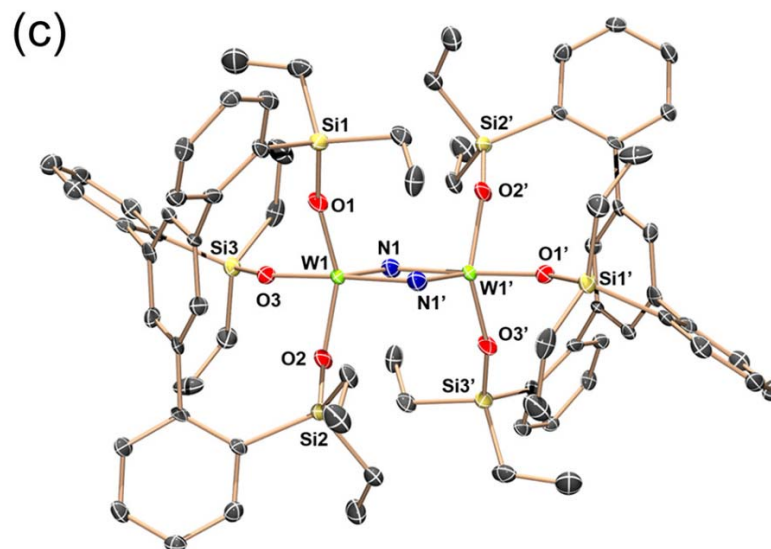
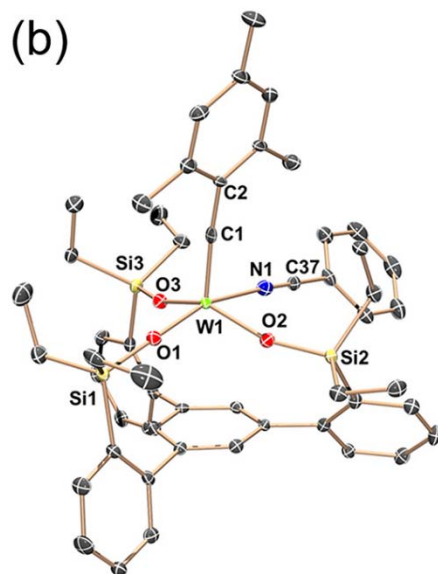
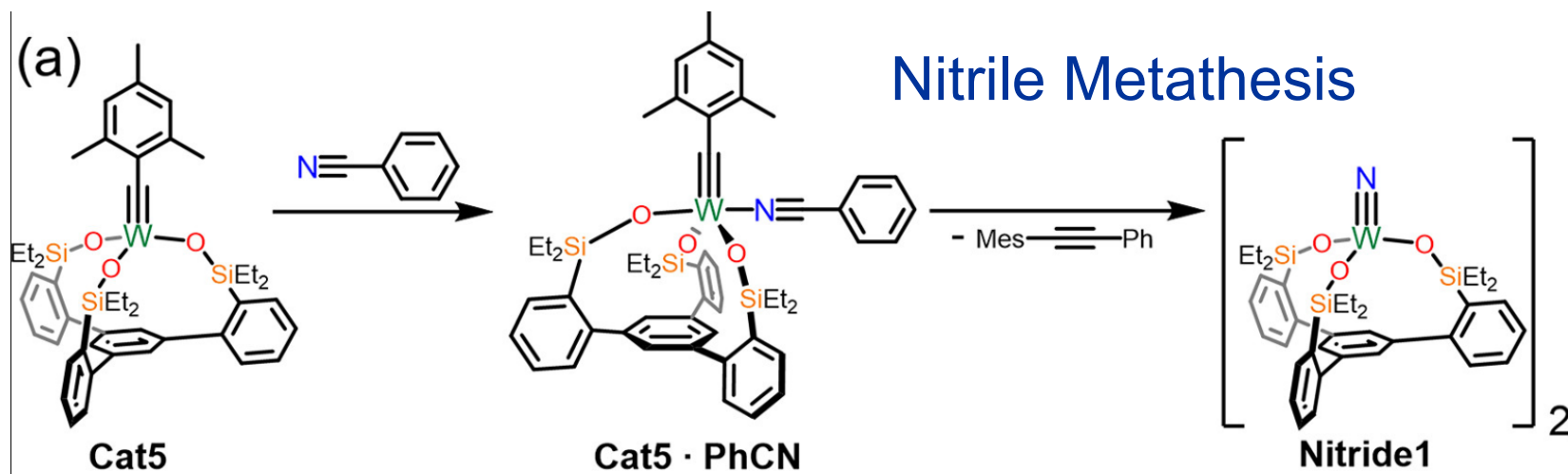
在基于[2+2]/反-[2+2]环加成的炔烃复分解机理中，金属杂环丁二烯已被认同是反应得以进行的关键中间体，反应中产生的MT_d物种被视为催化循环之外干扰有效产物形成的分解产物。最近的研究表明，从硅氧化物负载的钼基催化剂中分离出的中间体不是预期的金属杂环丁二烯，而是动态的金属四面体烷化合物，理论研究表明，这些物种在热力学上更稳定，并且在环加成中具有一定的能垒。

这篇文章里报道了VI族金属卡拜配合物的合成，以及中心金属性质、配体的柔性、二级配位环境和底物的性质对可分离出的中间体的影响。此外，也报道了配体位阻和金属选择对催化剂性能的影响。并通过DFT计算，阐明配体和金属在生成金属杂环丁二烯和金属四面体烷中间体中的机理和作用，以及它们对炔烃复分解的影响。

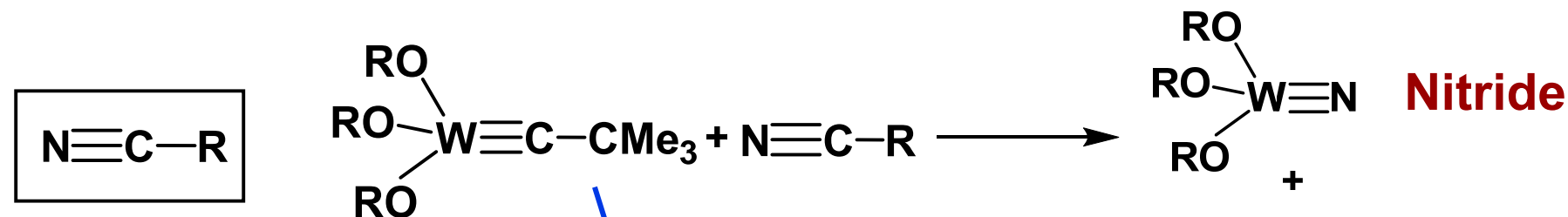
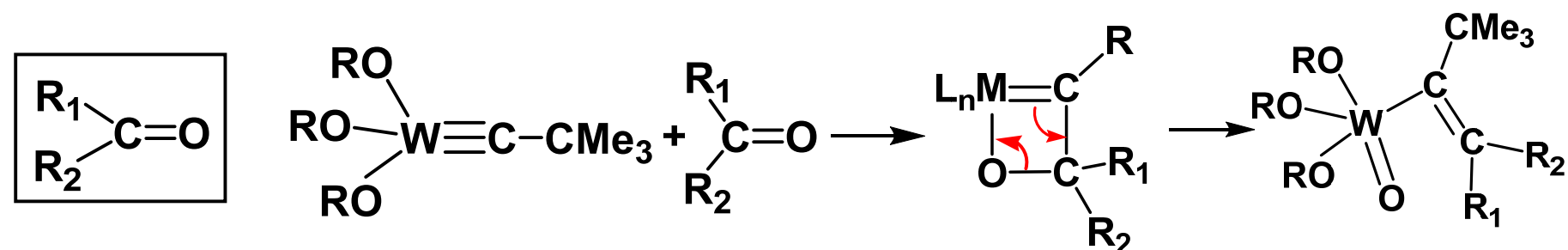
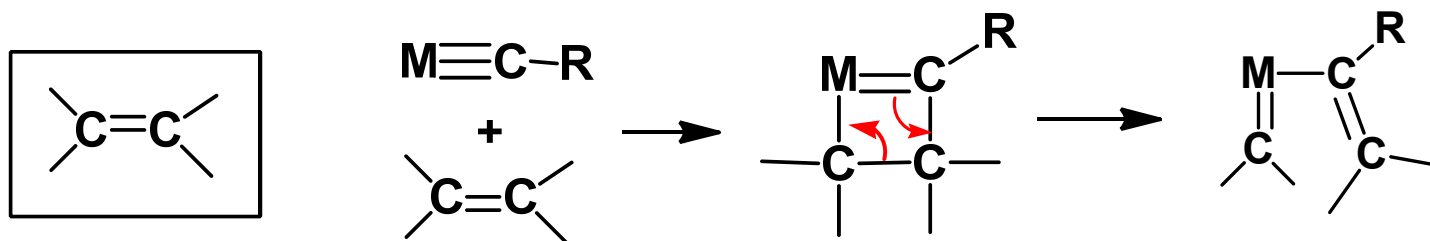
Formation of a pentylidyne (Cat7) and a metallatetrahedrane (MT_d2) using 5-decyne.



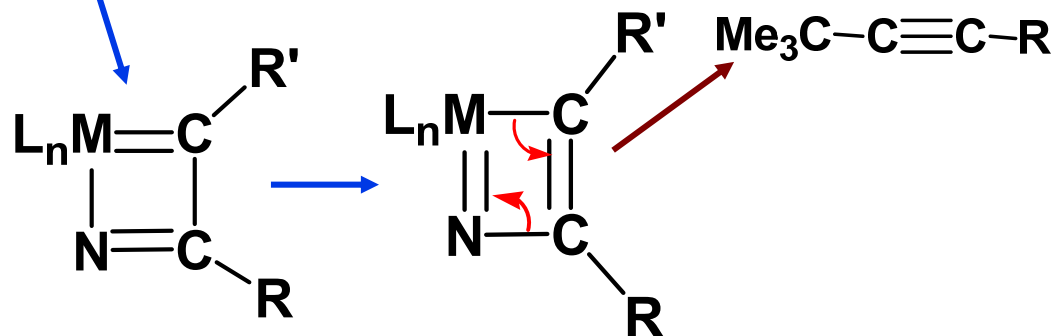
Synthesis of Nitride1 via nitrile metathesis with Cat5.



(3) Reactions of $M\equiv CR$ with $X=Y$ or other $X\equiv Y$ bonds

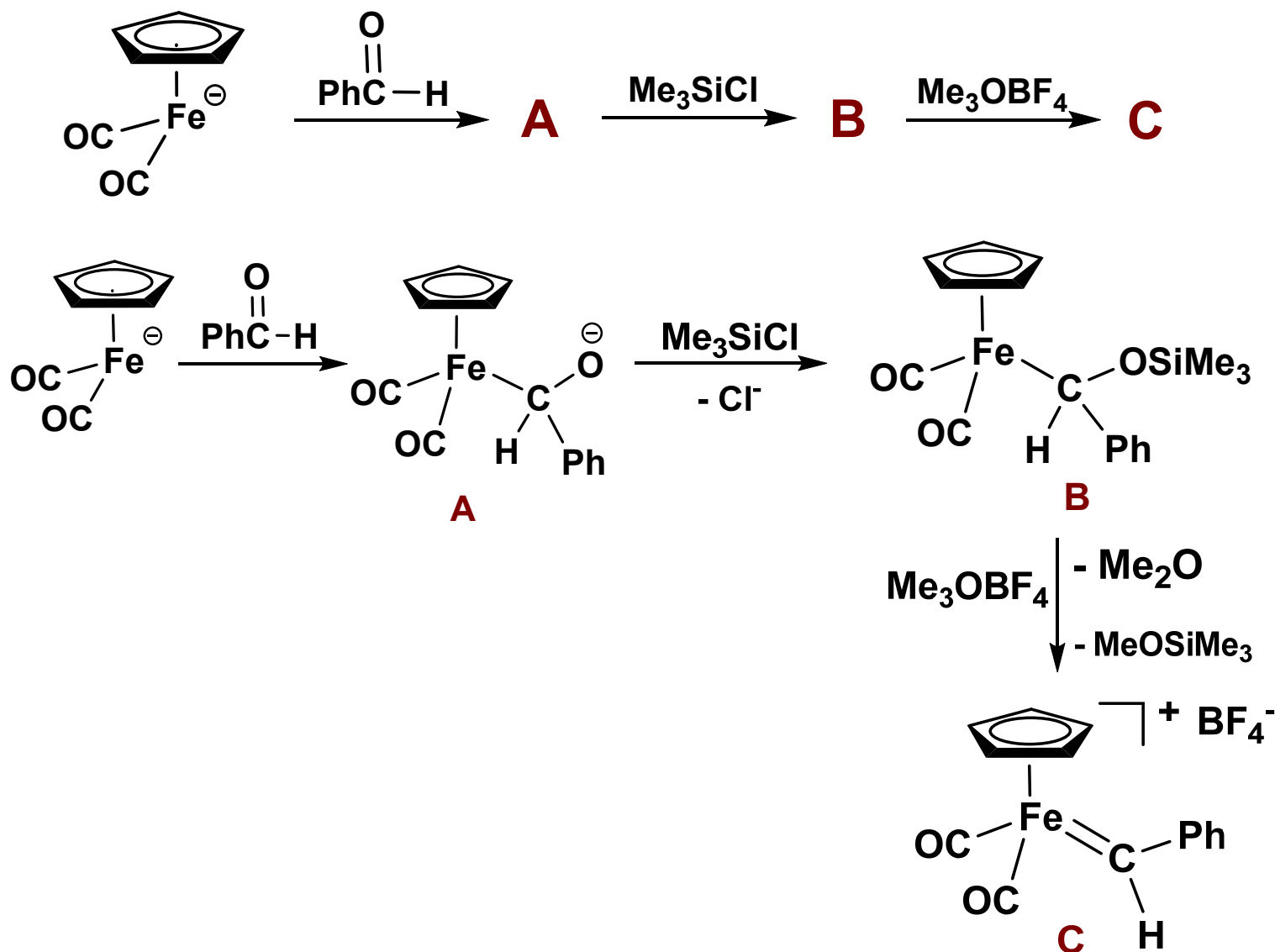


Nitrile Metathesis

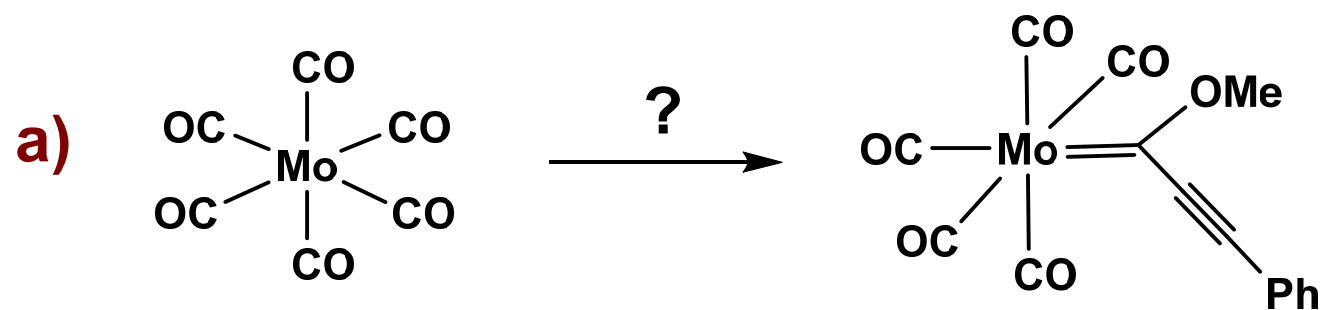


Practice Problems

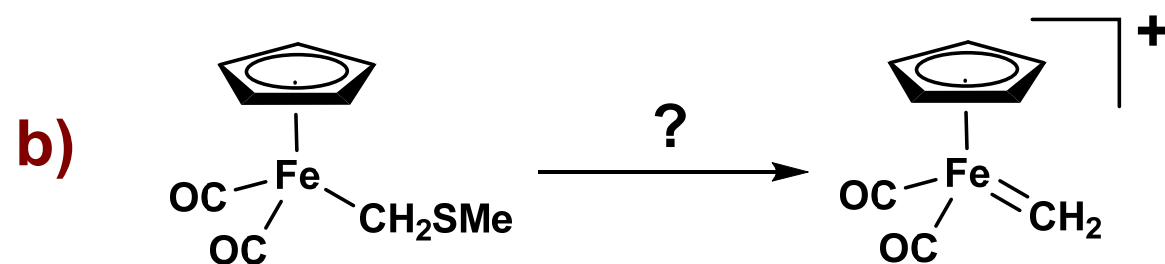
1. Predict the products



2. Suggest reagents



1) $\text{LiC}\equiv\text{CPh}$, 2) MeI



HBF_4 , or Me_3OBF_4